GNAMMA RESEARCH · PRIMARY TEXTS

المنطق

al-Manṭiq

Muḥammad Riḍā al-Muẓaffar

Reading edition

Contents

About this edition

Introduction to Logic

- Why logic is needed
- Definition and instrumental character of logic
- From sensation to intellectual knowledge
- Knowledge as the presence of a form
- Conception and assent
- What can be conceived, and what can be assented to?
- Certainty, opinion, doubt, and counter-opinion
- Simple and compound ignorance
- Immediate and acquired knowledge
- Thought as the intellect's movement between known and unknown
- The six principal divisions of logic

Book I · Expressions and Signification

- Why logic must investigate expressions
- Signification: rational, natural and conventional
- Verbal signification: total, partial, and concomitant
- The condition for implicative signification
- Five kinds of single expressions: specific, equivocal, transferred, coined, literal and figurative
- Synonymy and difference: words and conceptual meanings
- Three relations of distinct concepts: similarity, difference, and opposition
- The four kinds of opposition
- Simple and compound expressions, complete and incomplete compounds
- Assertion and performance: what can be true or false
- The three kinds of simple expression: verb, noun, and particle

Book II · Universals and Their Predication

- The universal and the particular
- The relative particular
- Univocal and graduated universals
- Concept and instance
- Concept as title, its referent, and two modes of predication
- The four relations between universal concepts
- Relations between complements of universals: equality and inclusion
- Complements of overlapping and disjoint universals
- The five predicables: species, genus, and differentia
- Relative species, higher and lower genera, and near and remote differentiae
- Essential and accidental predicates; proprium and common accident
- Relative predication, compound properties, and subspecies
- The three distinctions of predication
- Accidental attribution means predication, not mere inherence
- Divisions of accidental predicates: necessary, evident, demonstrable, and separable
- Logical, natural, and intellectual universals

The Four Questions

- The first question: explaining a word
- What is it? The explanatory question
- Does it exist? Simple and composite hal
- Why? The cause of assent and the cause of being
- Summary of the inquiries and their subsidiary forms

The Defining Account

- What is a defining account?
- The complete essential definition
- The incomplete essential definition
- Complete and incomplete descriptions
- Why definitions are difficult
- Definitions by example and by comparison

Conditions of Definition

- A definition must be coextensive and clearer
- No self-definition and no circularity
- Unambiguous language in definitions

Book IV · Propositions and Their Properties

- A proposition is intrinsically apt for truth or falsity
- Categorical and conditional propositions

Connected and disjunctive; affirmative and negative

Parts of categorical and conditional propositions

Categoricals by subject: singular, natural, indefinite, quantified

Why logic privileges quantified propositions

Quantifier expressions and symbols of four categorical forms

Personal, indefinite and quantified conditionals

Quantifying connected and disjunctive propositions

Mental, external, and real categorical propositions

Positive and negative terms: muhassala and maʿdūla

Real modality: necessity, impossibility, special and general possibility

Expressed modal qualifier versus a proposition's real matter

Eight simple modal proposition forms

Compound modal propositions: the implicit opposite-quality judgment

Two special compounds: qualified necessity and customary permanence

Actual yet not necessary; actual yet not permanent

Occurrent but nonpermanent; genuinely two-sided possibility

Seven authorial exercises on modal relations and compounds

Necessary versus coincidental connected conditionals

Disjunctive incompatibility: essential versus coincidental; genuine division

At-most-one versus at-least-one disjunctions

Conditionals made of conditionals: nested and mixed propositions

Displaced propositions and reconstructing their logical form

Four authorial exercises in classifying and restoring conditionals

Why relations between propositions are needed

Contradiction: need, definition, and the force of “in itself”

The eight unities and unity of predication

Differences in quantity, quality, and modality

Subalternation, contrariety, and subcontrariety

Simple conversion and its conditions

Conversion of universal and particular affirmatives

Conversion of the universal negative and its proof

Why particular negatives and disjunctives have no simple converse

Two methods of contraposition and their quantity rules

Two proofs for the contraposition of the universal negative

Proofs of the two contrapositives of the particular negative

Proofs of the two contrapositives of the universal affirmative

Why the particular affirmative has neither form of contraposition

Exercises in consequence, conversion and juristic contraposition

Complementation and the four proofs of predicate obversion

Transforming the original and handling complemented predicates

Seven exercises on obversion and methods of proof

Complete complementation and subject-complementation of the universals

Why neither particular admits complete or subject complementation

The author’s table of relations among quantified propositions

The logical axiom of immediate transformation

Book V · Argument and Inference

Book V: argument, its aim, and the three methods of inference

Syllogism: definition and the five qualifications

Syllogisms divided by matter and form

Categorical conjunctive syllogism: its three terms

The four figures and the conditions of the first

The second figure and its conditions

The third figure and its conditions

The fourth figure: conditions, moods, and proofs

Conditional conjunctive syllogism: terms, divisions, and connected premises

Disjunctive premises: converting affirmative disjunctions to connected conditionals

A connected and a disjunctive premise

Exceptional syllogism: composition, division, and conditions

Implicit syllogism and enthymeme

Compound syllogisms: definition and two forms

Induction: definition, complete and incomplete

Analogy: definition and its four components

Final exercises on syllogisms and the close of Part Two

Glossary

About this edition

140 source-anchored Arabic–English working passages from Muẓaffar’s six-book logic. Book V is available in full, from its preface through syllogism, induction, analogy and the final exercises. Earlier selection gaps in Books I–IV and all of Book VI remain.

Passage identifiers and chapter links remain stable across the online reader and these editions. The online dictionary and “Across the books” links connect specific discussions in Muẓaffar, Bidāya and Nihāya, with reasons and cautions where their treatments differ. Source page references belong to the cited printed edition; the PDF page numbers belong to this layout.

Sources and bibliographic references

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al-Manṭiq, searchable Arabic text

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al-Manṭiq: Logic, English translation by Shaykh ʿAbdul-Ḥakīm Carney

75-page user-provided partial English comparator, not republished. Its final substantive pages cover induction and analogy through Book V; the earlier assertion that it ended before Book III was incorrect. Continuous coverage of the whole six-book work is not established.

INTRODUCTION

Introduction to Logic

المدخل

M-P-01

INTRODUCTION

Why logic is needed

الحاجة إلى المنطق

God has made the human being naturally capable of speech and has given him a tongue as its instrument. Yet he still needs something to regulate and correct his speech, so that it accords with the language he learns in both the forms and materials of its expressions. He therefore needs, first, a teacher who habituates him to its use and, second, rules to consult that guard his tongue against error: these are grammar and morphology.

Likewise, God has endowed the human being with an intellective power and an aptitude for thought. Even so, people make many mistakes in thinking: they take something for a cause that is not a cause, regard as a consequence something that does not follow, and mistake what is not a demonstration for a demonstration. One may even hold a true conclusion on the strength of unsound premises. There is therefore a need for something that corrects thought, directs it towards valid inference, and trains it in the proper organisation of its operations.

Logicians describe logic as an instrument by which one can protect oneself against error and correct one's thinking. Grammar does not give one the ability to speak; it guides one to speak correctly. Similarly, logic does not impart the ability to think; it guides thought towards correctness.

Our need for logic, then, is the correction of our thinking—and how great that need is. If someone objects that people learn logic yet still reason badly, we answer that people learn grammar and morphology yet still make mistakes in speech. This happens because they have not acquired a settled mastery of the discipline, have failed to observe its rules when required, or have applied them incorrectly.

1. **Translator's note:** The distinction between the possession of a rational faculty and a trained habit (malaka) of correctly exercising it is central here. Logic is an instrumental discipline, not a substitute for intellectual attention.

2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Arabic text, Dār al-Taʿāruf (1402 AH) · p. 8–9 https://lib.rafed.net/view.php?b_id=1604&page=9&type=c_fbook

M-P-02

INTRODUCTION

Definition and instrumental character of logic

تعريف علم المنطق — المنطق آلة

Logic is accordingly defined as “a rule-governed instrument whose observance safeguards the mind against error in thinking.” Attend especially to the word *observance*. Learning logic does not automatically protect someone from mistakes, just as learning grammar does not automatically prevent errors of speech. The rules must be applied when they are needed.

Consider also the word *instrument*. Logic belongs to the instrumental sciences, which are employed for an end other than simply knowing their own propositions. It sets out the general, correct procedures by which thinking may attain truths not yet known, much as algebra investigates procedures for solving equations and arriving at unknown quantities.

More precisely, logic teaches general rules of correct reasoning applicable across the sciences. It investigates the structure and arrangement by which the mind moves from the intelligible contents already present to matters as yet unknown. Hence it has been called the *balance*, the *standard*, and the *servant of the sciences*. Even algebra, which served as our comparison, depends upon logical ordering in solving its problems.

A student of logic must therefore work exercises and actually employ its operations during study, as students of mathematics and the natural sciences must do.

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1. **Terminology:** Āla qānūniyya: an instrument governed by rules. The verb rāʿā (‘to observe, heed, apply’) is part of the definition; logic does not promise that mistakes become impossible merely by having studied a textbook.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Arabic text, Dār al-Taʿāruf (1402 AH) · p. 9–10 https://lib.rafed.net/view.php?b_id=1604&page=10&type=c_fbook

M-P-03

INTRODUCTION

From sensation to intellectual knowledge

العلم — تمهيد

We said that the human being is naturally disposed towards thinking and is capable of acquiring knowledge through an intellectual faculty that distinguishes him from non-rational animals. To clarify what we mean by knowledge and how logic is related to it, we may distinguish stages of

cognitive development.

First, the newly born child's soul possesses an innate capacity but no acquired, actual knowledge. As sight, hearing, taste, smell, and touch begin to operate, the soul is affected by the things presented to these senses. This is sensory cognition, the first degree of knowledge and a beginning from which further knowledge develops. Other animals share some or all of these sensory powers.

Second, the growing child begins to work upon retained sensory forms, comparing one with another—this object is taller, this light is brighter—and combining images into things not encountered externally. A town the child has only heard described may be imagined from previously seen towns. This is imaginative cognition (*ʿilm khayālī*), which some other animals also possess.

Third, cognition extends to particular meanings that are neither material nor measurable: the love shown by parents, the hostility of an enemy, the fear of a frightened person, the grief of someone bereaved, or the joy of a delighted person. This is estimative cognition (*ʿilm wahmī*). It is still not the distinctive perfection of the human intellect.

Fourth, the human being exercises intellect and reasoning over sensible, imaginative, and estimative contents. The intellect discriminates the true from the false, abstracts universal notions from particulars, compares these notions, and proceeds from what it knows to further judgments and conclusions. This is intellectual knowledge in its fullest sense.

The sciences and disciplines aim to cultivate this rational power. Logic, in particular, orders its activity and guards against its being misled by imagination or estimation.

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1. **Translator's note:** This passage is the author's graduated account of acquired cognition. The distinction between the faculty of estimation (wahm), imagination (khayāl), and intellect (ʿaql) is preserved rather than flattening all three into ‘thinking’.
2. **Authorial footnote — source:** In the printed source Muẓaffar explicitly restricts the present discussion to ʿilm ḥuṣūlī (knowledge through an acquired form). Knowledge by presence (ʿilm ḥuḍūrī) differs because the known is present by its very existence, not through an impressed representation. Only the former is here divided into conception and assent.
3. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Arabic text, Dār al-Taʿāruf (1402 AH) · p. 10–12 https://lib.rafed.net/view.php?b_id=1604&page=11&type=c_fbook

M-P-04

INTRODUCTION

Knowledge as the presence of a form

تعريف العلم

One may ask how these acts of cognition arise in a human being. To make the preceding explanation clearer, look at something before you, then close your eyes and direct your attention towards it. You will find something in yourself as though you were still seeing it. Likewise, after hearing a clock ticking, block your ears but turn your attention to the sound; something of the hearing seems to remain. The experiment can be repeated with the other senses.

On examining these experiences, the author invites us to understand cognition as the reception of the forms of things in the soul, by analogy with the way an image is impressed in a mirror. On this account, knowledge is defined as “the presence of a thing's form to the intellect,” or, equivalently for present purposes, “its being impressed upon the intellect.”

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1. **Interpretive note:** This is Muẓaffar's introductory presentation of representational or acquired knowledge, not a claim that every kind of knowledge is reducible to a material impression. See the preceding author's footnote on the distinct case of knowledge by presence.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Arabic text, Dār al-Taʿāruf (1402 AH) · p. 12–13 https://lib.rafed.net/view.php?b_id=1604&page=13&type=c_fbook

M-P-05

INTRODUCTION

Conception and assent

التصور والتصديق

Draw a triangle: its form comes to be present in the mind. This cognition is called *conception* (taṣawwur), considered here simply as an apprehension without affirmation or judgment. The same is true when you attend to its angles, or picture a horizontal line intersected by a perpendicular and thereby picture two right angles.

Now compare those two right angles with the sum of the triangle's angles. At first you may ask whether they are equal and remain uncertain. Even apprehending the *relation of equality* between them at this point is still mere conception.

Once you demonstrate their equality, however, a further cognitive state arises: you apprehend that the relation accords with reality, and this apprehension entails the soul's judgment and assent. This is called *assent* (taṣdīq), the cognition being named by its inseparable concomitant, the act of affirming that the relation obtains.

Thus the triangle's angles, the two right angles, and their proposed equality can all be conceived without judging. Assent occurs when the mind apprehends the relation as actually corresponding to reality. Recognising a proposition as *not* corresponding to reality is also a kind of assent—namely, a negative judgment.

Notice the qualification. In the broadest usage, conception, apprehension, and knowledge can all refer to the presence of a form to the intellect. Assent itself presupposes such an apprehension; but in the restricted contrast here, *conception* denotes apprehension without judgment, and *assent* denotes apprehension that entails judgment. Consequently, “conception” in the unrestricted sense may comprehend both.

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1. **Authorial context:** Muẓaffar's accompanying note relates this account of taṣdīq to Avicenna's discussion in the Ishārāt and to Ṣadr al-Dīn al-Shīrāzī's independent treatise on conception and assent. He does not simply identify taṣdīq with an extra pictorial representation.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf Arabic witness · p. 13–15 https://lib.rafed.net/view.php?b_id=1604&page=14&type=c_fbook

M-P-06

INTRODUCTION

What can be conceived, and what can be assented to?

بماذا يتعلق التصديق والتصور؟

Assent has only one immediate object: the relation expressed in an assertoric sentence, insofar as the mind judges that relation to correspond—or not correspond—to reality.

Conception, by contrast, can concern four things. First, a single expression: a noun, verb, or particle. Second, the relation asserted in a statement when we are uncertain of it or merely entertain it, as when we consider the possibility that Mars is inhabited without affirming it. Third, the relation expressed by an injunction, prohibition, wish, question, or other non-assertoric utterance; as such, these are not descriptions to be affirmed or denied as corresponding to reality. Fourth, an incomplete expression, such as a genitive phrase, a noun and its adjective, a relative clause, or one member of a conditional construction.

Take the Qurʾānic conditional, “If you count God's blessings, you cannot enumerate them.” Each of its components is apprehended as a component of the conditional, rather than as a separately asserted proposition. Likewise, “God's blessings” is an incomplete genitive expression. The conditional statement taken as a whole, however, is something to which assent may be given.

The logical distinction here concerns the *role in which* an expression is apprehended, not merely whether it contains words capable of forming a statement outside that construction.

1. **Translator's note:** Muẓaffar uses ‘maʿlūm taṣawwurī’ for a known content apprehended without assent. The conditional's antecedent and consequent do not become independently asserted merely by occurring within a larger proposition.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf Arabic witness · p. 15–16 https://lib.rafed.net/view.php?b_id=1604&page=16&type=c_fbook

M-P-07

INTRODUCTION

Certainty, opinion, doubt, and counter-opinion

أقسام التصديق

Assent is divided into *certainty* (yaqīn) and *probable opinion* (ẓann). It consists in preferring one of the two sides of a report—its obtaining or not obtaining. If the mind excludes the alternative altogether, this is certainty; if the alternative remains possible, though less likely, it is opinion.

Consider any report. Either you admit only one possibility, or you regard both alternatives as possible. The first condition is certainty. The second admits three possibilities: the alternatives may be equally balanced, in which case the state is *doubt* (shakk); the content of the report may be preferred, yielding probable opinion (ẓann); or the opposite alternative may be preferred, in which case the report's content is only a weaker apprehension (wahm).

There are thus four states, not five. (1) **Certainty:** you affirm the report and rule out its falsity, or deny it and rule out its truth. (2) **Opinion:** you favor one side while admitting the other as possible. (3) **Counter-opinion:** you entertain one side while favoring its opposite. (4) **Doubt:** you judge the alternatives equally possible.

Two consequences follow. In the classificatory sense used here, doubt and the weaker opinion are not themselves assent; and probable opinion about one side is always the weaker opinion concerning the opposing side. The corresponding terms are relational to whichever side of the proposition is being considered.

1. **Terminology:** Wahm here denotes the weaker side of a propositional assessment; it is not simply the psychological faculty of estimation described earlier. The identical word is used in two technical contexts.
2. **Authorial note:** The printed apparatus distinguishes a narrower sense of yaqīn: firm, factually correct assent that is not merely accepted by imitation. The chapter's initial division uses the broader sense of firm conviction.
3. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf Arabic witness · p. 16–17 https://lib.rafed.net/view.php?b_id=1604&page=17&type=c_fbook

M-P-08

INTRODUCTION

Simple and compound ignorance

الجهل وأقسامه

Ignorance is the absence of knowledge in a subject capable of knowing. We do not call a stone blind, because blindness is the lack of sight in something naturally fitted to see. Similarly, knowledge and ignorance are opposites in the manner of a possession and its privation.

Since knowledge is distinguished into conception and assent, one may speak correspondingly of ignorance affecting conception and ignorance affecting assent. The further distinction between *simple* and *compound* ignorance, however, strictly concerns the latter: ignorance in conception can only be simple.

Simple ignorance is not knowing something while recognising that one does not know it. Compound ignorance occurs when a person does not know something yet believes that he does know it. It comprises two deficiencies: ignorance of the thing and ignorance of that ignorance. Because it includes a mistaken conviction, it belongs to the sphere of assent.

Compound ignorance, Muẓaffar argues, should not be counted as genuine knowledge merely because it includes a firm belief. On his definition, knowledge presents the form of the very thing known. Mistaking a different form for it does not make that form the thing itself. A distant silhouette taken for a human being does not become a human being through the observer's conviction.

The point is about the truth-condition implicit in this account of knowledge, not an assertion that everyone who holds a mistaken belief knows nothing whatever.

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1. **Context:** This passage is useful for distinguishing doxastic firmness from knowledge: a firmly held false conviction remains compound ignorance rather than certainty in the narrower epistemic sense.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf Arabic witness · p. 17–18 https://lib.rafed.net/view.php?b_id=1604&page=18&type=c_fbook

M-P-09

INTRODUCTION

Immediate and acquired knowledge

العلم ضروري ونظري — توضيح في الضروري

Both conception and assent are divided into two kinds. The first is *immediate* or self-evident (*ḍarūrī*, *badīhī*): it does not require a further process of inquiry or discursive thought. Examples include apprehending the concepts of being and non-being and assenting that the whole is greater

than its part or that contradictories cannot both obtain.

The second is *discursive* or acquired (*naẓarī, kasbī*): it requires a process of thought. A conclusion about the movement of the earth, or a conceptual account of an unfamiliar phenomenon, may need prior knowledge to be arranged and used in an inquiry.

For immediate cognition it is sufficient that the mind attend to the relevant object without an intervening inferential process. For discursive cognition the mind must perform intellectual operations, much as one manipulates the terms of an algebraic problem, moving from what is already known to the initially unknown.

Self-evident, however, does not mean that every person actually knows every self-evident thing. The disposition of the subject and the conditions of attention matter. Muẓaffar distinguishes five conditions or occasions for the soul's attending to such knowledge: (1) attention rather than distraction, (2) a sound cognitive faculty, (3) functional senses when sense perception is involved, (4) freedom from a misleading sophism, and (5) some non-inferential operation appropriate to the kind of knowledge, such as receiving convergent testimony or undertaking an experiment.

The classification therefore concerns whether the knowledge depends on an *inferential act of thought*, not simply whether attaining it took time or effort.

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1. **Translator's note:** Do not translate ḍarūrī in this context as ‘necessary’ in the metaphysical modal sense. The distinction is between epistemic immediacy and discursive acquisition.
2. **Translator's note:** Muẓaffar's fourth condition is especially significant for kalām: a specious argument may obscure an otherwise evident principle, such as the impossibility of simultaneous affirmation and denial. He describes this as a sophism in opposition to an evident truth.
3. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf Arabic witness · p. 18–20 https://lib.rafed.net/view.php?b_id=1604&page=19&type=c_fbook

M-P-10

INTRODUCTION

Thought as the intellect's movement between known and unknown

تعريف النظر أو الفكر

By inquiry (*naẓar*) or thinking (*fikr*), Muẓaffar means the performance of an intellectual operation upon what is already present to the mind for the sake of attaining the object sought—knowledge of something previously unknown. More precisely, thought is “the movement of the intellect between the known and the unknown.”

When faced with a problem, one first identifies what kind of unknown it is, then turns to the stock of knowledge relevant to that kind of problem. The intellect searches and compares these contents, arranging them until it finds a combination capable of resolving the difficulty. It then moves from

this ordered knowledge to the desired conclusion.

The process may be analysed into five moments: (1) encountering the problem; (2) identifying its type; (3) moving from the problem to stored knowledge; (4) examining and combining that knowledge appropriately; and (5) moving from the assembled premises to the answer. The last three movements, strictly speaking, constitute discursive thought.

These stages often occur so swiftly that we are not consciously aware of them. In someone gifted with strong *ḥads* (intuitive insight), the first two discursive movements can be bypassed: the relevant knowledge yields the conclusion in a single immediate movement. This helps explain why a proposition may be self-evident for one mind but require discursive inquiry for another.

Here ‘movement’ names the ordered activity of cognition; it should not be confused with physical locomotion.

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1. **Terminology:** Ḥads in this passage signifies an immediate intellectual grasp that dispenses with some intermediate search. It should not be assimilated without qualification to the separate notion of mystical witnessing.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf Arabic witness · p. 20–22 https://lib.rafed.net/view.php?b_id=1604&page=21&type=c_fbook

M-P-11

INTRODUCTION

The six principal divisions of logic

أبحاث المنطق

Logic is needed above all to acquire discursive knowledge, because it provides the rules governing inquiry and thought. Immediate truths are available without discursive proof and serve as the initial capital from which further knowledge is obtained. As knowledge increases, the stock of what is already known grows, and further investigation becomes possible.

Logic therefore studies how to combine available cognitions so as to gain cognitions not yet possessed. When the unknown is conceptual, the route is through a *defining account* (muʿarrif); when the unknown is a judgment, the route is through an *argument* (ḥujja). The investigation of arguments may concern either the form of their composition or the character of their premises—the latter treated under the five argumentative arts.

The book is accordingly arranged into six parts: (1) expressions and signification; (2) universals; (3) definition and division; (4) propositions and their properties; (5) arguments and their formal composition; and (6) the five arts.

The architecture matters: the first three parts concern how to attain a previously unknown conception; the later parts explain how to attain or evaluate previously unknown judgments.

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1. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf Arabic witness · p. 23–25 https://lib.rafed.net/view.php?b_id=1604&page=24&type=c_fbook

BOOK 1

Book I · Expressions and Signification

الباب الأول — مباحث الألفاظ

M-L-01

READING SECTION 1

Why logic must investigate expressions

الحاجة إلى مباحث الألفاظ

The logician's principal concern is with intelligible meanings themselves. Nevertheless, the examination of expressions cannot be dispensed with, because expressions provide a means of reaching those meanings. People ordinarily communicate and exchange thoughts through one language or another, and words can undergo alteration, ambiguity, or confusion in ways that obstruct understanding. Logic therefore considers the conditions of expressions in general, rather than the peculiarities of one language, so that we can assess our own speech and that of others by an appropriate standard.

The qualification *in general* is important: logic is not a discipline restricted to speakers of a particular language. Sometimes a logical inquiry may require a question belonging to the grammar of the language in which it is expressed—for example, whether the Arabic definite article has a universal or exhaustive import, the logical role of *kāna* and related expressions, or the particles expressing generality and negation. But such matters can often be left to the sciences of language rather than incorporated into logic itself.

Thus inquiry into words is instrumental: it safeguards the passage between our linguistic expressions and the meanings that are the proper objects of logical investigation.

1. **Translator's note:** The distinction is between the logician's primary object (intelligible meanings) and the necessary instrumental investigation of language. Rules peculiar to Arabic grammar are not automatically universal laws of logic.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Arabic text · p. 29–32 https://lib.rafed.net/m/?b_id=1604&type=c_fbook

Signification: rational, natural and conventional

تعريف الدلالة وأقسامها

When you hear a knock at your door, your mind passes to the thought that someone outside is calling for you. The sound of the knock has disclosed the presence of that person. We say that it *signifies* the person's presence.

The knocking is the **sign** (*dāll*), the presence of the person is the **signified** (*madlūl*), and the sign's having this property is **signification** (*dalāla*). In general, whenever knowing one thing leads the mind to the existence of another, the first is a sign and the second what is signified. Signification is thus the condition of a thing whereby awareness of it leads the mind to another thing.

This intellectual transition cannot occur without an explanation. It depends upon a firmly established connection between the two items in the mind, itself grounded in awareness of a connection or concomitance between them. Because the connection can be essential, natural, or instituted by convention, signification is divided into three kinds.

1. Rational signification (dalāla ʿaqliyya). The sign and the signified have an intrinsic connection in their external existence, as effect to cause. When one knows that daylight follows the rising sun and sees the morning light on a wall, one infers that the sun has risen. Similarly, hearing a voice behind a wall signifies the presence of someone speaking. The inference rests upon a known causal or existential connection.

2. Natural signification (dalāla ṭabʿiyya). The connection arises from a disposition or customary reaction of the human constitution and need not occur invariably in every person. Some utter ‘ouch’ when in pain, ‘ah’ when distressed, or a sound of disapproval when annoyed; others crack their knuckles, stretch, fidget with an object or their beard, make certain gestures while thinking, or yawn when drowsy. One who has learned such patterns may infer a person's pain or mental preoccupation from an utterance or gesture. Unlike the causal concomitance invoked in rational signification, these dispositions can vary across individuals.

3. Conventional signification (dalāla waḍʿiyya). Here the connection is established through an agreement or convention that one thing is to serve as a sign of another. Examples include writing as a sign of spoken words, gestures used instead of speech, communication signals, mathematical and geometrical symbols, symbols in other sciences, and words instituted to express intentions. When the convention is known and the sign is apprehended, the mind passes to what it signifies.

The division therefore distinguishes **the ground of signification**, not merely the physical kind of sign. A sound or gesture may function differently according to whether the relevant link is causal, natural, or conventional.

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1. **Translator's note:** A ‘natural’ sign (ṭabʿī) is a disposition-dependent cue, potentially varying by person; a ‘rational’ sign (ʿaqlī) is inferred from an intrinsic causal or concomitant relation. This should not be conflated with the modern linguistic contrast between arbitrary and iconic signs.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Arabic text · p. 32–34 https://lib.rafed.net/m/?b_id=1604&type=c_fbook

M-L-03

READING SECTION 3

Verbal signification: total, partial, and concomitant

أقسام الدلالة اللفظية: المطابقية والتضمنية والالتزامية

Conventional signification is divided into **verbal** signification, where the instituted sign is an expression or word, and **non-verbal** signification, where the instituted sign is something other than a word—gestures, written marks, scientific symbols, road signs indicating distance or direction, and similar conventions.

The reason a word signifies a meaning is the established intellectual connection between expression and meaning. That connection arises from convention, insofar as the speakers know the convention. Verbal signification is therefore the condition of an expression through which knowing that the speaker has uttered it gives us knowledge of the meaning intended by the speaker.

A word signifies in three different respects.

1. Total or complete signification (dalāla muṭābaqiyya). The word signifies the whole of the meaning for which it was instituted. ‘Book’, for instance, signifies the complete object including its pages, markings and cover. ‘Human’ signifies the whole defining account ‘rational animal’. This is the basic signification for which the expression was instituted.

2. Inclusive or partial signification (dalāla taḍammuniyya). The word signifies an essential part included within its full instituted meaning. ‘Book’ signifies its pages or its cover; ‘human’ signifies animality or rationality as a constituent of its definition. A purchaser of a book understands its cover to be included. If the seller later attempted to exclude it, the purchaser could appeal to what was contained in the meaning of ‘book’. Partial signification depends upon total signification because the part is understood through the prior meaning of the whole.

3. Concomitant or implicative signification (dalāla iltizāmiyya). The expression leads the mind to a meaning **outside** its instituted meaning that is nevertheless connected to it as a necessary or familiar concomitant. ‘Inkwell’ may imply a pen. If someone requested an inkwell and the

respondent brought no pen, the requester might appeal to the understood implication. This implicative signification also depends upon total signification: to infer something associated with a meaning, the mind must first understand the meaning actually signified by the expression.

The distinctions concern whether the signified content is the **whole**, an **internal constituent**, or an **external concomitant** of the word's instituted meaning. They should not be confused with the speaker's intention to assert some particular content.

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1. **Authorial apparatus · paraphrase:** The author explains why the verbal/non-verbal division is emphasised for conventional signification: although rational and natural signs may themselves involve sounds or words, the distinction has special logical utility for conventional verbal signification and its three forms.
2. **Terminological note:** Taḍammun is the signification of a part included within the whole instituted meaning; iltizām is signification of what is outside the instituted meaning yet connected to it. Do not render both simply as implication.
3. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Arabic text · p. 34–36 https://lib.rafed.net/m/?b_id=1604&type=c_fbook

M-L-04

READING SECTION 4

The condition for implicative signification

شرط الدلالة الالتزامية

Implicative signification requires a **mental** connection between the instituted meaning of the expression and the further meaning that follows from it. A concomitance existing only externally is insufficient if it has not been established in the mind: without that, the mind would not pass from the expression's meaning to the further meaning.

The connection must also be evident or immediately apparent. On apprehending the meaning of the expression, the intellect must be able to pass to its concomitant **without needing a further mediating step**.

Thus a logical or physical association, however strong, does not by itself constitute signification by implication for someone who has never grasped the relation. The necessary condition is an intelligible and evident mental concomitance.

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1. **Translator's note:** The particle ‘also’ (أيضاً) adds a separate condition: not only must the relation be mental, it must be sufficiently evident to produce the transition without further mediation. In usūl al-fiqh and later logical discussions the precise scope of ‘evident concomitance’ may be further distinguished.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Arabic text · p. 36 https://lib.rafed.net/m/?b_id=1604&type=c_fbook

Five kinds of single expressions: specific, equivocal, transferred, coined, literal and figurative

المختص والمشترك والمنقول والمرتجل والحقيقة والمجاز

After the analysis of signification, we turn to general divisions of expressions according to their meanings—divisions not peculiar to any one language. There are three perspectives: an expression considered singly, several expressions compared with one another, and expressions considered in general, whether single or multiple.

The five classes of a single expression. An expression that signifies a meaning through one of the three previously discussed forms of signification either has one meaning, in which case it is *specific*, or has several meanings. Expressions with several meanings fall into four further classes, giving five in all.

1. Specific (mukhtaṣṣ): an expression assigned to only one meaning, such as “iron” or “animal.”

2. Equivocal (mushtarak): an expression conventionally assigned to several distinct meanings, each independently, with no one assignment prior to the others. The Arabic word *ʿayn*, for example, may signify the eye, a spring of water, gold, and other things; *jawn* may signify black or white. Arabic has many such expressions.

3. Transferred (manqūl): like an equivocal expression, it has several conventionally established meanings, but one assignment precedes another and an affinity between the meanings is considered in the later assignment. Thus *ṣalāh* originally means supplication but was transferred in the Islamic law to specific acts of standing, bowing, prostration, and the like. *Ḥajj* originally means intention or setting out, then became the designation of the specific pilgrimage to Mecca performed with particular acts at an appointed time. Similar transfers occur in the terminology of law, sciences, and arts, including modern terms for the motorcar, aeroplane, telephone, and radio.

A transferred expression may be named after the community whose usage effects the transfer: a legal, logical, grammatical, or philosophical transfer.

4. Coined without semantic connection (murtajal): this resembles the transferred expression except that no relationship between the earlier and later meanings is observed. Many personal names belong to this category.

5. Literal and figurative (ḥaqīqa and majāz): a word is conventionally assigned to one meaning but used for another through an affinity between them, without the second meaning becoming established by a separate convention. It is literal in its first meaning, figurative in the second. Figurative use invariably requires a contextual indication (*qarīna*) that diverts the hearer from the literal sense and identifies which figurative sense is intended.

Two cautions. First, equivocation and metaphor should not enter definitions or demonstrations unless a contextual indication fixes the intended meaning. The same is true of transferred or newly coined uses while their original sense remains current; when the old sense has been abandoned, that fact itself can indicate the later sense. Even where contextual indication is supplied, scientific prose should preferably avoid metaphor.

Second, transfer of meaning occurs either **by explicit assignment (taʿyīnī)**, when some identifiable party intentionally determines a new usage—as often happens in scientific nomenclature—or **by established usage (taʿayyunī)**, when a community repeatedly uses a word in a nonliteral meaning without expressly intending to establish a new convention. The usage becomes widespread until the new sense is grasped without a contextual clue, and the figurative usage thereby becomes literal through an established mental association.

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1. **Textual scope:** Main authorial prose of pp. 38–41 rendered in sequence. Exercises after this division are not in this reading unit; punctuation is normalized from the searchable source. https://lib.rafed.net/view.php?b_id=1604&page=39&type=c_fbook
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH searchable Arabic text (normalized punctuation; printed 1968 witness uncollated) · p. 38–41 https://lib.rafed.net/view.php?b_id=1604&page=38&type=c_fbook

M-L-06

READING SECTION 6

Synonymy and difference: words and conceptual meanings

الترادف والتباين

When one expression is compared with another, or with several others, the expressions fall into one of two classes.

Synonyms (mutarādifa) are several expressions assigned to one and the same meaning. Examples include the Arabic words *asad*, *sabʿ*, and *layth* (“lion”), *hirra* and *qiṭṭa* (“cat”), and *insān* and *bashar* (“human being”). Synonymy is the sharing of one meaning by multiple expressions.

Different expressions (mutabāyina) are those each assigned a meaning distinctive to it—for example, book, pen, heaven, earth, animal, inanimate object, sword, and *ṣārim* (a cutting sword).

This difference means that the meanings multiply with the expressions. It must not be confused with **disjointness of extensions**, which will be examined later. Two meanings can be conceptually different even if their instances overlap partly or entirely. Thus *sword* and *ṣārim* differ in meaning because the latter specifically signifies a cutting sword, yet their instances overlap. Likewise *human being* and *rational* differ conceptually, although under the traditional account all humans are rational and all rational beings are human. This is difference in **intension**, not necessarily difference in **extension**.

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1. **Distinction:** The author explicitly warns against confusing tabāyun of conceptual signification with tabāyun of external instances; this anticipates the four extensional relations in Book II.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH searchable Arabic text (normalized punctuation; printed 1968 witness uncollated) · p. 42–43 https://lib.rafed.net/view.php?b_id=1604&page=42&type=c_fbook

M-L-07

READING SECTION 7

Three relations of distinct concepts: similarity, difference, and opposition

قسمة الألفاظ المتباينة: المثلان، المتقابلان، المتخالفان

Expressions with different meanings are *mutabāyina*, whose signified contents are diverse. Their meanings can differ in three ways: **similarity, difference, and opposition**. When we attend to the two things' participation in one nature, we speak of similars (*mithlān*). When we do not attend to such shared nature, the things are opposites (*mutaqābilān*) if their meanings cannot coexist in the same subject, in the same respect, and at the same time; otherwise they are merely different (*mutakhālifān*).

1. Similars. Two things are similar insofar as they share a nature *under the consideration of that shared nature*. Muḥammad and Jaʿfar, for example, share humanity. Human being and horse share animality. Yet, considered in their personal distinctness rather than in what is common, Muḥammad and Jaʿfar are different; likewise human being and horse when considered as those two kinds. Where the shared nature is specific, the two are called similars in the strict sense. Where it is generic, they are of the same genus (*mutajānisān*); where quantitative, equal (*mutasāwiyān*); and where qualitative, resembling (*mutashābihān*). Similarity is the general name embracing these cases. Two numerically distinct similars do not, as such, become one and the same individual.

2. Things that differ. These are considered in their diversity, without taking account of what they may share. Two qualities may be different while capable of belonging to the same subject: blackness and sweetness, length and thinness, courage and generosity, whiteness and heat. The difference may be personal (Muḥammad versus Jaʿfar), specific (human versus horse), or generic,

as in cotton and snow considered as different things despite their common whiteness. The very same pair can accordingly be *similar* under one consideration and *different* under another. Nor is the category limited to things that can physically coexist: many mutually separate things are different without being logical opposites. Sometimes “different” is also used broadly for everything that is not similar, in which case it includes opposites.

3. Opposites. These are mutually excluding meanings that cannot be realized in one subject, in one respect, at one time: human and non-human, blind and sighted, fatherhood and sonship, blackness and whiteness. The condition “one subject” is needed because whiteness of paper and blackness of ink can both exist, but not as the opposed properties of the **same** subject. “One respect” is needed because a person may be a father to one individual and a son to another. “One time” is needed because the same object may be cold now and hot later. The three restrictions prevent equivocal claims of contradiction.

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1. **Technical caution:** The source uses 'two similars do not coexist' as a logical rule about numerical sameness, not the implausible claim that two people of the same species cannot be present in the same place.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH searchable Arabic text (normalized punctuation; printed 1968 witness uncollated) · p. 43–45 https://lib.rafed.net/view.php?b_id=1604&page=43&type=c_fbook

M-L-08

READING SECTION 8

The four kinds of opposition

أقسام التقابل: النقيضان والملكة والضدان والمتضايفان

Opposition has four species.

1. Contradictory opposition (taqābul al-naqīḍayn), or affirmation and negation: human/non-human, black/non-black, luminous/not luminous. Contradictories consist of a positive determination and its negation. By the fundamental law of thought, they can neither obtain together nor both be absent, and no intermediary is possible.

2. Opposition of possession and privation (malaka wa-ʿadamuhā): sight and blindness, being married and being unmarried. Blindness applies only to something naturally suited to possess sight; it is not non-sight in every possible subject. Similarly, unmarried status applies only to something capable of marriage. Unlike contradictories, both terms can be inapplicable to a subject unsuited to the positive disposition. A stone is neither blind nor sighted, neither married nor unmarried. The pair is positive and privative: the terms cannot belong together, but both may be absent from a subject incapable of the possession.

3. Contrary opposition (taqābul al-ḍiddayn): hot/cold, black/white, virtue/vice, rashness/cowardice, lightness/heaviness. Contraries are two positive attributes successively capable of belonging to one subject but incapable of existing together in that same subject. Conceiving one does not require conceiving the other. The phrase “successively belonging to one subject” indicates that contraries are attributes, not two distinct substances such as human and horse or animal and stone. The final clause distinguishes contraries from correlatives, whose conceptual grasp is reciprocal.

4. Correlative opposition (taqābul al-mutaḍāyifayn): father/son, above/below, earlier/later, cause/effect, creator/creature. Three points explain their distinctive structure. First, conceiving either requires conceiving its correlate: to understand something as a father or cause is to understand it as having a son or effect. Second, one thing cannot occupy the two relational positions in the same respect: a person cannot be father and son to the same person in that very relation, although he can be father to one and son to another. Likewise a thing cannot stand both above and below itself in a single comparison. Third, the two terms need not apply to every possible subject: the Necessary Existent has neither an above nor a below, and a stone is neither father nor son. Where a pair such as cause/effect is held to exhaust existents, that follows from something specific to that pair, not from correlation merely as correlation.

Correlatives are thus two positive descriptions necessarily conceived together, incapable of belonging to the same subject in the same respect, and capable of both being inapplicable.

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1. **Translation:** The source treats reciprocal conception as part of correlative opposition. Do not confuse this with the claim that the relata are the same concrete individuals, or with equal extensions.
2. **Source:** al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH searchable Arabic; punctuation normalized, 1968 printed witness not yet fully collated · p. 45–47 https://lib.rafed.net/view.php?b_id=1604&page=45&type=c_fbook

M-L-09

READING SECTION 9

Simple and compound expressions, complete and incomplete compounds

المفرد والمركب؛ المركب التام والناقص

An expression considered without regard to whether it is one word or several falls into two kinds.

A. Simple expression (mufrad). For logicians, this means either an expression that has no parts, such as the prefixed Arabic letter *bāʾ* in “I wrote with a pen,” or the one-letter imperative *qi* from *waqā, yaqī*; or an expression with parts whose parts do **not**, considered as parts, independently signify components of its meaning. Examples include Muḥammad, ʿAlī, *qaraʾa* (“he read”), ʿAbd

Allāh, and ʿAbd al-Ḥusayn. When the last two are personal names, the names' internal elements are not used to signify distinct constituents of the person's individual identity, any more than a letter in Muḥammad's name does.

The same sounds can have a different logical status in another context. In “Muḥammad is God's servant and messenger,” *ʿabd Allāh* means “servant of God”: its components carry separate significations and the phrase is compound, not simple. But in “Muḥammad, son of ʿAbd Allāh,” the father's proper name is a single designation.

Grammarians sometimes call that proper name *compound* because the parts may have different grammatical inflections. The grammatical criterion is morphology and case construction; the logician considers **signification**. Thus a simple expression is one *with no part that signifies a part of its meaning while functioning as a part of the expression*.

B. Compound expression (murakkab), also called a *qawl*: its components, considered as components, signify components of the complete meaning. In “Wine is harmful,” each term contributes its own part. Likewise, “Backbiting is the effort of the incapable” contains a compound predicate, while “The worst of friends is one for whom pretence is necessary” contains internal compounds.

Compound expressions are either **complete** or **incomplete**. A complete compound supplies the hearer with a finished meaning, so the speaker may stop speaking: “Patience is courage”; “A person's worth is what they do well”; “When you know, act.” It is defined as **that on which the speaker may properly stop**.

An incomplete compound leaves the hearer awaiting something further: “The worth of every person ...” or “When you know ...” without the consequent. It is accordingly **that on which the speaker cannot properly stop**. Complete compounds are further divided into assertions and performative or nonassertoric utterances.

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1. **Editorial:** The proper-name examples are given in the Arabic and explicitly distinguish logicians' semantic criterion from grammarians' inflectional criterion.
2. **Source:** al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH searchable Arabic; punctuation normalized, 1968 printed witness not yet fully collated · p. 49–50 https://lib.rafed.net/view.php?b_id=1604&page=49&type=c_fbook

READING SECTION 10

Assertion and performance: what can be true or false

الخبر والإنشاء

Every complete compound contains a relation among its components, called a **complete relation** (*nisba tāmma*). Such a relation can be of two different kinds.

1. Assertion (khabar). Sometimes the relation possesses a determinate reality independently of the words: the expression describes or discloses that reality. For example, if rain has fallen or is expected tomorrow, one may say “It rained” or “It will rain tomorrow.” This is an assertion, also called a proposition (*qaḍiyya*) or statement (*qawl*). It need not correspond to what actually obtains: if it corresponds it is **true**, and if it does not it is **false**. An assertion is therefore a **complete compound properly describable as true or false**. The logician chiefly investigates these expressions, since assent concerns them.

2. Performance or nonassertoric utterance (inshāʾ). Sometimes the complete relation has no independent reality to which the utterance could correspond: the speaker establishes the intended relation through the expression itself. Its meaning, in this technical sense, comes into being through the act of speaking. Thus it is not an independently subsisting state of affairs of which the words are a true or false report.

Examples are: **command**, “Memorize the lesson”; **prohibition**, “Do not accompany advocates of evil”; **question**, “Is Mars inhabited?”; **calling**, “O Muḥammad!”; **wish**, “If only we could return and be among the believers!”; **exclamation**, “How great is the human being's danger!”; **contractual enactment**, such as formulae establishing sale, lease, or marriage—“I sold,” “I leased,” “I gave in marriage”; and **unilateral legal enactment** (*īqāʿ*), such as formulae for divorce, manumission, or endowment—“So-and-so is divorced,” “My slave is free.”

In the author's logical analysis these expressions do not describe independently established propositions to which they either correspond or fail to correspond. Their significance is constituted by the speaker's utterance and intention; consequently they are not properly characterized as true or false. Hence a performative utterance is **a complete compound not properly describable as true or false**.

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1. **Source footnote:** The printed note says that the qualification «لذاته» ('in itself') will later be added to the definition of assertion and performance in the treatment of propositions. The later refinement matters where grammatical form and pragmatic force diverge.
2. **Terminology:** A legal formula is classified here by logical signification; the note is not a ruling on whether a given utterance actually effects a valid legal contract or divorce.
3. **Source:** al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH searchable Arabic; punctuation normalized, 1968 printed witness not yet fully collated · p. 50–51 https://lib.rafed.net/view.php?b_id=1604&page=50&type=c_fbook

READING SECTION 11

The three kinds of simple expression: verb, noun, and particle

أقسام المفرد: الكلمة والاسم والأداة

Simple expressions divide into three kinds: **word (kalima)**, **noun (ism)**, and **particle (adāh)**. The logician's terminology here differs somewhat from that of the grammarian.

1. The word (kalima), or verb. Consider the three Arabic verbs *kataba*, *yaktubu*, and *uktub* (“he wrote,” “he writes,” “write!”). First, they share the same root letters—kāf, tāʾ, and bāʾ—and the same independent lexical meaning of writing. Second, their verbal patterns differ, and each pattern signifies a distinct **complete temporal relation** between that meaning and an indefinite agent. *Kataba* relates writing to an agent in past time; *yaktubu* relates the event's renewed occurrence to an agent in the present or future; *uktub* expresses a request that someone perform writing now.

The shared lexical material therefore signifies the shared autonomous meaning; the differing forms signify the differing temporal relations. A verb can thus be defined as **a simple expression whose lexical material signifies an independent meaning, while its form signifies a complete temporal relation of that meaning to an unspecified agent**.

The specification “complete relation” excludes derived nouns, such as active and passive participles or names of time and place. They too signify an independent lexical content through the root and a relation through their form, but the relation is incomplete rather than a complete temporal predication.

2. Noun (ism). A simple expression signifying a meaning that is independent in itself, without a form signifying a complete temporal relation. Examples are Muḥammad, human being, writer, and question. It may carry a form signifying an **incomplete** relation, as participles and nouns of time do, because it signifies a bearer of the lexical content.

3. Particle (adāh). Corresponding to the grammarians' *ḥarf*, this signifies a relation between two terms: *fī* (“in”) expresses containment, *ʿalā* (“upon”) elevation or superposition, and *hal* introduces interrogation. A relation is not independently intelligible in the same way as a substance or free-standing meaning because it requires its two terms. A particle is therefore **a simple expression signifying a meaning that is not independent in itself**.

A final note: the logician may classify the “defective verbs” *kāna* and its sisters with particles, because in this analysis they indicate only a temporal relation rather than an independently signified event. In “Muḥammad was standing,” the word *standing* supplies the event-content.

Grammarians, using different criteria, class them as verbs; some logicians call them “existential words” (*kalimāt wujūdiyya*).

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1. **Conceptual note:** This chapter supplies a precise logical parallel to Ṭabāṭabāʾī’s discussions of relational or connective existence: a particle signifies a dependent relation, whereas noun and verb signify an autonomous meaning to differing degrees.
2. **Source:** al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH searchable Arabic; punctuation normalized, 1968 printed witness not yet fully collated · p. 51–53
https://lib.rafed.net/view.php?b_id=1604&page=51&type=c_fbook

BOOK 2

Book II · Universals and Their Predication

الباب الثاني — مباحث الكلي

M-K-01

READING SECTION 1

The universal and the particular

الكلّي والجزئي

A human being apprehends concepts of the individual things he perceives: Muḥammad, this book, this pen, this rose, Baghdad, Najaf, and so forth. On considering them, he finds that each applies to no other individual and is true only of that particular existent. This is a *particular concept* (juzʾī), which may be defined as a concept incapable of applying to more than one individual.

When a person observes several particulars and compares them, however, he may discover a feature shared by them. He then abstracts a conceptual form that embraces all of them and applies to each. Such an encompassing concept, or abstracted form, is a *universal concept* (kullī): a concept whose application to more than one individual is not impossible.

Examples include human being, animal, mineral, white, apple, stone, learned, ignorant, sitting in the house, and confessing one's wrongdoing.

1. **Translator's note:** The distinction concerns a concept's aptitude to be predicated of multiple individuals, not the number of instances presently existing.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf (1402 AH), checked against the digitized searchable text; early al-Nuʿmān edition catalogued by Arabic Collections Online as public domain · p. 56–57 https://lib.rafed.net/view.php?b_id=1604&page=56&type=c_fbook

M-K-02

A universal need not have individuals that actually exist. The intellect may form a universal concept capable of applying to more than one individual without abstracting it from presently existing particulars; it may instead suppose individuals of which it would be true. Indeed, even a single actual instance may be impossible, as with the concepts of a partner to the Creator or the conjunction of contradictories. This does not impair the universality of the concepts.

Conversely, a universal may have only one existent instance, with any second instance being impossible. The *Necessary Existent* is an example, since its uniqueness is established by demonstration. Yet the intellect can consider supposed individuals of which this concept would be predicated *if they existed*. Were the concept of the Necessary Existent intrinsically particular, there would be no need for a demonstration of divine unity: apprehending the concept itself would suffice to exclude multiplicity. Its restriction to one existent instance therefore follows from something other than the concept considered just as a concept; the concept's own signification does not prevent its supposition of multiple instances.

Consequently, both definitions must include the qualification **“even by supposition.”** A particular is a concept that cannot be true of many, even hypothetically; a universal is one for which such application is not impossible, even hypothetically.

Note that what particles signify is particular. Verbs signify particular meanings through their morphological forms, but universal meanings through their lexical materials. Nouns differ: common nouns may signify universals, whereas proper names, demonstratives and pronouns may signify particulars.

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1. **Translation and argument:** The impossibility of a second Necessary Existent is a metaphysical result of demonstration, not the same as the logical incapacity of a deictic or singular concept to apply to more than one. The distinction anticipates the discussions of existence in the Ḥikma books.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf (1402 AH), checked against the digitized searchable text; early al-Nuʿmān edition catalogued by Arabic Collections Online as public domain · p. 57–57 https://lib.rafed.net/view.php?b_id=1604&page=57&type=c_fbook

M-K-03

READING SECTION 2

The relative particular

الجزئي الإضافي

The particular previously discussed is called a **real particular** (al-juzʾī al-ḥaqīqī). There is another use of “particular”: the **relative particular** (al-juzʾī al-iḍāfī). It is so called in relation to something more general than itself. Although narrower in extension than a wider universal, it may itself be genuinely universal.

For clarification, straight line is a universal concept abstracted from numerous individuals, and curved line is likewise a universal abstracted from others. When we bring the two groups together and disregard their differences, we obtain a more comprehensive universal, applying to the members of both: the concept *line*.

The relation of this wider, third concept to the two narrower concepts resembles the relation of either narrower universal to its own individuals. Just as an individual is particular relative to its narrower universal, the narrower universal is as a particular relative to the wider one. It is thus called a relative particular, not a real particular, for considered in itself it remains genuinely universal.

A real particular is likewise a relative particular when considered with reference to the universal above it.

Any concept, then, when taken in relation to a concept wider in extension, is a relative particular. Zayd, for instance, is a real particular in himself and a relative particular with respect to animal. Animal is relatively particular with respect to growing body, and growing body with respect to body without qualification.

A relative particular may therefore be defined as **what is narrower than something else**, or as a concept considered in relation to one of wider extension.

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1. **Source normalisation:** The searchable Rafed witness appears to read «فزي مثلا» in the illustrative example. The present working transcription reads «فزيد مثلا», supplying the expected name Zayd; this should be checked against the 1968 printed witness before certification.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf (1402 AH), checked against the digitized searchable text; early al-Nuʿmān edition catalogued by Arabic Collections Online as public domain · p. 58–58 https://lib.rafed.net/view.php?b_id=1604&page=58&type=c_fbook

M-K-04

READING SECTION 3

Univocal and graduated universals

المتواطئ والمشكّك

A universal is divided into the *mutawāṭiʾ* (univocal or uniformly predicated) and the *mushakkik* (graded or analogically predicated).

First, consider a universal such as human being, animal, gold or silver, and apply it to its individuals. There is no variation among those individuals in the very truth of the concept as predicated of them. Zayd, ʿAmr and Khālid, and every other human being, are on an equal footing with respect to their humanity. The humanity of one is not more properly human, more intense or greater than that of another. If the individuals differ, it is in other respects—height, colour, strength, health, sincerity, soundness of thought and the like.

The same applies to individual animals and pieces of gold, and similar cases. A universal whose instances agree equally in the concept predicated of them is called *mutawāṭiʾ*: its instances concur in that respect, for *tawāṭuʾ* denotes agreement or equality.

Second, consider a universal such as whiteness, number or existence. When applied to its instances, unlike the previous type it admits variation in the mode of predication: through intensity, multitude, priority or precedence. The whiteness of snow is more intense than the whiteness of paper, yet both are white; a thousand exceeds a hundred, yet both are numbers. The Creator has priority in existence over the creature, and the cause is prior to its effect by its very existence, not by an additional thing; yet each is said to exist.

A universal whose instances thus vary in the way its concept applies to them is called *mushakkik*, and the variation is called *tashkīk*.

1. **Terminology:** The Arabic distinction is tawāṭuʾ/tashkīk. 'Univocal' and 'graded' are contextual equivalents, not an unqualified identification with every Western account of univocity or analogy. The phrase 'by its very existence' is essential to the causal-priority example.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf (1402 AH), checked against the digitized searchable text; early al-Nuʿmān edition catalogued by Arabic Collections Online as public domain · p. 59–59 https://lib.rafed.net/view.php?b_id=1604&page=59&type=c_fbook

M-K-05

READING SECTION 4

Concept and instance

المفهوم والمصداق

A **concept** (*mafhūm*) is the meaning itself considered as such: the mental form abstracted from the realities of things.

An **instance** (*miṣdāq*) is whatever the concept applies to, or the reality of the thing from which the mental form—the concept—is abstracted.

Thus the mental form signified by “Muḥammad” is a particular concept, whereas the actual person outside the mind is its instance. The mental form signified by “animal” is a universal concept, whose instances include existing individual animals and such universals falling under it as human being, horse and bird. The mental form signified by “non-existence” is a universal concept; what it applies to—non-existence itself—is its instance, and so on.

A point to notice: the first example shows that a concept can be both a real and a relative particular. The third shows that an instance need not be a real, existing extramental thing. An instance is whatever a concept applies to, even if that is a privative or non-existent matter with no concrete realization.

1. **Conceptual note:** Miṣdāq is logical instantiation or what falls under a concept; it is not in every case an independently existing extra-mental object. Muẓaffar explicitly uses ʿadam as an example.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf, 1402 AH, searchable Arabic transcription; 1968 NYU ACO scan registered for separate printed collation · p. 61 https://lib.rafed.net/view.php?b_id=1604&page=61&type=c_fbook

Concept as title, its referent, and two modes of predication

العنوان والمعنون أو دلالة المفهوم على مصداقه

When predicating something of a subject, one may have in view only its concept, making the concept itself what the judgment concerns. Thus, when one says, “A human being is a rational animal,” the expression “human being” is taken by **primary predication** (*al-ḥaml al-awwalī*).

One may instead look beyond the concept to what it represents. In saying “The human being laughs” or “The human being is in loss,” the concept of humanity points to individual humans, who are the real subjects intended by the judgment. The concept is introduced into the proposition only to enable a judgment about those individuals. In this use the concept is termed a **title** (*ʿunwān*) and its instance the **thing entitled** (*muʿanwan*); the predication is **commonly accepted or common predication** (*al-ḥaml al-shāʾiʿ*).

Three examples make the distinction clearer.

1. Grammarians say, “One cannot predicate something of a verb.” An objector may reply: “But you have just predicated something of the verb!” The answer is that although the concept *verb* appears as the grammatical subject of the proposition, the judgment concerns not the concept as such but the individual verbs it signifies, such as *ḍaraba* (“he struck”) and *yaḍribu* (“he strikes”). It is a verb under common predication that has the property in question.

2. A logician says, “A particular cannot apply to many.” An objector replies that the term “particular” applies to this book, Muḥammad, ʿAlī and Mecca. The reply is that the *concept of particularity*, particular under primary predication, is itself universal, and so applies to many things. But an *instance* of the particular is incapable of applying to many. The impossibility concerns the particular under common predication, not the universal concept of particularity under primary predication.

3. A theorist of legal principles says that an ambiguous expression (*mujmal*) is one whose meaning is not clear. An objection arises: how could we define something whose meaning is unclear, since definition presupposes a clear meaning? The concept “ambiguous expression,” considered under primary predication, is itself clear and intelligible. What is obscure is an instance under common predication, such as a homonymous expression given without any contextual indicator selecting its intended meaning. The definition concerns the latter.

1. **Translation note:** The authorial source presents three numbered objections and replies; all three are preserved. The passage distinguishes first/primary predication from common predication and should not be flattened into a generic distinction between mental and physical objects.

2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf, 1402 AH, searchable Arabic transcription; 1968 NYU ACO scan registered for separate printed collation · p. 61–62 https://lib.rafed.net/view.php?b_id=1604&page=61&type=c_fbook

M-K-07

READING SECTION 6

The four relations between universal concepts

النسب الأربع

Book I distinguished synonymous expressions from expressions with different meanings. There, difference (*tabāyun*) was **difference in concept**: the meanings of the terms were distinct. Here, difference is one of the four relations, but it concerns **difference in extension or instances**.

Conceptually distinct terms may therefore have any of four relations according to whether they apply to the same instances. This examination makes sense only for concepts already different in meaning: one cannot posit a relation between a concept and itself. The four relations are **coextensiveness** (*tasāwī*), **absolute generality and specificity**, **generality and specificity in some respects**, and **complete disjointness** (*tabāyun*).

1. Coextensiveness: Two concepts apply to precisely the same individuals. “Human being” and “capable of laughter” are the author's example: all human beings laugh in the relevant predicable sense, and all beings characterized as laughing are human. They resemble equal, coincident lines. Symbolically, *B* = *J*.

2. Absolute generality and specificity: All instances of one concept fall under the other, but not conversely. Thus *animal* is absolutely more general than *human*, and *mineral* more general than *silver*. As a geometrical analogy, a longer line covers the whole shorter line and extends beyond it. Thus *B* > *J* reads “B is absolutely more general than J”; conversely *J* < *B* reads “J is absolutely more specific than B.”

3. Generality and specificity in some respects: The two concepts overlap in some instances while each also has instances excluded from the other. *Bird* and *black* coincide in a crow; a pigeon is a bird but not black, while black wool is black but not a bird. Each concept is broader in one respect and narrower in another. They are like intersecting lines, meeting at a common point but diverging elsewhere. The author's notation is B ∗ J.

4. Complete disjointness: The concepts do not share even one instance. This includes opposed meanings treated earlier, and also some different concepts such as *stone* and *animal*. They resemble parallel lines that never meet, however far extended. The author's notation is *B // J*.

The classification is about the extension of different concepts, not just the spelling or semantic difference of their terms.

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1. **Editorial symbols:** The symbols =, >, <, *, and // retain the author's conventional notation but are not used in place of his prose. The example 'human being / laughing' is a traditional logical proprium, not a claim about manifest laughter at every moment.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf, 1402 AH, searchable Arabic transcription; 1968 NYU ACO scan registered for separate printed collation · p. 63–65 https://lib.rafed.net/view.php?b_id=1604&page=63&type=c_fbook

M-K-08

READING SECTION 8

Relations between complements of universals: equality and inclusion

النسب بين نقيضي الكليين — المساواة والعموم المطلق

Whenever two universals stand in one of the four relations previously discussed, their *complements* must also stand in a relation of this kind. Determining which requires demonstration. The method used here is **exhaustive enumeration** (*istiqṣāʾ*), also called exhaustive alternation: set out every conceivable alternative and establish the falsity of all but one, so that the remaining alternative is established. This method will recur in the treatment of exceptional syllogism.

1. The complements of coextensive universals are coextensive. If *human being* and *rational being* are coextensive, *non-human* and *non-rational* are likewise coextensive. Suppose B = J. The claim is ¬B = ¬J.

Demonstration. If ¬B and ¬J were not coextensive, one of the remaining three relations would hold, so that one expression would apply without the other in at least some instance. Suppose ¬B applied without ¬J. Then J would hold, since contradictories cannot both be false; and B would fail, since contradictories cannot both be true. But then J would hold where B fails, contrary to the supposition that B = J. The converse case is symmetrical. Hence only coextensiveness remains: ¬B = ¬J.

2. The complements of universals related by absolute generality and specificity are likewise absolutely general and specific, but in the inverse order. If B is more general than J, ¬B is more specific than ¬J. For example, non-human is more general than non-animal: every non-animal is non-human, but a horse, monkey or bird is a non-human that is an animal.

Suppose B > J; we must establish ¬B < ¬J. If the latter did not hold, we would have equality, complete disjointness, partial overlap, or the reverse inclusion. Equality of complements would entail equality of B and J by the preceding result, contradicting the hypothesis. Each of the other three alternatives would allow ¬J to hold without ¬B. The searchable editions then appear to argue that ¬J could hold without ¬B, before concluding that J would hold without B. Read literally, those

symbolic steps do not follow. A valid demonstration requires that ¬*B cannot hold where J holds*, because J is included in B. Hence every ¬B is a ¬J. Since B also has instances outside J, ¬J need not imply ¬B. Thus ¬B is strictly included in ¬J.

Textual caution: the negation symbols and direction of inclusion in this proof are inconsistent in the searchable 1402 AH and 1421 AH witnesses. The conclusion is correct, but the transmitted demonstration should be checked against the earliest printed witness, not tacitly normalized.

These derivations concern the extension of a universal and its negation in the same universe of discourse, not negation of the word considered merely as a string.

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1. **Critical translation caution:** At 1402 AH printed p. 67 the proof runs «لا ب بدون لا جـ» followed by «لا ب مع جـ» and «جـ بدون ب». These steps, as printed, are not logically consistent; a valid contradiction requires «لا ب بدون لا جـ» giving «جـ بدون ب». A second searchable 1421 AH edition reproduces the same sequence. This is an editorial *conjecture*, not an attested corrected manuscript reading, and the 1968 scan remains uninspected. Do not label as authorially corrected. https://lib.rafed.net/view.php?b_id=1728&page=98&type=c_fbook
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf (1402 AH) searchable transcription; 1968 al-Nuʿmān printed scan still requires checking · p. 65–67 https://lib.rafed.net/view.php?b_id=1604&page=65&type=c_fbook

M-K-09

READING SECTION 9

Complements of overlapping and disjoint universals

النسب بين نقيضي الكليين — التباين الجزئي

3. Complements of universals overlapping in some respects exhibit partial disjointness. *“Partial disjointness” (*tabāyun juzʾī*) means that the two do not coincide in some cases, leaving open whether or not they coincide in others. It covers both complete disjointness and partial overlap, since either situation permits one to say that the concepts fail to coincide in at least some instances.

The complements of partially overlapping universals need not always have exactly the same one of those two relations. Take *animal* and *non-human*: they overlap in horses; animal occurs without non-human in a human being, and non-human occurs without animal in a stone. Their complements—*non-animal* and *human*—are completely disjoint. By contrast, take *bird* and *black*: their complements, non-bird and non-black, overlap in white paper, yet non-bird includes a black garment, and non-black includes a white pigeon. Thus they overlap in some respects.

Both examples fall under “partial disjointness.” To demonstrate the generalization, suppose B and J overlap in some respects, and ask about ¬B and ¬J. They cannot be equal, since then B and J would be equal. They cannot stand in absolute inclusion, since then B and J would stand in the corresponding inverse inclusion. Nor can they be declared *always* partially overlapping, since the first example gives complete disjointness; nor *always* completely disjoint, since the second gives partial overlap. The appropriately general conclusion is therefore partial disjointness.

4. Complements of entirely disjoint universals also exhibit partial disjointness in this broad sense. The same reasoning applies with different examples. Existing and nonexistent, and their respective complements, may be treated as completely opposed. But human and stone are completely disjoint, whereas non-human and non-stone overlap in a horse; each also applies where the other does not. Hence the relation of their complements must not be fixed universally as either full disjointness or partial overlap alone.

Summary. Equality remains equality under complementation; absolute inclusion reverses; partial overlap and total disjointness lead only to the broader classification of partial disjointness. The last two conclusions are underdetermined until the actual universe of discourse is specified.

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1. **Source caution:** The searchable Arabic has broken or omitted symbolic relation marks in the proof and an apparent typographic omission in its final animal/human example. The prose above resolves the intended set relations but must be compared against the early scan before certification.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf (1402 AH) searchable transcription; 1968 al-Nuʿmān printed scan still requires checking · p. 67–69 https://lib.rafed.net/view.php?b_id=1604&page=67&type=c_fbook

M-K-10

READING SECTION 10

The five predicables: species, genus, and differentia

الكليات الخمس — النوع والجنس والفصل

The five predicables divide as follows: a universal predicate is either essential or accidental. The essential comprises species, genus, and differentia; the accidental comprises proprium and common accident.

One may ask about an individual, “**Who is he?**” or “**What is he?**” The first asks for features distinguishing him personally: so-and-so's son, the author of a certain book, the holder of a particular post, or a bearer of some identifying description. Answering “human being” would be wrong because it does not distinguish him from others. In modern administrative usage, the intended reply concerns *personal identity*.

“What is he?”, however, asks about the reality through which this individual shares an essence with others. The desired answer establishes his complete nature among natures, not his individual identity among people. The answer is “human being”; this is the **species** (*nawʿ*), the first of the five predicables.

Suppose we ask, “What are Zayd, ʿAmr, and Khālid?” We seek the whole nature shared by individuals differing only numerically, and answer “human beings.” Suppose we instead ask about Zayd, ʿAmr, Khālid, this horse and this lion. Their natures differ as well as their numbers, so we answer “animals.” The shared essential account is the **genus** (*jins*), the second predicable.

Accordingly, **species** is the whole nature common to particulars differing only in number, in response to “What is it?”; **genus** is the whole nature common to particulars differing in essence, in response to the same question. Diversity of essence also entails numerical plurality.

Genus also answers “What are human being, horse, and monkey?”—a question concerning diverse *universals* rather than their individual instances. Conversely, asked “What is human being?” taken as a single universal, we respond “a rational animal.” The reply analyzes the quiddity into the entire nature shared with other things and a feature that distinguishes it within that nature. Together these yield the **complete essential definition** (*ḥadd tāmm*). Its common constituent is the genus; its differentiating constituent is the **differentia** (*faṣl*), the third predicable.

The differentia is therefore the part of a thing's quiddity specific to it, just as the genus is its shared part. If we know a distant figure to be an animal but not which animal, we naturally ask, “What sort of animal is it *in itself*?” The distinguishing answer might be “rational” for a human being or “neighing” for a horse. If only that it was a body were known, asking “What kind of body is it in its own nature?” might be answered “sentient,” the differentia of animal.

Thus **the differentia is the specific part of the quiddity answering “What kind of thing is it in itself?”** The “thing” in this formula refers back to a genus already known before asking the further question.

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1. **Conceptual note:** These are traditional logical accounts of genus/species and rational/sentient differentiae, not biological taxonomy in the modern evolutionary sense. The differentia need not be directly observable in each individual.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf (1402 AH) searchable transcription; 1968 al-Nuʿmān printed scan still requires checking · p. 70–73 https://lib.rafed.net/view.php?b_id=1604&page=70&type=c_fbook

M-K-11

READING SECTION 11

Relative species, higher and lower genera, and near and remote differentiae

تقسيمات النوع والجنس والفصل

There are four sets of divisions: species may be real or relative; genus may be near, remote, or intermediate; relative species may be higher, lower, or intermediate; and differentia may be near or remote, constitutive or divisive.

1. Real and relative species. “Species” has two senses. The first is real species, one of the five predicables already described. The other is **relative species**: any universal above which stands a genus, whether or not it is itself a real species. Thus human being is a species relative to animal; animal relative to growing body; growing body relative to body without qualification; and body relative to substance.

2. Near and remote genus. Universals can form an ordered series under one another, such as human being → animal → growing body → body → substance. Moving upward from human being, the next term, animal, is the **proximate genus**, also called the lowest genus in the series. Above it lie successively more general genera, until the highest genus is reached with none above it. Those between the extremes are intermediate genera. “Near,” “remote,” and “intermediate,” or “low,” “high,” and “intermediate,” describe positions within such an ordering.

3. Higher and lower relative species. Moving downward from the highest genus, the first species beneath it is the **highest relative species**; the final one, having no species below it, is the **lowest species** or “species of species.” Terms between them are intermediate species, which are species of those above them and genera of those below them.

There may be only one intermediate stage in a four-term series, such as water, liquid, body, substance; or white, colour, sensible quality, quality. Longer series admit several, such as human being up to substance, or isosceles triangle → triangle → rectilinear figure → plane figure → figure → quantity. A middle term accordingly may function as both a genus and a relative species.

4. The constitutive and dividing differentia. Every relative species has a specific differentia constituting its quiddity and distinguishing it from other species within its genus. Thus “sentient” constitutes animal and divides growing body into sentient and non-sentient kinds. A differentia constituting a higher species must also enter into the constitution of the subordinate species, for a part of a part belongs to the whole. Hence the rule: **what constitutes the higher constitutes the lower, but not conversely**.

A differentia is **proximate** relative to the species coextensive with it—“sentient” for animal, “rational” for human being—and **remote** relative to a subordinate species—“sentient” relative to human being. The same differentia is called *constitutive* with respect to what it constitutes and *divisive* with respect to the genus it partitions.

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1. **Editorial reading:** The series are pedagogical logical taxonomies; 'growing body' and 'sentient' belong to the historical genus–differentia framework. No modern scientific identification is implied.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf (1402 AH) searchable transcription; 1968 al-Nuʿmān printed scan still requires checking · p. 73–75 https://lib.rafed.net/view.php?b_id=1604&page=73&type=c_fbook

M-K-12

READING SECTION 12

Essential and accidental predicates; proprium and common accident

الذاتي والعرضي؛ الخاصة والعرض العام

The terms **essential** (*dhātī*) and **accidental** (*ʿaraḍī*) have different technical uses in logic. Here they are taken in the sense employed in the *Isagoge*, the account of the five predicables. The three essential predicables were introduced first in order to make the distinction clear.

1. Essential predicate. This belongs to the constitution of a subject's essence and is not extrinsic to it: the subject's quiddity cannot be constituted without it. The essential predicate may be the whole quiddity, as *human being* is predicated of Zayd or ʿAmr, or an essential part, as *animal* and *rational* are predicated of human being. Thus the species, genus, and differentia are essential: the species is the complete quiddity of its individuals, while genus and differentia enter into the quiddity as parts.

2. Accidental predicate. This is extrinsic to the subject's essence and attaches to it after all its essential constituents are given. Being capable of laughter in relation to human being and walking in relation to animal provide examples.

Accidental predicates divide into **proprium** (*khāṣṣa*) and **common accident** (*ʿaraḍ ʿāmm*). A proprium belongs exclusively to its subject in the relevant comparison. It may be coextensive with the entire subject, as *laughing* is traditionally said to be with human being, or occur only in some of its individuals, as poet, orator, or legal expert do among humans. It may characterize a real species or a genus—such as occupying a place with respect to body, walking with respect to animal, or existing not-in-a-subject with respect to substance as the highest genus.

A **common accident** applies both to the subject under consideration and to something else, as walking does in relation to human being, being a bird in relation to crow, or spatial extension in relation to animal and growing body.

Accordingly, a proprium is **an extrinsic universal predicate peculiar to its subject**; a common accident is **an extrinsic universal predicate shared by the subject and something else**.

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1. **Historical example:** The proprium 'capable of laughter' does not imply that every human actually laughs at each instant; it is the inherited example of a natural aptitude belonging to a species in Porphyrian predication.
2. **Source:** al-Muẓaffar, al-Manṭiq, searchable Dār al-Taʿāruf 1402 AH text (preliminary collation, not 1968 printed-scan verified) · p. 75–77 https://lib.rafed.net/view.php?b_id=1604&page=75&type=c_fbook

READING SECTION 13

Relative predication, compound properties, and subspecies

تنبيهات وتوضيحات؛ الصنف

Four clarifications concern the five predicables.

First, a single property may be a proprium in relation to one subject and a common accident in relation to another. Thus walking is taken as peculiar to animal but common with respect to human being. The same applies to existence not-in-a-subject, spatial extension, and comparable properties attributed to genera.

Second, a characteristic may be accidental in relation to one subject but essential in relation to another. *Coloured* is accidental with respect to body but is a genus of white and black. Likewise, dispersing vision is accidental with respect to body but serves as the differentia of white, since white is traditionally defined as “colour that disperses vision.”

Third, a proprium or differentia may consist of a simple or a compound expression. *Laughing* and *rational* are examples of simple expressions; *upright in stature and manifest in skin* is given as a compound proprium of human being, and *sentient and moving by volition* as a compound differentia of animal.

*Fourth: the subclass (*ṣinf*). A differentia constitutes a species and separates it from other species of its genus: it divides the genus. A proprium does not constitute the essential nature that it characterizes, but it distinguishes that nature from others, and hence divides a more general universal. A proprium can divide a genus, or even a common accident—for example, existence in no subject divides beings into substance and non-substance. When a proprium differentiates some members* within a species, *as poet does among humans, the division is called classification (*taṣnīf*) and its members subclasses (*aṣnāf*).

A subclass is any universal narrower than a species whose members share the entire essential nature of that species but differ by an extrinsic feature. Division into species (*tanwīʿ*) is by differentiae intrinsic to a genus's subkinds; classification into subclasses is by accidents outside the shared essence. Examples include humans categorized as Eastern or Western, learned or unlearned, male or female; horses of different breeding; and varieties of palm tree.

These logical examples express the author's historical theory of classification; whether a given characteristic is genuinely essential, accidental, or a valid modern taxonomic criterion is a separate question.

1. **Conceptual note:** Do not confuse relative subclass (ṣinf) with differentia-defined species (nawʿ). The distinctions here are logical and depend on a historical conception of essence, not modern genetic taxonomy.
2. **Source:** al-Muẓaffar, al-Manṭiq, searchable Dār al-Taʿāruf 1402 AH text (preliminary collation, not 1968 printed-scan verified) · p. 77–78 https://lib.rafed.net/view.php?b_id=1604&page=77&type=c_fbook

M-K-14

READING SECTION 14

The three distinctions of predication

الحمل وأنواعه: الطبيعي والوضعي، الأولي والشائع، المواطاة والاشتقاق

Calling all five predicables **predicates** raises three objections. A species can apparently be predicated of its genus—"an animal is a human being, horse or camel"—without thereby being essential or accidental to the wider genus. A complete definition can be predicated of its subject —"a human being is a rational animal"—although the entire definition is neither a further species, genus, nor differentia of what it defines. Finally, we say that laughter is a proprium of humanity and walking a common accident, but cannot properly say "a human being is laughter" or "a human being is walking" when using those words as abstract nouns. These difficulties are resolved through three distinctions.

1. Natural and stipulated predication. A genuine particular, as particular, is not predicated of something else; what is predicated is universal. A universal concept broader in signification is naturally predicated of a concept narrower in signification: animal of human being, human being of Muḥammad, rational of human being. This is **natural predication** (*ḥaml ṭabīʿī*). Reversing the relation and predicating the narrower concept of the wider requires an artificial stipulation, **stipulated predication** (*ḥaml waḍʿī*).

Breadth of concept is not always breadth of extension. *Rational* signifies "something possessing rationality" without explicitly specifying *human being*, so is broader **in concept** even if it is coextensive with human being **in instances**. Similar remarks concern derivative words such as laughing, neighing, and walking. The essential/accidental distinction in the five predicables refers to predicates naturally applicable to their subjects, not every imaginable predication.

2. Primary essential and commonly used predication. Predication requires identity in one respect and difference in another; otherwise there are not two conceptions to relate. **Primary essential predication** (*ḥaml dhātī awwalī*) rests on unity of concept or meaning, with difference only in the mode of consideration, as in "human being is rational animal," or "human being is human being." **Common predication** (*ḥaml shāʾiʿ ṣināʿī*) rests on unity in existence or instance

despite distinct concepts, as in “human being is capable of laughter.” Thus a complete essential definition has the same conceptual content as its definiendum under primary predication, without constituting a fourth essential predicable.

3. Predication by identity and predication by derivation. “Human being is laughing” (in the sense of having the capacity) uses **predication by identity** (*ḥaml muwāṭāh*, or *huwa huwa*): the subject is the very thing picked out by the predicate. All five predicables are predicated in this manner. In contrast, “human being has laughter” expresses **predication by derivation** (*ḥaml ishtiqāq*, or *dhū huwa*). We cannot strictly predicate the abstract noun *laughter* directly of a human being without deriving the adjective *laughing* or adding “possessing.” Thus the five predicables include *laughing* and *sentient*, not the abstract nouns *laughter* and *sensation*. The occasional use of an abstract noun where a derived adjective is meant is loose language that may confuse beginners.

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1. **Editorial source warning:** This is a normalized explanatory transcription of the continuous argument on pp. 78–81, including the main authorial distinctions and first/second objections. The raw source has additional micro-examples and typographic line breaks; check exact wording against the 1968 ACO scan before calling it a diplomatic transcription.
2. **Source:** al-Muẓaffar, al-Manṭiq, searchable Dār al-Taʿāruf 1402 AH text (preliminary collation, not 1968 printed-scan verified) · p. 78–81 https://lib.rafed.net/view.php?b_id=1604&page=78&type=c_fbook

M-K-16

READING SECTION 16

Accidental attribution means predication, not mere inherence

العروض معناه الحمل

A further confusion must be avoided. You might object: “You said that an extrinsic universal predicated of its subject alone is a proprium, and otherwise a common accident. Laughter belongs only to human beings, so it must be a proprium.” The answer depends upon the intended meaning of *ʿurūḍ* (“occurrence” or “attribution”) in this discussion: it means **being predicated accidentally rather than essentially**. In this technical sense, the abstract noun *laughter* is not predicated of the human being. When we say that laughter occurs in the human being, “occurs” means something else: that it **exists in the subject**.

A second ambiguity arises because logicians call an extrinsic universal a special or common *ʿaraḍ*, and then also call such a thing as laughter an *ʿaraḍ*. In the first usage, the term means an **accidental predicate**, in contrast to an essential predicate. In the second, it means an **accident inhering in a subject**, in contrast to a substance that does not inhere in a subject.

Colour, for example, is an accident in the latter sense because it exists in a subject. It is not an accidental predicate of body in the first sense, because the abstract term colour is not predicated of body by direct identity predication (*ḥaml muwāṭāh*). Considered instead in relation to the kinds

below it, such as blackness and whiteness, colour is their genus, so it is essential to them rather than accidental.

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1. **Technical distinction:** The author here distinguishes (1) ʿaraḍī as an extrinsic *predicable* and (2) ʿaraḍ as an ontological accident inhering in a subject. This distinction is critical to avoiding confusion between Avicennian category theory and Porphyrian predicables.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH searchable transcription; 1968 ACO scan not yet independently collated · p. 82–83 https://lib.rafed.net/view.php?b_id=1604&page=82&type=c_fbook

M-K-17

READING SECTION 17

Divisions of accidental predicates: necessary, evident, demonstrable, and separable

تقسيمات العرضي: لازم ومفارق، بيّن وغير بيّن

Accidental predicates fall primarily into two divisions: **necessary** (*lāzim*) and **separable** (*mufāriq*).

1. A necessary accident is one that cannot rationally be separated from its subject, such as oddness from three, evenness from four, or—in the author's traditional natural-philosophical example—heat from fire.

2. A separable accident is one whose separation from the subject is not rationally impossible, as with descriptions of a person as standing, sitting, sleeping, healthy, or ill. The feature may in practice never disappear: an eye may remain blue throughout its existence, yet blueness is still separable, since one can intelligibly suppose its removal without the eye ceasing to be an eye. This contrasts with oddness and three: a three stripped of oddness could no longer remain three. Likewise, by the author's historical example, remove heat from fire and fire ceases to be fire. Such examples explain **rational inseparability**, not merely constant empirical association.

Necessary accidents may be **evident** (*bayyin*) or **non-evident** (*ghayr bayyin*). An evident necessary accident is either evident in the narrower or broader sense.

Evident in the narrow sense (*bayyin bi-l-maʿnā al-akhaṣṣ*) means that simply conceiving the underlying subject brings with it conception of what necessarily belongs to it, without a mediating consideration.

Evident in the broad sense (*bayyin bi-l-maʿnā al-aʿamm*) means that conceiving the two terms and their relation is sufficient to establish their necessary connection. For example, two is half of four and a quarter of eight. Someone might conceive two but not spontaneously think of its relation to eight. Once eight and the appropriate numerical relation are brought to mind, the

conclusion becomes evident. Similarly, considering the obligation of prayer, ritual ablution, and the dependence of an obligatory prayer upon it leads to acknowledgment of the obligation of its prerequisite.

This broader evidential class includes both what follows upon conceiving the subject alone and what becomes evident only when the other term and their relation are conceived. Repeated familiarity can make the consequence spring to mind immediately whenever the underlying subject is conceived.

Non-evident necessity requires a demonstration beyond conceiving the two terms and their relation. The familiar example is the proposition that the angles of a triangle equal two right angles: assent to this relation depends upon geometrical demonstration, not merely forming the three concepts. In short, evident necessity is immediate, whereas non-evident necessity is discursively established.

Separable accidents are further classified as **permanent in practice**, **quickly passing**, or **slowly passing**. Historical examples include the motion attributed to the sun, a persistently blue eye, blushing with embarrassment, pallor from fear, and the passing stage of youth.

These classifications concern the *logical mode of attribution*, not a guarantee that every traditional natural-scientific example survives modern empirical scrutiny.

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1. **Historical example:** The author uses fire's heat, apparent motion of the Sun, and blue eyes as examples within traditional pedagogy. Their empirical status must not be promoted into timeless demonstrations of physical necessity.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH searchable transcription; 1968 ACO scan not yet independently collated · p. 83–84 https://lib.rafed.net/view.php?b_id=1604&page=83&type=c_fbook

M-K-15

READING SECTION 15

Logical, natural, and intellectual universals

الكلي المنطقي والطبيعي والعقلي

When we say “the human being is universal,” there are three distinct things to consider: (1) humanity as such; (2) the concept **universal**, considered without attending to which nature it characterizes; and (3) humanity **as characterized by universality**. In other words, we can attend to the subject alone, the attribute alone, or subject and attribute together.

*1. The natural universal (*al-kullī al-ṭabīʿī*). The intellect considers the thing's nature without attending to universality as an additional description. Thus it considers human being* as human being*, as when it predicates “rational animal” of it. The natural universal is the nature itself taken

in this manner; it exists outside the mind in and through the existence of its individual instances.

*2. The logical universal (*al-kullī al-manṭiqī*).** The intellect considers the abstract characteristic of being a universal—"that which may be supposed to apply to many"—without attaching it to any particular nature, such as human, animal, or stone. It exists only in the intellect, for it is a purely mental concept formed by abstraction.

*3. The intellectual universal (*al-kullī al-ʿaqlī*). The intellect considers the nature* precisely as characterized by universality*: human being as a concept capable of application to many. This composite of a nature and a mental attribute exists only in the intellect, since anything existing externally is in fact an individual particular.

Consider the proposition "the surface is above." One can attend to the surface itself, with its bricks or timber, disregarding whether it is above or below—analogous to the natural universal. One can attend to *aboveness* abstractly and without any particular thing above—analogous to the logical universal. Or one can attend to the surface as *above*—analogous to the intellectual universal.

The distinction separates the nature, the logical property of universality, and the nature insofar as the property is predicated of it. This is essential when comparing the later Ḥikma discussion of *quiddity considered in itself* and its mental predication.

These three modes of consideration apply not only to universality in general but to each of the five predicables and their subdivisions, and even to the particular. Thus one can speak of a *natural species*, a *logical species*, and an *intellectual species*, and similarly of genus and the rest. Natural species is human being considered as such; logical species is the abstract definition "the complete nature common to particulars that differ numerically, in answer to 'What is it?'"; and intellectual species is human being precisely as so described. The remaining cases follow the same pattern.

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1. **Cross-book reference:** The later Ṭabāṭabāʾī reader discusses the external reality of natural quiddity in Nihāya stage V, but that should not be equated uncritically with every ontological implication of al-Muẓaffar's introductory logical terminology.
2. **Source:** al-Muẓaffar, al-Manṭiq, searchable Dār al-Taʿāruf 1402 AH text (preliminary collation, not 1968 printed-scan verified) · p. 84–86 https://lib.rafed.net/view.php?b_id=1604&page=84&type=c_fbook

BOOK 3

The Four Questions

مطلب ما وأي وهل ولم

M-Q-01

READING SECTION 1

The first question: explaining a word

المرحلة الأولى: التعريف اللفظي

When an expression from any language confronts you, there are five successive stages through which you must pass to attain knowledge. Some seek knowledge by way of conception, and others knowledge by way of assent.

The first stage is to seek an initial, general conception of the expression's meaning. If you do not know which meaning the word has been assigned, you ask a purely linguistic question. The answer is another word signifying that meaning. Thus, if you ask what ghaḍanfar means, the answer is lion; or what sumaydaʿ means, the answer is chief. An answer of this kind is called a verbal definition (taʿrīf lafẓī). Providing such definitions is the responsibility of dictionaries.

1. **Translator's note:** A verbal definition establishes customary signification. It need not yet explain the thing's essential constituents.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Book III (al-Muʿarrif) · p. 110 https://lib.rafed.net/view.php?b_id=1604&type=c_fbook

M-Q-02

READING SECTION 2

What is it? The explanatory question

المرحلة الثانية: مطلب ما الشارحة

Once you have conceived the word's meaning in a general way, you turn naturally to the second stage: seeking a conception of the meaning's quiddity. That is, you seek an articulation of what the name had indicated only generally, so that the object may be distinguished completely from other objects in the mind. You therefore ask “What is it?” (mā huwa?).

The interrogative mā here is called explanatory (shāriḥa), because it asks for an explanation of what a word signifies. Its answer is called an explanation of the name (sharḥ al-ism), or nominal definition. The primary form of answer combines the proximate genus with the proximate differentia; this is the complete nominal essential definition. One may also answer with the differentia alone, the proprium alone, either combined with a more remote genus, or the proprium together with the proximate genus. According to the constituents supplied, such accounts are called incomplete essential definitions or complete/incomplete descriptions; but at this stage all remain nominal.

Suppose, however, that a respondent gives only the proximate genus. Asked “What is a date-palm?”, he answers “A tree.” This will not satisfy the questioner, who still wants to know what distinguishes this tree from others. He asks “Which kind of tree is it, essentially?” or “Which kind is it by its distinctive attribute?” The former seeks the differentia; the latter seeks a proprium. This is the proper place of the interrogative “which?” (ayy).

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1. **Translator's note:** Muẓaffar's ‘nominal’ does not mean an arbitrary stipulation: the same account can become a real definition once existence has been established.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Book III (al-Muʾarrif) · p. 110–111 https://lib.rafed.net/view.php?b_id=1604&type=c_fbook

M-Q-03

READING SECTION 3

Does it exist? Simple and composite hal

المرحلتان الثالثة والرابعة: هل البسيطة وهل المركبة

After learning what the expression means, you move to the third stage: seeking assent to the thing's existence. You ask with hal, the simple existential question (hal basīṭa): “Does such-and-such exist?” or “Is it existent?”

A qualification is necessary. The second and third stages can occur in either order. Their natural order is the one just given, but the questioner may already know that a thing exists, or may ask about its existence first. If existence is known before the general expression has been explained in detail, a subsequent question “What is it?” is called the real or quiddative ‘what?’ (mā ḥaqīqiyya). Its answer is the same as to the explanatory mā. The only difference is that the explanatory question precedes knowledge of the thing's existence, while the real question follows it. Hence an essential definition can be nominal before the simple existential judgment and real afterwards.

Once these stages are completed, one naturally passes to the fourth stage, seeking assent that some attribute or condition belongs to the thing. This is again asked by hal, but now it is the composite question (hal murakkaba), because it asks whether one thing is established of another,

presupposing the subject's existence. By contrast, simple hal asks only whether the subject exists. “Does God exist?” is a simple existential inquiry; “Is the existent God willing?” is composite.

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1. **Translator's note:** ‘Simple’ and ‘composite’ distinguish asserting that a subject exists from predicating something further of an existent subject. Neither question by itself provides an essential definition.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Book III (al-Muʿarrif) · p. 111–113 https://lib.rafed.net/view.php?b_id=1604&type=c_fbook

M-Q-04

READING SECTION 4

Why? The cause of assent and the cause of being

المرحلة الخامسة: مطلب لم

If a respondent has answered either the simple or composite hal, you proceed to the fifth stage: seeking the cause. Sometimes you seek only the ground of the judgment—namely, a demonstration of what the respondent affirmed. At other times you seek both the ground of the judgment and the cause of the fact's existence, so that you know why the thing is actually so.

For either purpose you ask “Why?” (limā?). To seek the ground of an affirmation, for instance, you may ask, “Why is God willing?” To seek both the reason for the judgment and the real cause, you may ask, “Why does a magnet attract iron?” Once someone has answered “Yes” to whether a magnet attracts iron, you are entitled to ask the further question, “Why?”

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1. **Translator's note:** The two senses anticipate the difference between demonstrating that a proposition is true and demonstrating why the fact it reports obtains.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Book III (al-Muʿarrif) · p. 112–113 https://lib.rafed.net/view.php?b_id=1604&type=c_fbook

M-Q-05

READING SECTION 5

Summary of the inquiries and their subsidiary forms

تلخيص وتعقيب وفروع المطالب

We may summarise the foregoing as follows. “What?” (mā) seeks a conception of the thing's quiddity; it may be explanatory or real. From this interrogative the technical expressions māʾiyya and māhiyya are formed.

“Which?” (ayy) asks how a thing is distinguished, essentially or accidentally, from others sharing its genus, once that genus is known.

“Whether?” (hal) can be simple, seeking assent to the existence or nonexistence of the thing, or composite, seeking assent that something belongs—or does not belong—to something else.

“Why?” (limā) asks either for the cause of the assent alone, or for both the cause of the assent and the cause of the thing's existence.

These are the principal inquiries, whose questions have a universal scope across the sciences. Other inquiries employ “how?”, “where?”, “when?”, “how many?”, and “who?”. These are subsidiary, because they are not applicable to every object: what has no spatial location cannot be asked “where?”, and what has no time cannot be asked “when?”. Moreover, many such questions can be recast as composite hal questions.

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1. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Book III (al-Muʿarrif) · p. 113–114 https://lib.rafed.net/view.php?b_id=1604&type=c_fbook

BOOK 3

The Defining Account

المعرّف وأقسامه

M-D-01

READING SECTION 1

What is a defining account?

التعريف وأقسامه

Disputes in scientific and other matters frequently arise because the concepts conveyed by the words used remain indeterminate. Mutual understanding is disturbed when the disputants have not agreed on the boundaries of the words' meanings.

Anyone pursuing realities must therefore express the elements of his thought in language that defines and explains them clearly, preserving the intended meaning without exceeding or falling short of it. Sound thinking is built on this basis.

A defining account may be an essential definition (ḥadd) or a description (rasm); either may be complete or incomplete. We have already mentioned verbal definition. Here we are not primarily concerned with it, since it tells us only the meaning assigned to a word. The logician's subject is the previously known conception by which we arrive at a conception not yet known—the kind of account given in answer to explanatory or real mā.

1. **Translator's note:** The Arabic contrast is between ḥadd (essential definition), rasm (description through distinguishing attributes), and taʿrīf lafẓī (lexical explanation).
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Book III (al-Muʿarrif) · p. 115–116 https://lib.rafed.net/view.php?b_id=1604&type=c_fbook

M-D-02

READING SECTION 2

The complete essential definition

الحد التام

A complete essential definition states all the essential constituents of what is being defined. It is given through the proximate genus and proximate differentia, because together they encompass all those constituents.

If one asks, “What is a human being?”, the answer may be “A rational animal.” This is complete: it articulates what the word “human” signified without detail and includes all the essential constituents. For “animal” already implies substance, growing body, sensation, and movement by will.

One might instead answer, “A growing body that senses, moves voluntarily, and reasons.” This is also a complete definition, identical in meaning to the first but more fully expanded: the defining account of “animal” has been substituted for the word itself. Such expansion is unnecessary unless the questioner does not understand the quiddity of an animal.

Likewise, an unfamiliar genus may be unfolded by its own definition, if one is available, until the explanation reaches primary and self-evident concepts that require no definition—for example being and thing.

Hence the proximate genus and proximate differentia encompass every essential constituent without omission. A longer complete definition differs from a shorter one in explicitness, not in its intelligible content. The complete definition is conceptually equivalent to what it defines and signifies it directly, by correspondence (muṭābaqa).

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1. **Translator's note:** ‘Being’ (mawjūd) is named here as an immediately understood concept, not as a constituent that must be added to the essential definition of each contingent quiddity.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Book III (al-Muʿarrif) · p. 117–118 https://lib.rafed.net/view.php?b_id=1604&type=c_fbook

M-D-03

READING SECTION 3

The incomplete essential definition

الحد الناقص

An incomplete essential definition gives only some of the essential constituents of the thing, but must contain at least its proximate differentia. It therefore may be expressed through a remote genus together with the proximate differentia, or through that differentia alone.

In the first case, one might define a human being as “a growing body … rational,” omitting the intervening determination “sentient and moving by will.” In the second, one might say simply “rational,” dropping the entire proximate genus. The second is even less complete.

Three points follow. First, the incomplete definition is not conceptually equivalent to what it defines, because it states only part of its essential content, although its extension may coincide with the thing being defined. Second, it does not supply a complete mental conception of the

thing's quiddity. At most it distinguishes the thing essentially from everything else. Third, it signifies the whole by necessary implication (iltizām) rather than by direct correspondence, since a distinctive essential part points to the whole.

1. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Book III (al-Muʿarrif) · p. 118 https://lib.rafed.net/view.php?b_id=1604&type=c_fbook

M-D-04

READING SECTION 4

Complete and incomplete descriptions

الرسم التام والرسم الناقص

A complete description (rasm tāmm) combines the genus with a proprium (khāṣṣa): for example, “a laughing animal” as an account of the human being. It includes an essential constituent—the genus—and a distinctive but nonessential attribute—the proprium. For this reason it is called complete.

An incomplete description (rasm nāqiṣ) employs the proprium alone, as when a human being is described as “laughing” or “capable of laughter.” Because it contains only an accidental predicate, it is incomplete.

According to another formulation, genus plus proprium counts as complete only when the genus is the proximate one; a remote genus plus proprium remains an incomplete description.

In either case, a description, like an incomplete essential definition, serves to distinguish its subject from all other things. But it does so through what is extrinsic to the quiddity. It agrees with what it describes in the things to which it applies, not in the meaning it articulates, and points to it by implication rather than by a statement of the complete essence.

1. **Translator's note:** Here ‘accidental’ (ʿaraḍī) means extrinsic to the quiddity. A proprium need not be temporally separable or contingent in the ordinary English sense.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Book III (al-Muʿarrif) · p. 119 https://lib.rafed.net/view.php?b_id=1604&type=c_fbook

M-D-05

READING SECTION 5

Why definitions are difficult

إنارة

The primary form of definition is the complete essential definition, for the aim of defining is twofold: first, to conceive the thing through its true quiddity, so that a clear and fully articulated representation of it arises in the soul; second, to distinguish it completely from everything else in the mind. Only a complete essential definition achieves both aims.

Where the first aim is unattainable, one must be content with the second. An incomplete essential definition or either kind of description may provide it. If an essential distinction is available, an incomplete essential definition is preferable to a description; among descriptions, the complete is preferable to the incomplete.

Yet scholars acknowledge that grasping things' true essences and differentiæ is extraordinarily difficult and may be unattainable. Many of the differentiæ customarily given are really necessary proprieties that disclose the true differentia without themselves being that differentia. Thus most—or perhaps all—of the definitions available to us are descriptions resembling essential definitions.

Accordingly, one should select an evident and necessary distinctive attribute, one that best reveals the thing's reality and most closely resembles its differentia. An obscure attribute will not serve to explain the thing to an ordinary questioner.

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1. **Translator's note:** The passage itself qualifies the theory of ḥadd: satisfying the formal schema ‘genus + differentia’ does not mean the alleged differentia has actually been shown to be the real one.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Book III (al-Muʿarrif) · p. 120 https://lib.rafed.net/view.php?b_id=1604&type=c_fbook

M-D-06

READING SECTION 6

Definitions by example and by comparison

التعريف بالمثال والتشبيه

Scholars, especially those concerned with language and literature, often explain a thing by citing one of its instances as an example. This is definition by example, and it is particularly accessible to beginners seeking to understand and distinguish things.

A related educational procedure is inductive instruction: before stating the definition or general rule, the author or teacher supplies many examples and exercises, allowing the student to infer the common conception or rule independently. Only then is the result expressed clearly, so that the

learner can compare the explicit statement with what he has discovered.

Definition by example is not a fifth fundamental kind of definition. It is subordinate to explaining something through a distinctive feature, and hence falls under incomplete description.

Similarly, definition by resemblance explains an unfamiliar thing by comparing it with something already known through a relevant likeness. An example is the comparison of existence with light: each is manifest in itself and makes other things manifest.

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1. **Translator's note:** The light comparison illustrates a point; it is not itself a proof that existence is literally a species of light.
2. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Book III (al-Muʿarrif) · p. 121–122 https://lib.rafed.net/view.php?b_id=1604&type=c_fbook

BOOK 3

Conditions of Definition

شروط التعريف

M-D-07

READING SECTION 1

A definition must be coextensive and clearer

شروط التعريف: الأول والثاني

The point of a definition is to make its subject intelligible and distinguish it from everything else. Five conditions are therefore required.

First, the defining account must be coextensive with what it defines. It must be both exclusive (māniʿ) and inclusive (jāmiʿ): exclusive because it applies to nothing outside the subject's extension, and inclusive because it applies to every instance of the subject without exception.

Accordingly, an account too broad is invalid: “a two-legged animal” does not exclusively identify the human being, since other animals walk on two legs. An account too narrow is likewise invalid: “an educated animal” excludes many human beings. Nor may a completely disparate description define the subject, because the two do not apply to the same instances.

Second, the defining expression must be clearer and better known to the addressee than what is being defined. Otherwise it fails to explain. One should not explain an unknown term through something equally obscure—for example by defining one of two correlatives solely through the other when both are equally unfamiliar—nor through something still more obscure, as though explaining light by saying that it is a power resembling existence.

1. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Book III (al-Muʿarrif) · p. 123–124 https://lib.rafed.net/view.php?b_id=1604&type=c_fbook

M-D-08

READING SECTION 2

No self-definition and no circularity

شروط التعريف: الثالث والرابع

Third, the defining account must not be identical in meaning with its subject, when a real rather than a merely verbal definition is intended. To explain “motion” by the synonymous “transition,” or “human being” by the synonymous “person,” provides no new intelligible determination. The two expressions must differ either as an implicit content differs from its explicit analysis, as in a complete essential definition, or by a distinct conception which makes the subject known. Otherwise knowledge of the subject would be presupposed in the very attempt to acquire it.

Fourth, the account must not be circular. Circularity arises when the expression offered to explain the subject is itself unintelligible except through that subject. The subject then becomes part of its own explanation.

Explicit circularity occurs when the sun is defined as a star that rises during the day, while day is defined as the time during which the sun rises. Knowing the sun presupposes knowing day, which in turn presupposes knowing the sun.

Concealed circularity may involve several mediating steps. Suppose “two” is explained as the first even number, “even” through division into equals, “equals” through two matching things, and “two things” through the number two. The circuit eventually returns to the very notion it was meant to explain.

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1. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Book III (al-Muʿarrif) · p. 124–126 https://lib.rafed.net/view.php?b_id=1604&type=c_fbook

M-D-09

READING SECTION 3

Unambiguous language in definitions

شروط التعريف: الخامس

Fifth, a definition must be expressed in clear, unambiguous language. One should not employ obscure, rare, or unfamiliar expressions, nor equivocal or figurative terms without an indication fixing the intended sense. Where such an indication is present, the use of these terms is not formally prohibited; nevertheless, scientific definitions and argument should generally avoid metaphor when greater precision is available.

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1. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Book III (al-Muʿarrif) · p. 126 https://lib.rafed.net/view.php?b_id=1604&type=c_fbook

BOOK 4

Book IV · Propositions and Their Properties

الباب الرابع — القضايا وأحكامها

M-C-01

READING SECTION 1

A proposition is intrinsically apt for truth or falsity

القضية — الصدق والكذب لذاته

It was established in Book I that an assertoric statement (khabar) is a proposition (qaḍiyya), defined as a complete expression capable of being described as true or false.

“Complete expression” is the proximate genus embracing the two kinds of complete expression: assertion and non-assertoric utterance (inshāʾ). The remainder of the definition is a property (khāṣṣa) distinguishing the former, since being describable as true or false is peculiar to assertions, as explained earlier. The definition is therefore a complete descriptive account (rasm tāmm), not a complete essential definition.

For precision we add “in itself” (li-dhātihi): a proposition is a complete expression that is, in itself, capable of being described as true or false. The corresponding qualification should also enter the definition of non-assertoric utterance. Without it, someone may suppose that the definition of assertion includes certain non-assertoric expressions, so that it does not exclude everything else, and that the definition of non-assertoric expression excludes those cases, so that it is not comprehensive.

The mistaken impression arises because we sometimes call certain non-assertoric acts truthful or false: for instance, asking about what one already knows, requesting help in the manner of a poor person when one is rich, or wishing for something already in one's possession. Conversely, an uninformed questioner, needy petitioner, or deprived and despairing person may be described as truthful. Questions, requests, and wishes nevertheless belong to non-assertoric expression.

On examination, genuine questioning proceeds from ignorance, genuine petition from need, and genuine wishing from deprivation and despair. Such acts consequently signify, by implication (dalāla iltizāmiyya), a report of ignorance, need, or despair. It is this implicitly signified *report*, not

the act of questioning, requesting, or wishing in itself, that can properly be called true or false. Thus even the shorter definition does not intrinsically include those utterances; “in itself” makes the boundary explicit and prevents confusion.

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1. **Textual status:** Consecutive authorial Book IV passage, digitally transcribed with normalized punctuation and Arabic kāf. Printed pp.125–127 in the searchable 1402 AH copy, whose web page locator begins at page=127. Independent 1968 scan collation and exercise verification are pending.
2. **Terminology:** The contrast is truth-aptness li-dhātihi versus truth/falsity attributable only to an implicated khabar. Here rasm tāmm is a definition through proximate genus and proprium, not a ḥadd.
3. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH searchable Arabic, Book IV; 1968 ACO scan not yet independently collated · p. 125–127 https://lib.rafed.net/view.php?b_id=1604&page=127&type=c_fbook

M-C-02

READING SECTION 2

Categorical and conditional propositions

القضية حملية وشرطية

Propositions divide into categorical (ḥamliyya) and conditional (sharṭiyya).

Examples of categorical propositions are: “Iron is a metal”; “Usury is forbidden”; “Truthfulness is praised”; “A liar is not trustworthy”; and “The miser does not attain pre-eminence.” Each has two terms and a relation between them: the union of the terms, expressed as the second being affirmed of the first, or the denial of such union and affirmation. In short, a categorical proposition says either “this is that” or “this is not that.” It may therefore be defined as a proposition in which something is judged to be affirmed of, or denied of, something else.

Examples of conditional propositions are: “If the sun rises, it is daytime”; “It is not the case that if a person spreads malicious gossip, he is trustworthy”; “An expression is either simple or composite”; and “It is not the case that a human is either a writer or a poet.” Each contains two sides that were originally propositions in their own right. In the first example, without “if” and the consequential *fa*, “the sun rose” and “daytime exists” would each be a complete assertoric statement. By combining them and relating the one to the other, the speaker makes a single proposition: stopping after “If the sun rose ...” leaves an incomplete expression, as previously explained.

The relation between these originally separate propositions is not the categorical relation of predication or union; two propositions are not united as subject and predicate are. It is instead either a relation of connection, accompaniment, and dependence—making the second dependent on the first, or denying that dependence—as in the first two examples; or a relation of mutual exclusion, disjunction, and separation, or its denial, as in the last two.

1. **Source status:** Book IV opening; consecutive Arabic selected from printed pp.127–128, punctuation and kāf normalized, not independently collated against the 1968 printed witness.
2. **Logic:** The negative examples negate an entire connection or disjunction; do not mistake them for a positive conditional with a negative consequent.
3. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH searchable Arabic, Book IV; 1968 ACO scan not yet independently collated · p. 127–128 https://lib.rafed.net/view.php?b_id=1604&page=129&type=c_fbook

M-C-03

READING SECTION 3

Connected and disjunctive; affirmative and negative

المتصلة والمنفصلة والموجبة والسالبة

Several conclusions follow.

First, a conditional proposition is one in which one judges that a relation exists between one proposition and another—or that such a relation does not exist.

Second, conditionals divide into connected (muttaṣila) and disjunctive (munfaṣila). If the relation is connection and the dependence of one proposition on the other, or the denial of that connection, it is called connected, as in the first pair of examples. If the relation is exclusion, disjunction, or separation between the propositions, or its denial, it is called disjunctive, as in the second pair.

Third, all three kinds—categorical, connected conditional, and disjunctive conditional—divide into affirmative and negative. Affirmation (ījāb) judges that predication, connection, or disjunction obtains; negation (salb) denies predication, connection, or disjunction. Strictly speaking, it is improper to call the negative proposition categorical, connected, or disjunctive in the same unqualified sense as the affirmative, since it denies precisely the relation in question. These names are nevertheless extended to negatives by analogy with the affirmatives.

Affirmation and negation together are called the *quality* (kayf) of a proposition: one asks “how?” to determine whether the relation is affirmed or denied.

1. **Terminology:** Kayf al-qaḍiyya is affirmative/negative quality, distinct from the category of qualitative accidents. A negative connected conditional denies the connection, not necessarily its antecedent or consequent.
2. **Textual status:** Arabic digitally transcribed from the 1402 AH searchable text; pp.128–129. No early print witness checked yet.
3. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH searchable Arabic, Book IV; 1968 ACO scan not yet independently collated · p. 128–129 https://lib.rafed.net/view.php?b_id=1604&page=129&type=c_fbook

M-C-04

READING SECTION 4

Parts of categorical and conditional propositions

أجزاء القضية: الموضوع والمحمول والمقدم والتالي

Every proposition has two terms or sides and a relation between them, and therefore three components.

In a categorical proposition, the first term—what the judgment concerns—is the *subject* (mawḍūʿ); the second—what is affirmed or denied of it—is the *predicate* (maḥmūl); and the expression indicating the relation is the *copula* (rābiṭa).

In a conditional proposition, the first side is the *antecedent* (muqaddam), the second the *consequent* (tālī), and the expression that signifies the relation is again the connective (rābiṭa).

Strictly, the sides of a disjunctive proposition do not deserve the names “antecedent” and “consequent,” because unlike a connected conditional they possess no inherent order. Either side can be placed first without changing the meaning. The labels are applied to disjunctions only by assimilation to connected conditionals, just as negative propositions inherit the conventional names of affirmative categorical, connected, or disjunctive propositions.

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1. **Terminology:** Rābiṭa is relational sign, not always the grammatical verb 'is'; muqaddam and tālī in disjunction are order labels by convention, not asymmetric logical positions.
2. **Textual status:** Consecutive Book IV source selection, printed pp.129–130 in 1402 AH transcription; 1968 ACO scan still awaiting independent visual comparison.
3. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH searchable Arabic, Book IV; 1968 ACO scan not yet independently collated · p. 129–130 https://lib.rafed.net/view.php?b_id=1604&page=131&type=c_fbook

M-C-05

READING SECTION 5

Categoricals by subject: singular, natural, indefinite, quantified

أقسام القضية باعتبار الموضوع

We begin by classifying categorical propositions according to their subjects, then turn to an analogous classification of conditionals. A categorical subject is either a true particular or a universal. The resulting four types are personal (or singular), natural, indefinite, and quantified.

A. Personal (shakhṣiyya, makhṣūṣa). If the subject is a true particular, the proposition is singular: “Muḥammad is God's messenger”; “Shaykh al-Mufīd was the renewer of the fourth century”; “Baghdad is Iraq's capital”; “You are learned”; “He is not a poet”; and “This age offers little promise.”

B. Universal subject. When the subject is universal, three different modes of judgment are possible.

1. Natural (ṭabīʿiyya). The judgment attaches to the universal subject itself *as universal*, without regarding individual instances. It cannot correctly be redistributed to those individuals. Thus "human is a species," "rational is a differentia," "animal is a genus," and "capable of laughter is a proprium." An individual human is not itself a species, just as an individual is not a genus, differentia, or proprium in these predicational senses.

2. Indefinite (muhmala). The judgment concerns individual instances through the universal as their title or mirror, but leaves their quantity unspecified: neither "all" nor "some" is stated. Examples are "Man is in loss," "The leader of a people is their servant," "Swift reproach is not justice," and "A believer does not lie." These expressions alone do not determine whether the judgment extends to every member of the subject class.

Observation. The Chief Master, Avicenna, argues in the *Ishārāt* that, if the definite article universally quantifies and indefiniteness particularizes, Arabic would contain no indefinite propositions; this would have to be investigated in another language, and in any event the grammatical question belongs to grammar. Muẓaffar replies that Arabic does allow indefinite propositions: the article may designate the nature as instantiated without specifying all or some. If instead it denotes the genus in a generalizing use, the judgment has universal force. Context supplies the distinction; grammatical and rhetorical scholarship must settle its linguistic conditions.

3. Quantified (maḥṣūra, musawwara). Here the judgment concerns instances under the universal, *and* specifies whether all or only some are intended. A **universal** affirmative or negative concerns all: "Every imam is protected from error," "Every water is pure," "All usury is prohibited," "No ignorance is beneficial," and "There is nobody at home." A **particular** affirmative or negative concerns some: "Some people lie," "Few of My servants are grateful," "Most people, however eager you are, will not believe," "Not every human being is learned," and "Many a single meal has prevented later meals." The doctrinal illustrations express the examples of the original text, not editorial endorsements of their universal truth.

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1. **Source:** The Arabic follows the 1402 AH digitally transcribed reading with normalized hamza/kāf, punctuation and paragraph divisions; it is NOT a diplomatic transcription. Independent 1968 scanned-page collation and exercise checks pending.
2. **Citation caution:** The quotation attributed to Avicenna is preserved within Muẓaffar's argument but has NOT yet been independently checked word-for-word against the Ishārāt; doctrinal and Qurʾānic examples are source examples, not the translator's independent claims.
3. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH Arabic digital transcription; compare 1968 Najaf ACO scan pending · p. 130–132 https://lib.rafed.net/view.php?b_id=1604&page=130&type=c_fbook

READING SECTION 6

Why logic privileges quantified propositions

لا اعتبار إلا بالمحصورات

The propositions upon which logicians principally rely are quantified propositions, to the exclusion of the other divisions. The reason requires explanation.

A personal proposition lacks the generality appropriate to the general laws studied by logic. A natural proposition has the same standing for this purpose: its predicate belongs to the universal concept as such, not to its extension. “Human is a species” does not state a general rule about human individuals; none of them is itself a species.

An indefinite proposition, however, has the logical *force* of a particular proposition. The judgment might concern all the individuals or only some, without specifying which. “The leader of a people is their servant” could intend every leader, but it could also admit leaders whose authority does not depend upon the people. In either case the weaker statement, “Some leaders serve their people,” follows: if every leader does, then certainly some do; if only some do, it is true directly. The rest are simply not addressed.

Thus a particular proposition makes a judgment of some instances without deciding anything, affirmatively or negatively, about the remainder. “Some humans are animals” is true and remains true even when every human is in fact an animal. Its form does not imply that the other humans are not animals.

Once quantified propositions are recognized as the proper objects of logical analysis, combining their **quantity** (universal or particular) with their **quality** (affirmative or negative) yields the four classical forms: universal affirmative, universal negative, particular affirmative, and particular negative.

The author's note defines the *quantity of the proposition* (kamm al-qaḍiyya, with a doubled mīm) by reference to the interrogative “how many?” (kam); the abstract noun is *kammiyya*.

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1. **Source:** The Arabic follows the 1402 AH digitally transcribed reading with normalized hamza/kāf, punctuation and paragraph divisions; it is NOT a diplomatic transcription. Independent 1968 scanned-page collation and exercise checks pending.
2. **Logic:** Muẓaffar says muhmal is 'in the force of the particular,' not that the speaker explicitly intended exactly some rather than all; a particular proposition does not assert anything concerning unmentioned cases. The analysis presupposes a nonempty subject extension in its ordinary Aristotelian setting.
3. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH Arabic digital transcription; compare 1968 Najaf ACO scan pending · p. 132–133 https://lib.rafed.net/view.php?b_id=1604&page=132&type=c_fbook

Quantifier expressions and symbols of four categorical forms

السور وألفاظه

The word indicating the quantity of a categorical subject's instances is called the proposition's **quantifier** (sūr al-qaḍiyya). The expression compares the quantifier to a city's encircling wall (sūr), which delimits and encompasses it. Hence these propositions are called *maḥṣūra* and *musawwara*.

The universal affirmative uses terms such as “every,” “all,” “general,” “without exception,” and the Arabic lām of unrestricted inclusion: these attribute the predicate to every instance of the subject.

The universal negative uses “nothing,” “not one,” an indefinite noun under negation, and equivalent expressions: these deny the predicate of every subject instance.

The particular affirmative uses “some,” “one,” “many,” “few,” “sometimes,” “rarely,” and other forms attributing the predicate to at least some subject instances.

The particular negative uses “not some,” “some ... not,” “not every,” “not all,” and other forms denying the predicate of some subject instances.

To abbreviate, Muẓaffar writes **كل** for the universal affirmative, **لا** for the universal negative, **ع** for the particular affirmative, and **س** for the particular negative. With **ب** standing for the subject and **حـ** for the predicate, the four schemata are: **كل ب حـ**, **لا ب حـ**, **ع ب حـ**, and **س ب حـ**, respectively.

1. **Source:** The Arabic follows the 1402 AH digitally transcribed reading with normalized hamza/kāf, punctuation and paragraph divisions; it is NOT a diplomatic transcription. Independent 1968 scanned-page collation and exercise checks pending.
2. **Logic:** The Arabic labels ع and س are the author's chosen abbreviations; do not silently replace them with the later Western A/E/I/O schema. “ليس كل” differs from “كل ... ليس”: keep negation scope explicit.
3. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH Arabic digital transcription; compare 1968 Najaf ACO scan pending · p. 133–134 https://lib.rafed.net/view.php?b_id=1604&page=133&type=c_fbook

Personal, indefinite and quantified conditionals

تقسيم الشرطية إلى شخصية ومهملة ومحصورة

Categorical propositions were classified by their subjects, but conditionals do not have a subject in the categorical sense. Their corresponding classification instead concerns the **states and times** at which a connection or incompatibility obtains. Thus they divide into personal, indefinite, and quantified conditionals—not natural propositions, since a natural proposition judges a universal concept as present in the intellect.

1. Personal (shakhṣiyya). A conditional affirms or denies a connection or disjunction at one determinate time or in one determinate state. Examples of connected conditionals are: “If ʿAlī comes while angry, I will not greet him”; “If it rains today, I will not leave home”; and “It is not the case that, if the teacher is present now, he is occupied with teaching.” Examples of disjunctives are: “The hour now is either one or two”; “Zayd, while at home, is either asleep or awake”; and “It is not the case that the student, while at school, is either standing or in the lesson.”

2. Indefinite (muhmala). The conditional affirms or denies the connection or incompatibility in a state or time without specifying whether this covers all states/times or only some. Connected examples: “If water reaches the measure of a kurr, it does not become impure by contact with an impurity”; “It is not the case that, if a human being lies, he is praised.” Disjunctive examples: “A proposition is either affirmative or negative”; and “It is not the case that a thing is either a metal or gold.” These are the author's original examples, not modernized replacements.

3. Quantified (maḥṣūra). This specifies whether the judgment holds in all relevant states/times or some of them. **Universal** conditionals affirm or deny the relation across every state/time: “Whenever a community cares for virtue, it walks the path of happiness”; “Never is it the case that someone patient amid hardship is thereby unsuccessful in his actions.” For disjunctions: “An integer is always either even or odd”; and “It is never the case that an integer is either even or divisible by two” (a denied exclusive disjunction between equivalent conditions).

Particular conditionals affirm or deny the relation in some unspecified states/times: “It can happen that, if someone is learned, he is happy”; “Not whenever someone acts prudently is he successful.” For disjunctions: “It may happen that a person is either lying down or sitting,” as in a vehicle where standing is impossible; and “It may fail to be the case that a person is either lying down or sitting,” as when standing is possible.

1. **Source:** The Arabic follows the 1402 AH digitally transcribed reading with normalized hamza/kāf, punctuation and paragraph divisions; it is NOT a diplomatic transcription. Independent 1968 scanned-page collation and exercise checks pending.
2. **Scope of negation:** ‘ليس أبداً’ applies negation to the connection/disjunction over all relevant states, while ‘ليس كلما’ denies universality without denying every instance. The example involving evenness and divisibility is a denial of exclusive disjunction, not a claim that neither property holds.
3. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH Arabic digital transcription; compare 1968 Najaf ACO scan pending · p. 134–136 https://lib.rafed.net/view.php?b_id=1604&page=134&type=c_fbook

M-C-09

READING SECTION 9

Quantifying connected and disjunctive propositions

السور في الشرطية

In a categorical proposition, the quantifier specifies how many of the subject's instances are included. In a conditional, it instead specifies whether the connection or disjunction holds across **all** relevant conditions and times or only **some**. Each of the four combinations of quantity and quality has its appropriate expressions.

The **universal affirmative** uses “whenever” (*kullamā*), “whatever” (*mahmā*), “when” (*matā*), and similar expressions for connected conditionals, and “always” (*dāʾiman*) for disjunctions.

The **universal negative** uses “never” (*laysa abadan*) or “not at all / under no circumstances” (*laysa al-batta*) in either kind of conditional.

The **particular affirmative** uses “it may happen” (*qad yakūn*) in both kinds.

The **particular negative** uses “it may not happen” (*qad lā yakūn*) in both, and “not whenever” (*laysa kullamā*) specifically with connected conditionals. The latter denies an unrestricted connection; it does not assert the connection's universal failure.

1. **Source:** The Arabic follows the 1402 AH digitally transcribed reading with normalized hamza/kāf, punctuation and paragraph divisions; it is NOT a diplomatic transcription. Independent 1968 scanned-page collation and exercise checks pending.
2. **Logic:** The occurrence of ‘ليس كلما’ constitutes a quantifier-scope distinction, not merely a negative consequent. Keep this in later contradiction/conversion rules.
3. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH Arabic digital transcription; compare 1968 Najaf ACO scan pending · p. 136–137 https://lib.rafed.net/view.php?b_id=1604&page=136&type=c_fbook

Mental, external, and real categorical propositions

تقسيمات الحملية — الذهنية والخارجية والحقيقية

Divisions of categorical propositions. The earlier divisions of categorical judgments were by quality—affirmative and negative—and by subject—singular, natural, indefinite, and quantified, the last subdividing into universal and particular. Conditional propositions share these divisions to a degree. Three further divisions belong especially to categorical propositions: (1) according to the existence of the subject in an affirmative, (2) according to whether subject and predicate are positive or contain negation, and (3) according to the modality of the predicative relation.

1. Mental, external, and real propositions. An affirmative categorical proposition affirms something of something. Affirmation presupposes that the thing of which a predicate is affirmed exists in the relevant domain: “the throne before the ornament,” as the proverb puts it. If Saʿīd did not exist, we could not truly affirm “Saʿīd is standing” of him.

A negative categorical proposition, by contrast, does not require its subject to exist. Something nonexistent can have every predicate denied of it. Thus logicians say that “a negative is true by the absence of its subject” (*taṣduqu al-sāliba bi-intifāʾ al-mawḍūʿ*). For example, “the father of Jesus son of Mary did not eat, drink, sleep, or speak” is true in that way, because no such father existed. This illustrates the distinction between a negative judgment and an affirmative predicate asserting an existent item.

The subject of a true affirmative must therefore be supposed to exist, but existence may be considered in three ways.

Mental (dhihniyya): the subject exists in the intellect alone. Examples are “Every conjunction of contradictories differs from a conjunction of similars” and “Every ruby mountain is possible in existence.” The relevant concepts are available for intellectual judgment even without the corresponding external objects.

External (khārijiyya): the proposition concerns the individuals of the subject actually existing in the external world at one of the three times—past, present, or future. Examples are “Every soldier in the camp is trained to carry a weapon,” “Some houses threatened with collapse in the town have been demolished,” and “Every pupil in the school is diligent.” The quantification is restricted to the actually instantiated domain.

Real (ḥaqīqiyya): the judgment covers both actual and suppositively existing instances. Anything that could be supposed to exist under the subject concept is included and would have the predicate when it exists. Examples are “The sum of every triangle's angles equals two right angles,” “Some triangles are right-angled,” “Every human is capable of higher education,” and “Every water is pure.” Whether particular instances are actual or merely supposed makes no difference to their inclusion in the judgment.

The last category must not be confused with an existential assertion that all its possible instances are externally present. It is a judgment extending over them under the supposition of their existence.

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1. **Source status:** Directly matched against the displayed 1402 AH Rafed Arabic source passage; editorial punctuation, orthography and paragraph breaks normalized. This is not a diplomatic 1968 ACO image collation or a fully checked set of exercises.
2. **Editorial distinction:** «الحقيقية» here is a judgment including both externally realized and merely supposed instances (when supposed existent), not a statement that all instances already exist. The examples reflect their authorial scientific/doctrinal setting.
3. **Source:** al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH searchable printed-text transcription (Rafed); independently scanned 1968 ACO witness NOT yet collated · p. 138–140 https://lib.rafed.net/view.php?b_id=1604&page=140&type=c_fbook

M-C-11

READING SECTION 11

Positive and negative terms: muhassala and maʿdūla

المحصّلة والمعدولة — سلب الحمل وحمل السلب

2. Propositions with positive or complemented terms. The subject or predicate of a categorical proposition may be *positive* or determinate (*muḥaṣṣal*): a term signifying an existent thing, such as human, Muḥammad, or lion, or a positive attribute, such as learned, just, generous, or learning. Alternatively, a negative particle can become part of the subject or predicate itself, yielding a *complemented* term (*maʿdūl*): non-human, non-learned, non-generous, or without sight.

This distinguishes two forms of propositions. A **muḥaṣṣala**, or one with two positive terms, has both subject and predicate positive, whether the *proposition* is affirmative or negative: “The air is pure”; “The air is not pure.” An affirmative proposition and a proposition with positive terms must therefore not be confused.

A **maʿdūla**, or proposition with a complemented term, has a complemented subject, predicate, or both, and can itself be either affirmative or negative. It is classified accordingly as a proposition with complemented subject, complemented predicate, or complemented terms on both sides. Complementing only one side leaves the other positive.

The author's examples with both terms complemented are “Every non-learned person is someone whose judgment is unsound” and “Every non-diligent person is not someone who is without failure in life.” Examples with complemented predicates are “The air is non-corrupt” and “The air is not non-corrupt”; examples with a complemented subject are “The non-learned person is held in contempt” and “The non-learned person is not happy.”

Critical distinction. A negative proposition whose predicate term is positive (*sāliba muḥaṣṣalat al-maḥmūl*) **negates predication** (*salb al-ḥaml*): “S is not P.” An affirmative with a complemented predicate (*maʿdūlat al-maḥmūl*) **predicates a negation** (*ḥaml al-salb*): “S is non-P.” The difference is logical, not merely stylistic. In the former, negation precedes the copula and removes the predication; in the latter, the copula precedes the negative sign, so the negative element belongs to what is predicated. Arabic commonly uses *laysa* for the first, *lā* or *ghayr* for the second.

The original concludes this subsection with a schematic summary; the digitally transcribed display renders it as an image rather than selectable prose. That figure is **not** transcribed into this reading unit and requires independent facsimile extraction before claiming full source coverage.

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1. **Source status:** Directly matched against the displayed 1402 AH Rafed Arabic source passage; editorial punctuation, orthography and paragraph breaks normalized. This is not a diplomatic 1968 ACO image collation or a fully checked set of exercises.
2. **Editorial distinction:** The figure following «الخلاصة» is an image in the 1402 digital source, so this unit explicitly does not claim its transcription. Keep negation scope distinct: predicating a complemented predicate ≠ simply denying a positive predicate, especially when the subject lacks an external instance.
3. **Source:** al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH searchable printed-text transcription (Rafed); independently scanned 1968 ACO witness NOT yet collated · p. 140–142 https://lib.rafed.net/view.php?b_id=1604&page=141&type=c_fbook

M-C-12

READING SECTION 12

Real modality: necessity, impossibility, special and general possibility

مادّة القضية — الوجوب والامتناع والإمكان الخاص والعام

3. The matter (mādda) of a proposition. For every predicate attributed to a subject, the predicative relation, considered as it actually obtains in reality (*nafs al-amr*), falls under exactly one of three modes by rational division.

Necessity (wujūb): the predicate must belong to the subject on account of what the subject is, so that its denial is impossible. Four must be even. The restriction “on account of the subject itself” excludes what is inseparable only through another condition. The author illustrates the latter with the traditional cosmological claim that the moon's motion follows from the celestial arrangement and its relation to the earth, rather than from the moon's own essence.

Impossibility (imtināʿ): a predicate cannot attach to the subject on account of what the subject is, so its denial is necessary. The coming together of contradictories is intrinsically impossible. A temporarily unavailable capacity—for example, the sleeper's inability to think while unconscious—is not thereby intrinsically impossible in the subject's essence.

Necessity and impossibility have in common the necessity of the judgment; necessity is the necessity of affirmation, whereas impossibility is the necessity of negation.

Special, or real, possibility (imkān khāṣṣ / ḥaqīqī): neither affirmation nor negation of the predicate is intrinsically necessary. One denies necessity **from both sides together** (*salb al-ḍarūra ʿan al-ṭarafayn maʿan*). This is privative relative to the opposed modes of necessity.

General possibility (imkān ʿāmm) is broader. Here only **one of the two opposing necessities** is denied. Saying that something "can exist" means merely that its nonexistence is not necessary; its existence might still be necessary. Saying that something "can fail to exist" means merely that its existence is not necessary; its nonexistence might still be necessary.

Logicians accordingly describe general possibility as denying necessity on the **opposite** side, without determining whether the same-side necessity is present. In the affirmative direction it therefore includes both necessity and special possibility; in the negative direction it includes both impossibility and special possibility.

The author gives two deliberately striking pairs. "God can exist" and "a human can exist" are both true in the *general* affirmative sense because in neither case is existence impossible: the first is necessary in the author's metaphysical account, the second possible in the special sense. "A partner of the Creator can fail to exist" and "a human can fail to exist" both deny necessity of existence: the first includes intrinsic impossibility of the supposed partner, the second special contingency. These are examples illustrating the **weak, one-sided general possibility**, not claims that the Necessary Existent is contingent or that a partner of the Creator is an independently possible existent.

General possibility can thus be applied, appropriately qualified, across the three real modes. It is **not itself a fourth kind of real matter** alongside necessity, impossibility, and special possibility. Those three exhaustive realities are the *materials of propositions* (*mawādd al-qaḍāyā*), also called the elements of assertions and fundamental modes of the predicative relation. General possibility instead belongs among the modalities of *expression*, which the author examines next.

1. **Source status:** Directly matched against the displayed 1402 AH Rafed Arabic source passage; editorial punctuation, orthography and paragraph breaks normalized. This is not a diplomatic 1968 ACO image collation or a fully checked set of exercises.
2. **Editorial distinction:** Preserve the asymmetry between one-sided general possibility (denying opposing necessity) and two-sided special possibility (denying both necessities). Source's traditional cosmology and theological examples are authorial illustrations; they are not updated scientific claims.

3. **Source:** al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH searchable printed-text transcription (Rafed); independently scanned 1968 ACO witness NOT yet collated · p. 142–144 https://lib.rafed.net/view.php?b_id=1604&page=144&type=c_fbook

M-C-13

READING SECTION 13

Expressed modal qualifier versus a proposition's real matter

جهة القضية — الفرق بين المادة والجهة

The modality of a proposition (jiha) as expressed. The *matter* (mādda) of a proposition is its actual predicative status in reality: necessity, impossibility, or possibility in the special sense. A different expression, the *modal qualifier* (jiha), concerns what the wording of the proposition makes us understand of that predicative status.

Both concern the manner of a predicative relation, but they must not be confused. The real matter obtains whether or not one thinks about it or expresses it. The qualifier is precisely the manner of the relation *as signified and understood* when one considers the stated proposition. If no qualification is conveyed, the proposition is without an expressed modality. Even when one is expressed, it need not coincide exactly with the real matter.

“Human is necessarily an animal” has necessity both as its real matter and as its expressed qualification. By contrast, “Human can be an animal” still has necessity as its real matter—the reality does not change when the expression changes—but now the qualifier is the broader *general possibility*: it says only that animality is not impossible for a human, leaving open whether it is necessary. Similarly, “Human can be a writer” could express this same general possibility while the underlying predication has special possibility as its matter.

“Human is always an animal” again has necessity as its underlying matter, but its expressed qualifier is only *permanence*. Permanence can characterize a predicate that is not essential or necessary; Muẓaffar gives the traditional examples of lunar movement and blue eyes. Thus the qualifier does not exhaust the real matter.

A proposition expressly stating its modal qualification is *modal* (*muwajjaha*, a passive participle). One omitting that qualification is *unqualified* (*muṭlaqa*) or *nonmodal* (*ghayr muwajjaha*).

One crucial restriction remains: a qualifier may be **less informative** than the real matter, but it must not **contradict** that matter if the proposition is to be true. If predication is intrinsically impossible, the proposition cannot truthfully assert that the predicate is permanently present or even possible in the affirmative sense. The truth-condition for a modal proposition is therefore that its expressed qualification be compatible with the real predicative relation.

1. **Source status:** Directly matched against the displayed 1402 AH Rafed Arabic source passage; editorial punctuation, orthography and paragraph breaks normalized. This is not a diplomatic 1968 ACO image collation or a fully checked set of exercises.
2. **Editorial distinction:** Technical distinction: mādda = ontic modality of the relation; jiha = expressed/cognized modality of the proposition. 'May fail to match' means may be weaker, not contradictory. The source's example of moon-motion/eye-colour permanence must be left as authorial historical illustration.
3. **Source:** al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH searchable printed-text transcription (Rafed); independently scanned 1968 ACO witness NOT yet collated · p. 144–146 https://lib.rafed.net/view.php?b_id=1604&page=144&type=c_fbook

M-C-14

READING SECTION 14

Eight simple modal proposition forms

أنواع الموجّهات — أقسام البسيطة

Simple and compound modal propositions. A compound modal proposition (*muwajjaha murakkaba*) resolves into two simpler modal propositions of opposite quality: one affirmative and one negative. A simple modal proposition does not resolve in that fashion. Muẓaffar lists eight principal kinds of simple modal proposition, although there are others.

1. Intrinsic necessity (ḍarūriyya dhātiyya). The predicate must belong to the subject, or must be denied of it, throughout the existence of that subject in itself and without an additional condition. Thus “A human is necessarily an animal” and “A tree is necessarily not a breathing creature.” In the affirmative, both real matter and expressed qualifier are necessity; in the negative, they are impossibility of affirmation. A stronger, *eternal necessity* (*ḍarūriyya azaliyya*) dispenses even with the condition “while the subject exists” and is exemplified in the author's theology by God's existence and essential attributes.

2. General conditional necessity (mashrūṭa ʿāmma). Necessity holds only as long as the *description* or title under which the subject is taken remains true. “A walker necessarily moves so long as he is walking.” Movement is not necessary to the underlying person without the qualification “walker.”

3. Unrestricted permanence (dāʾima muṭlaqa). The predicate is continuously affirmed or denied throughout the subject's existence, whether or not the attachment is intrinsically necessary. The traditional examples are “Every celestial sphere is always moving” and a historically contingent claim about a person's skin colour. These are the book's examples, not general scientific or anthropological assertions by the translator. The author distinguishes permanence in fact from impossibility of being otherwise.

4. General customary permanence (ʿurfiyya ʿāmma). Permanence holds while the qualifying title of the subject applies: “Every writer's fingers keep moving as long as the person is writing.” The predicate is not therefore permanent for the underlying individual irrespective of being a writer.

5. General actuality (muṭlaqa ʿāmma), also called the actual (fiʿliyya). The predicative relation has actually been realized, whether or not it is necessary or permanent, and whether its realization belongs to past, present, or future. The original illustrations include a human walking and a sphere moving. This is less restrictive than the preceding necessities or permanences.

6. Actuality during the qualified state (ḥīniyya muṭlaqa). The relation is realized *when* the subject possesses its stated description: “Every bird is actually beating its wings while it is flying.” The phrase “while it is flying” is logically operative, not merely narrative.

7. General possibility (mumkina ʿāmma). This excludes necessity of the opposite side of the affirmed or denied relation. It neither requires actuality nor permanence and does not itself exclude a stronger necessity of the same-side relation. For example, “Every human can be a writer in the general sense”: being a writer is not impossible for humans, even if it never occurs for some particular people. In this formal system it is broader than all six preceding simple forms.

8. Possibility during the qualified state (ḥīniyya mumkin(a)). General possibility obtains with regard to the subject while it has the qualifying description. “Every walking person can have hands that do not tremble while walking.” The form addresses the mistaken thought that the predicate could not possibly attach during the state specified.

All eight are descriptions of **what a proposition's modality says**; they are not eight additional intrinsic kinds of real modality beyond necessity, impossibility, and special possibility. The following section begins the distinct compound forms.

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1. **Source status:** Authorial Arabic from Dār al-Taʿāruf (1402 AH) digital transcription pp.146–148; independent 1968 printed image and all punctuation, examples and exercises are not diplomatically collated. List preserves eight source cases and eternal-necessity subdivision.
2. **Historical examples:** Celestial-sphere motion, fixed skin colour and social descriptions are retained to represent the original philosophical illustrations; they are not certified empirical facts.
3. **Source:** al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH digital transcription; independent 1968 printed witness check pending · p. 146–148 https://lib.rafed.net/view.php?b_id=1604&page=147&type=c_fbook

READING SECTION 15

Compound modal propositions: the implicit opposite-quality judgment

أقسام المركبة — القضية الصريحة والمشار إليها

Compound modal propositions. A compound proposition is analysable into two modal judgments, one affirmative and one negative. Muẓaffar now makes the structure precise. The **first component** is stated explicitly and may itself be positive or negative; the compound proposition receives its positive or negative classification from this expressly stated component. A **second component**, opposite to it in quality but the same in quantity, is indicated rather than fully expressed, commonly by “not always” (*lā dāʾiman*) or “not necessarily” (*lā bi-l-ḍarūra*).

The second qualification is needed because a general modal statement can leave open whether the predicate belongs by necessity or without necessity, or whether its attribution is permanent or nonpermanent. To exclude the stronger reading, one adds the corresponding negative qualification; it should not be confused with denying the original assertion.

For example, “Every person who prays in fact avoids indecency” is a universal affirmative of general actuality. It does not yet decide whether avoiding indecency follows **necessarily** from the description of praying. Adding “not necessarily” denies necessity while retaining the preceding actual assertion. Likewise, adding “not always” rejects perpetual attachment without canceling the positive first component.

The author expresses the implicitly signified second judgment in the corresponding **universal negative modal** form, with the same quantity as the first, but of opposite quality. His specific formal schemata are retained here rather than replaced by a simplified modern predicate-logic formula, because his use of scope, possibility and the modal negative requires separate analysis.

The six familiar compounds are now enumerated.

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1. **Source-critical status:** Working authorial-source transcription with normalized punctuation, spelling and paragraph breaks. The 1402 digital edition and separately published 1421 digital text were compared for the indicated argument and examples; no 1968 facsimile or manuscript comparison has been carried out. https://ablibrary.net/book_content/b/4628/178?lang=ar
2. **Logical/translation note:** The author says the two components agree in *quantity* and differ in *quality*. Do not read the implicit negative as if it simply erased the explicitly affirmed actual occurrence. His equivalent 'no S is P' forms are technical modal presentations, not unqualified factual denials; a formalization requires carefully locating the modal operator.
3. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH digital source (normalized transcription); compared at indicated locus with later Rafed 1421 AH digital edition, not a facsimile collation · p. 148–149 https://lib.rafed.net/view.php?b_id=1604&page=154&type=c_fbook

READING SECTION 16

Two special compounds: qualified necessity and customary permanence

المشروطة الخاصة والعرفية الخاصة

*1. The special conditioned necessity (*al-mashrūṭa al-khāṣṣa*). Begin with a universal conditioned-necessity proposition: the predicate belongs necessarily to the subject for as long as the specified description holds. This alone is compatible with two possibilities. The predicate might also belong throughout the existence of the underlying subject even after it no longer satisfies the description, or it might not. The* special *compound excludes the first alternative by adding a denial of permanence throughout the subject's existence as such (*al-lā-dawām al-dhātī*).

The proposition consequently has two components: an expressly asserted general conditional necessity, and an implicitly indicated general actuality of the opposite quality. Muẓaffar's example is that every tree necessarily grows so long as it is a tree, “but not always”—the qualification denying that growth lasts as long as the underlying subject exists regardless of its title. He also supplies an elliptical technical negative formula to express the second component. Its negation must not cancel the necessity under the stated description. The special form is narrower than the general conditioned form precisely because it removes the possibility of unrestricted permanence.

*2. The special customary proposition (*al-ʿurfiyya al-khāṣṣa*). Begin instead with general customary permanence: the predicate holds continuously while the subject satisfies its qualifying description, without asserting intrinsic necessity of the predicate. Add the same denial of* unrestricted* permanence through the subject's underlying existence. Thus “Every tree always grows while it is a tree—but not always [without regard to this description].” This combines an explicitly stated general customary proposition with an implicitly indicated general-actuality negative, and is narrower than the unrestricted general customary form.

The first compound asserts necessity *under the title* and denies unconstrained permanence; the second asserts permanence *under the title* and denies unconstrained permanence. The difference between **necessity** and **actual duration** remains essential.

1. **Source-critical status:** Working authorial-source transcription with normalized punctuation, spelling and paragraph breaks. The 1402 digital edition and separately published 1421 digital text were compared for the indicated argument and examples; no 1968 facsimile or manuscript comparison has been carried out. https://ablibrary.net/book_content/b/4628/179?lang=ar
2. **Logical/translation note:** The qualifier «لا دائماً» restricts duration in the *underlying subject*, not duration while the defining title holds. The 1421 parallel digital edition preserves both types and their tree examples. The short source schemata «لا شيء من الشجر بنام بالفعل» are presented as authorial abbreviated modal notation; their scope is not silently converted into the unconditional claim 'no tree grows'.
3. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH digital source (normalized transcription); compared at indicated locus with later Rafed 1421 AH digital edition, not a facsimile collation · p. 149 https://lib.rafed.net/view.php?b_id=1604&page=154&type=c_fbook

Actual yet not necessary; actual yet not permanent

الوجودية اللاضرورية والوجودية اللادائمة

*3. Actuality without intrinsic necessity (*al-wujūdiyya al-lā-ḍarūriyya*). An unrestricted general-actuality proposition tells us that the predication really occurs. It leaves open whether the predicate necessarily belongs to the subject's own essence. The compound form specifies not by intrinsic necessity**, thus combining an expressly affirmed general actuality with an implicitly indicated general possibility of the opposite quality.

The step is logical: general possibility denies the necessity of the **opposite** side. If the proposition explicitly affirms P of S, adding “not necessarily [P]” rules out necessary P; this in turn licenses the opposite side in the broad possibility sense. The author illustrates the distinction with “Every human is in fact breathing, but not necessarily.” He gives a technical negative formulation for the implied second component. The compound is called *wujūdiyya* because the explicitly stated form asserts actuality, and *lā-ḍarūriyya* because it excludes necessity of that actuality.

*4. Actuality without intrinsic permanence (*al-wujūdiyya al-lā-dāʾima*). General actuality also leaves open whether the predicate remains permanently true of the subject as such. Here one adds “not always” to exclude such permanence. The resulting compound has two general-actuality modal components** of opposite quality, one stated openly and the other indicated by the negative qualification.

Muẓaffar's example is expressed in the source with a negative first component about human breathing followed by “not always”; his stated explanatory equivalent then affirms breathing as actual. This polarity is retained as an **editorially unresolved example** rather than silently harmonized with a preferred affirmative template. The governing distinction is nonetheless clear: *lā-ḍarūriyya* excludes essential necessity, while *lā-dāʾima* excludes unrestricted duration. Neither qualifier by itself denies the first component's actual occurrence.

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1. **Source-critical status:** Working authorial-source transcription with normalized punctuation, spelling and paragraph breaks. The 1402 digital edition and separately published 1421 digital text were compared for the indicated argument and examples; no 1968 facsimile or manuscript comparison has been carried out. https://ablibrary.net/book_content/b/4628/180?lang=ar
2. **Logical/translation note:** The 1402 text's «لا شيء من الإنسان بمتنفس بالفعل، لا دائماً» followed by «أي أن كل إنسان متنفس بالفعل» is counterintuitive as a literal negative and apparent positive paraphrase. The 1421 digital witness has the same juxtaposition; retain both as an open **authorial/source example issue**, not a demonstrated printed variant. Do not silently replace the original initial negative with a positive.
3. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH digital source (normalized transcription); compared at indicated locus with later Rafed 1421 AH digital edition, not a facsimile collation · p. 150 https://lib.rafed.net/view.php?b_id=1604&page=154&type=c_fbook

Occurrent but nonpermanent; genuinely two-sided possibility

الحينية اللادائمة والممكنة الخاصة

*5. Actuality while qualified, without enduring actuality of the subject as such (*al-ḥīniyya al-lā-dāʾima*). The starting statement is an* actuality-in-a-specified-condition*: the predicate is present while the subject bears the stated title or description. Without further qualification, this is compatible both with permanence throughout the underlying subject's existence and with its absence. Adding "not always" rules out the unrestricted permanence. The compound consequently consists of an explicitly stated* ḥīniyya muṭlaqa *and an implicitly signified* muṭlaqa ʿāmma* of opposite quality.

The author's example is "Every bird is in fact beating its wings while it is [flying as] a bird, but not always [throughout its existence]." He supplies a negative technical expression for the hidden part, which must not be mistaken for saying that no bird ever beats its wings.

*6. Special possibility (*al-mumkina al-khāṣṣa*). The starting general-possibility proposition has already denied necessity of the opposite side of its assertion. This does not exclude necessity of the asserted side. The special compound adds "not necessarily" to deny that latter necessity as well. The result is two-sided absence of necessity,* salb al-ḍarūra ʿan al-ṭarafayn*: both affirmative and negative directions remain intrinsically possible.

For an affirmative compound this removes necessary affirmation; for a negative compound it removes necessary negation or impossibility. The fully expressed composite thus has two *mumkina ʿāmma* components of opposite quality, and its expressed qualification (*jiha*) coincides with the actual intrinsic possibility (*mādda*) when the proposition is true.

The author abbreviates the composite by saying "Every animal moves in the special possibility sense," meaning that motion is possible and its absence is possible. He also supplies the long form, "Every animal can move in the general sense, but not necessarily." The latter must be understood in the logic of intrinsic modality, not as an observation that an individual animal happens to be at rest.

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1. **Source-critical status:** Working authorial-source transcription with normalized punctuation, spelling and paragraph breaks. The 1402 digital edition and separately published 1421 digital text were compared for the indicated argument and examples; no 1968 facsimile or manuscript comparison has been carried out. https://ablibrary.net/book_content/b/4628/181?lang=ar
2. **Logical/translation note:** The 1421 digital source at p.181 also preserves both forms. Interpret «لا بالضرورة» as absence of *same-side* necessity supplementing the already absent *opposite-side* necessity. The example of moving animals is part of the historical author's logic, not proof of a universal zoological law.
3. **Source:** Muḥammad Riḍā al-Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH digital source (normalized transcription); compared at indicated locus with later Rafed 1421 AH digital edition, not a facsimile collation · p. 151 https://lib.rafed.net/view.php?b_id=1604&page=154&type=c_fbook

READING SECTION 19

Seven authorial exercises on modal relations and compounds

تمرينات على الموجهات والمركبة

Author's exercises — retained as exercises, not silently replaced with answers.

1. Determine which of the four relations between two universals holds between intrinsic necessity and unrestricted permanence. Do the same for intrinsic necessity compared respectively with general conditioned necessity and general customary permanence.

2. Determine the relation between unrestricted permanence and each of general actuality and general customary permanence.

3. Determine the relation between general conditioned necessity and general customary permanence, and between intrinsic necessity and the special conditioned compound.

4. If we qualify general conditioned necessity by the absence of intrinsic necessity, is the resulting compound well formed?

5. Can actuality limited to the period in which a description holds (*ḥīniyya muṭlaqa*) be qualified by absence of intrinsic necessity? If it can, what should the compound be called?

6. Can unrestricted permanence itself be qualified by absence of intrinsic necessity?

7. Invent one illustration of each simple modal proposition and convert it, where logically possible, into one of the six compound forms just described.

These seven questions test the inclusion, noninclusion, and possible joint satisfaction of *intrinsic necessity*, *permanence*, *actuality*, and *general/special possibility*. Some combinations may fail for reasons arising from the implicit second component.

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1. **Editorial status:** Digitally presented authorial prose and examples; spelling/punctuation normalized. Not a diplomatic facsimile or independently manuscript-collated passage.
2. **Logical note:** The 1402 edition inserts an authorial summary chart immediately before these exercises as a nontext image. The chart is NOT transcribed here; the later digital edition exposes a textualized chart at p.182, but it has not yet been independently matched to the earlier image. Do not treat this seven-exercise text as exhaustive of p.152's visual content.
3. **Source:** Muẓaffar, al-Manṭiq, 1402 AH Dār al-Taʿāruf digital printed-source text; parallel 1421 Qum digital page where specified; no early 1968 facsimile collation · p. 152 https://lib.rafed.net/view.php?b_id=1604&page=154&type=c_fbook

READING SECTION 20

Necessary versus coincidental connected conditionals

تقسيمات الشرطية الأخرى — اللزومية والاتفاقية

Further divisions of conditional propositions. The connected/disjunctive distinction concerns the relation, the affirmative/negative distinction its quality, and the singular/indefinite/quantified distinction the circumstances or times to which it applies. Further subdivisions concern the character of the relation itself.

A **connected conditional** is either *necessary in connection* (*luzūmiyya*) or *coincidental* (*ittifāqiyya*).

In a necessary connection, a genuine relation entails that one side follows from the other. This may take several causal forms: the antecedent causes the consequent, the consequent is a sign of a cause corresponding to it, or both statements describe effects of a common cause. Muẓaffar illustrates the first with water expanding when heated; the second with expansion indicating the presence of heat; and the third with boiling and expansion arising from heating to an appropriate level. These are the author's explanatory causal illustrations, not an independently verified modern physical model of all fluids.

In a coincidental connection, no intrinsic relation of entailment links the antecedent and consequent, although the latter happens when the former occurs. Suppose the student Muḥammad only arrives after the teacher has begun the lesson. We can say, “Whenever Muḥammad arrives, the teacher will already have begun the class.” The recurring sequence, however, may express an accidental pairing rather than a necessary relation between the student's arrival and the teacher's beginning.

Muẓaffar warns against treating **repeated coincidence alone** as proof of necessary connection. Frequent succession of two events does not establish that the first makes the second necessary.

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1. **Editorial status:** Digitally presented authorial prose and examples; spelling/punctuation normalized. Not a diplomatic facsimile or independently manuscript-collated passage.
2. **Logical note:** Logical point: al-luzūmiyya is an asserted genuine explanatory nexus, not merely the material conditional's actual truth; al-ittifāqiyya is repeated conjunction without that nexus. Examples and traditional physical causal descriptions preserved as historical authorial illustrations.
3. **Source:** Muẓaffar, al-Manṭiq, 1402 AH Dār al-Taʿāruf digital printed-source text; parallel 1421 Qum digital page where specified; no early 1968 facsimile collation · p. 153–154 https://lib.rafed.net/view.php?b_id=1604&page=154&type=c_fbook

READING SECTION 21

Disjunctive incompatibility: essential versus coincidental; genuine division

أقسام المنفصلة — العنادية والاتفاقية والحقيقية

Disjunctive conditionals: two classifications must be kept distinct.

The first asks whether the incompatibility between the alternatives is **intrinsic** (ʿ*ināḍiyya*) or merely **incidental** (*ittifāqiyya*). Intrinsic exclusion follows from the very relations asserted in the two components, as when an integer is either even or odd. Incidental exclusion arises from a circumstance external to the definitions: for example, the person sitting at home is either Muḥammad or Bāqir when we happen to know that nobody else is there; or a particular book is either about logic or belongs to Khālid, where circumstances ensure that Khālid owns no book on logic.

The second classification asks whether the two sides can both be true and whether they can both be false.

A **genuine or complete disjunction** (*ḥaqīqiyya*) affirms that its two alternatives **cannot both be true and cannot both be false**. This is exclusive and exhaustive disjunction. “An integer is either even or odd” provides the author's elementary illustration.

Its **negation** denies that two alternatives stand in this relationship of exhaustive mutual exclusion. The author illustrates it with “It is not the case that an animal is either rational or capable of being taught,” because both descriptions can occur together in a human and both can be absent in other animals. Here he explains the failure of exclusion and exhaustion, rather than asserting that every animal has neither property.

Genuine disjunction serves to establish exhaustive classifications into two or more alternatives. No modal claim about a mere accidental conjunction should be smuggled into an essential dichotomy.

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1. **Editorial status:** Digitally presented authorial prose and examples; spelling/punctuation normalized. Not a diplomatic facsimile or independently manuscript-collated passage.
2. **Logical note:** The first distinction (ʿināḍiyya/ittifāqiyya) concerns whether exclusion is intrinsic; the second (ḥaqīqiyya/māniʿat al-jamʿ/māniʿat al-khulūw) concerns the truth-conditions. Negative disjunction is the denial of the *disjunctive relation*, not necessarily a conjunction of denials.
3. **Source:** Muẓaffar, al-Manṭiq, 1402 AH Dār al-Taʿāruf digital printed-source text; parallel 1421 Qum digital page where specified; no early 1968 facsimile collation · p. 154–155 https://lib.rafed.net/view.php?b_id=1604&page=154&type=c_fbook

READING SECTION 22

At-most-one versus at-least-one disjunctions

مانعة الجمع ومانعة الخلو، الموجبة والسالبة

*2. Disjunction preventing conjunction (*māniʿat al-jamʿ*). In its affirmative form, this asserts that at most one of the two alternatives can be true. It does* not* require that one of them be true; both may fail. For example, a body may be either white or black under the intended exclusion, but a red body is neither, so the alternatives are not exhaustive. The negative form denies the at-most-one condition: the two descriptions can occur together. Muẓaffar's paired example uses “non-white” and “non-black”: both obtain together in a red body, although their double denial would demand that one body be simultaneously white and black.

The author applies the affirmative form to his doctrinal premise that an imam and a sinner cannot be the same person, although a person could be neither; the negative form to the claim that prophethood and imamate can coexist within one household. These are his school-specific illustrations of exclusive versus nonexclusive predication, not new historical arguments independently endorsed by the translation.

*3. Disjunction preventing joint absence (*māniʿat al-khulūw*). Its affirmative form asserts that at least one** of two alternatives is true; the two may also be true together. Thus every body is either non-white or non-black, and a red body satisfies both. Another source example is “Either the body is in water or it does not sink”: the body may be in water and not sink, but the author excludes the circumstance of sinking outside water, under his assumed conditions.

The negative form denies that at least one must always hold: both alternatives may fail. Consequently, “It is not the case that a body is either white or black” allows a body of some other colour while not asserting that white and black can coexist.

He applies the affirmative form to the claim that everything is a cause or an effect, potentially both relative to different things. His negative form rejects the supposed exhaustive opposition between an intelligent person without religion and a religious person without intelligence, because the two properties can belong together. The underlying logical distinction is between **at least one true** and **not necessarily at least one true**; the historical examples do not establish a timeless exhaustive ontology without additional premises.

1. **Editorial status:** Digitally presented authorial prose and examples; spelling/punctuation normalized. Not a diplomatic facsimile or independently manuscript-collated passage.

2. **Logical note:** Do not conflate inclusive 'at least one' with exclusive 'exactly one'. The water/sinking example depends upon implicit empirical assumptions; the theological examples are authorial illustrations. The original 1402 source text on pp.155–156 contains all named categories and examples, corresponding to later print's topical headings.
3. **Source:** Muẓaffar, al-Manṭiq, 1402 AH Dār al-Taʿāruf digital printed-source text; parallel 1421 Qum digital page where specified; no early 1968 facsimile collation · p. 155–156 https://lib.rafed.net/view.php?b_id=1604&page=154&type=c_fbook

M-C-23

READING SECTION 23

Conditionals made of conditionals: nested and mixed propositions

تنبيه أول — تأليف الشرطيات

First note: constructing conditionals. Ordinary writing frequently omits the explicit logical markers that would reveal a proposition's formal type. Beginners may therefore misclassify arguments or fail to see their inferential force, especially with conditionals. Muẓaffar directs the reader to reconstruct their actual structure.

Every conditional has two sides that, taken on their own, were propositions. A disjunctive proposition may have three or more alternatives. The component propositions may themselves be categorical, connected conditional, disjunctive, or a mixture. There is no useful upper limit to the arrangements.

A connected conditional can have a **categorical antecedent and connected-conditional consequent**: “If knowledge causes happiness, then if a human is learned, that person is happy.” The outer antecedent is “knowledge causes happiness,” while the outer consequent is itself the full conditional concerning a learned human.

A connected conditional may also have a **categorical antecedent and disjunctive consequent**: “If an expression is simple, it is either a noun, a verb, or a particle.” The consequent here consists of three mutually distinguished alternatives.

A disjunctive proposition can combine a **categorical alternative and a conditional alternative**: “Either the earth's interposition is not a cause of the lunar eclipse, or, if the earth intervenes between moon and sun, the moon is eclipsed.” The latter is an entire conditional serving as one side of the disjunction; Muẓaffar's astronomical example is quoted as an example of structure, not independently endorsed geometry.

By repeating this operation, one obtains connected conditionals made of two connected conditionals, two disjunctives, or mixed components; similarly for disjunctive propositions. **A conditional nested within another is still one component of the outer proposition**, not a separate unrelated outer proposition. A correct reconstruction must respect the scope and hierarchy of every connective.

1. **Source status:** Authorial prose from 1402 Rafed printed pp.156–157, with normalized punctuation; no independent 1968 scanned-page collation.
2. **Syntax note:** Do not flatten “if A, then (if B, then C)” into two unrelated clauses or substitute a different formula without proving equivalence.
3. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH authorial Arabic digital text; normalized, not independently image-collated · p. 156–157
https://lib.rafed.net/view.php?b_id=1604&page=154&type=c_fbook

M-C-24

READING SECTION 24

Displaced propositions and reconstructing their logical form

تنبيه ثانٍ — المنحرفات والتطبيقات

Second note: propositions displaced from their normal form. A proposition may depart from its most transparent logical syntax, so that its category becomes difficult to recognize. Muẓaffar calls such a proposition *munḥarifa*, “displaced” or “deviant” in its expression, without implying that its meaning is inherently false.

A categorical can be displaced by attaching the quantifier to the **predicate** instead of the **subject**: “Human is part of animal,” or “Human is not the whole of animal.” He restores the normal categorical arrangement as “Some animals are humans” and “Not every animal is human,” respectively. The reconstructions display the intended quantity of the subject.

A conditional may omit the usual signs of entailment or incompatibility and resemble a categorical. The sentence “Either the sun is not shining or it is daytime” can express a connected form (“Whenever the sun shines, it is daytime”) or a disjunctive form (“Either the sun is not shining or it is daytime”), depending on the intended relation. Likewise, “Daytime does not occur without the sun shining” can express one of the foregoing structures.

The moralizing expression “Wealth accumulates only through avarice or illicit gain” can be analysed either as a disjunction of the two stated sources, or as a connected conditional: “If wealth accumulates, its accumulation results from either avarice or illicit gain.” In the latter reconstruction the antecedent is categorical while the consequent is disjunctive. The content is the author's example, not an independent claim about the causes of wealth.

Muẓaffar applies the procedure to three further statements.

First, “A human has only what he has striven for” expresses restriction and resolves into an affirmative and a negative categorical: what a human has comes from the results of effort, and what a person has not striven for is denied in the indicated respect.

Second, “Whoever becomes possessed by greed has debased himself” is an implicit connected conditional: “Whenever a person becomes possessed by greed, he has debased himself.”

Third, “One who clung to you has not been disappointed” is a displaced **universal affirmative with negative predicate content**: “Every person who clings to you does not fail.” Its status as affirmative is at the level of the principal copular assertion, not the absence of the word *not* in its predicate.

The underlying rule is to recover **quantity, quality, connective, and scope** before drawing inferences; surface grammar alone can obscure the proposition's structure.

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1. **Source status:** Authorial examples from the 1402 Rafed text printed pp.157–158; spelling/punctuation normalized; no independent early facsimile comparison.
2. **Technical distinction:** The analysis of 'Every S is non-P' versus 'Not every S is P' depends on the earlier distinction between complemented terms and negated predication; this passage preserves the author's own source transformations.
3. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH authorial Arabic digital text; normalized, not independently image-collated · p. 157–158 https://lib.rafed.net/view.php?b_id=1604&page=154&type=c_fbook

M-C-25

READING SECTION 25

Four authorial exercises in classifying and restoring conditionals

تمرينات على الشرطيات والمنحرفات

Author's exercises at the end of the initial treatment of propositions. Answers have deliberately not been supplied: the author intends independent logical analysis.

1. Classify two connected conditionals as intrinsically necessary (*luzūmiyya*) or coincidental (*ittifāqiyya*): “Whenever an animal ruminates, it has a split hoof”; and “Whenever a human being is short, that person is intelligent.” The exercise tests whether a constant correlation implies an essential nexus.

2. Identify the forms of these statements and restore any displaced forms: (a) “When answers crowd one another, correctness becomes hidden”; (b) “When power or capacity increases, desire diminishes”; (c) “Whoever gains becomes proud”; (d) “Whoever reveals his harm has accepted humiliation”; (e) “Only those of His servants who possess knowledge fear God.” Preserve the original special restriction in the final Qurʾānic example rather than turning it into an unrestricted affirmative.

3. Identify the logical kind of “Time has two days: a day for you and a day against you.” If its expression is displaced, reconstruct the explicitly qualified form and give its type.

4. Identify the logical kind of the attribution to ʿAlī: “The earth is never without someone who stands as God's proof, either well known and openly visible or fearful and concealed.” If it is a displaced proposition, reconstruct it in standard form and classify its disjunction.

This ends the original first-chapter material on propositions in the 1402 searchable witness. The final *summary chart* appearing as an image is not independently transcribed or certified in this reader.

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1. **Source status:** Working original Arabic with all four end-of-chapter exercise questions, including the five lettered subitems, copied from 1402 digitally presented pp.159. The closing summary is an image in this edition and is not part of this textual unit; 1968 scanned print and transmitted attribution in question 4 not independently verified.
2. **Pagination correction · Stage VIII:** This exercise text is under Rafed’s displayed p.160. The inherited p.159 locator is corrected. The earlier summary chart remains an explicit omission; this pagination repair does not add a facsimile transcription.
3. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH authorial Arabic digital text; normalized, not independently image-collated · p. 160 https://lib.rafed.net/view.php?b_id=1604&page=160&type=c_fbook

M-C-26

READING SECTION 26

Why relations between propositions are needed

في أحكام القضايا أو النسب بينها

Chapter Two: The rules governing propositions, or the relations between them

Introduction. An investigator frequently encounters difficulty in proving the proposition sought directly; at times this may even be impossible. He therefore turns to proving another proposition that bears a relation to the proposition sought, so that he can compare the two. Knowledge that the proved proposition is true may give him knowledge that the proposition sought is false, or conversely. This happens when the truth of either necessarily accompanies the falsity of the other. Alternatively, knowledge that the proved proposition is true may give him knowledge that the proposition sought is true; or knowledge that the first is false may give him knowledge that the second is false. This happens when the first proposition's truth entails the second's truth, or its falsity entails the other's falsity.

Before beginning the study of inference, therefore, and after becoming acquainted with a number of propositions, the student of logic must know the relations between them. He can then sometimes prove what he seeks by proving another proposition related to it, and his mind can pass from the proposition whose truth or falsity has been demonstrated to the truth or falsity of the proposition about which he is seeking knowledge.

The investigations through which the relations between propositions are known are those of contradiction, simple conversion, contraposition, and their supplementary topics. These are called “the rules governing propositions.” God willing, we shall now begin these investigations in the order just given.

1. **Source status:** Complete consecutive authorial prose in this logical division, transcribed from Rafed displayed page labels 162–162; Arabic letter-shape normalization only (Persian kaf/ya to Arabic equivalents), with source spelling anomalies retained. This displayed-page locator is verified from HTML page containers and is not silently identified with inherited printed-page metadata. English is an independent working translation, not a copied published translation.
2. **Digital-source anomaly:** Rafed displayed p162 inserts فيلتجيء between يعاني and الباحث in the opening sentence. The Arabic here preserves it; English follows the intelligible clause and records the extra word as unresolved digital corruption. The source also has minor spacing/spelling anomalies; no independent 1968 facsimile comparison is claimed.
3. **Source:** Muẓaffar, al-Manṭiq, Rafed digital presentation catalogued as Dār al-Taʿāruf 1402 AH; displayed-page text, not independently facsimile-collated · p. 162 https://lib.rafed.net/view.php?b_id=1604&page=162&type=c_fbook

M-C-27

READING SECTION 27

Contradiction: need, definition, and the force of “in itself”

التناقض والحاجة إليه وتعريفه

Contradiction

Why this investigation is needed. We said in the introduction that it is often necessary to establish a proposition other than the proposition sought. Knowledge of its falsity, however, entails knowledge of the truth of the proposition sought, or conversely, when the truth of either entails the falsity of the other.

Two propositions with this property are contradictory propositions. Suppose, for example, that you wish to demonstrate the truth of “The soul exists,” but cannot do so directly. It is sufficient to demonstrate the falsity of its contradictory, “The soul does not exist.” Once you know that this contradictory is false, you must know that the first proposition is true, because contradictories cannot both be false. If you demonstrate the truth of the contradictory, you must know that the first proposition is false, because contradictories cannot both be true. [The digital Arabic repeats an incomplete sentence here; see the textual note.]

One might suppose that identifying a proposition's contradictory is obvious, like identifying the contradictories of simple terms, such as human and nonhuman, where a difference of affirmation and negation suffices. The matter is not so simple, however. An affirmative and a negative can both be true: “Some animals are human” and “Some animals are not human.” They can also both be false: “Every animal is human” and “No animal is human.”

Accordingly, to identify the contradictory of each proposition, the investigator cannot dispense with the rules of contradiction stated in logic.

Definition of contradiction. You have already learned what contradiction means as one of the kinds of opposition. Let us now express it in a comprehensive technical formula specifically concerning propositions: “Contradiction between propositions is a difference between two

propositions that, in itself, requires one of them to be true and the other false.”

The qualification “in itself” is indispensable. A difference between two propositions may require them to differ in truth and falsity, but because of something else, rather than because of that difference itself. Consider “Every human is an animal” and “No human is an animal.” Because the subject is narrower than the predicate, one of these two universal propositions is true and the other false. If the subject were broader than the predicate, however, both would be false: “Every animal is human” and “No animal is human,” as above.

By the difference that requires them to differ in truth, we mean a difference that requires this whatever the propositions' matter may be, and whatever the relation between subject and predicate: for example, the difference between a universal affirmative and a particular negative.

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1. **Source status:** Complete consecutive authorial prose in this logical division, transcribed from Rafed displayed page labels 163–164; Arabic letter-shape normalization only (Persian kaf/ya to Arabic equivalents), with source spelling anomalies retained. This displayed-page locator is verified from HTML page containers and is not silently identified with inherited printed-page metadata. English is an independent working translation, not a copied published translation.
2. **Digital-source anomaly:** Displayed p163 repeats the incomplete sequence وإذا برهنت على صدق النقيض لا بد أن تعلم صدق لأن النقيضين لا يكذبان معا between the two complete correct sentences. Arabic preserves this duplication; English translates the complete reasoning once and explicitly flags the intrusion. Displayed p164 has حويان and شيسء, contextually read in English as animal and thing; these are not silently emended in Arabic.
3. **Source:** Muẓaffar, al-Manṭiq, Rafed digital presentation catalogued as Dār al-Taʿāruf 1402 AH; displayed-page text, not independently facsimile-collated · p. 163–164 https://lib.rafed.net/view.php?b_id=1604&page=163&type=c_fbook

M-C-28

READING SECTION 28

The eight unities and unity of predication

شروط التناقض والوحدات الثمان

Conditions of contradiction

For two propositions to contradict one another, they must agree in eight respects and differ in three.

The eight unities. The respects in which two contradictory propositions must agree are called “the eight unities.” They are as follows.

1. **Subject.** If the subjects differ, there is no contradiction: “Knowledge is beneficial”; “Ignorance is not beneficial.”

2. **Predicate.** If the predicates differ, there is no contradiction: “Knowledge is beneficial”; “Knowledge is not harmful.”

3. **Time.** There is no contradiction between “The sun is shining,” meaning during the day, and “The sun is not shining,” meaning during the night.

4. **Place.** There is no contradiction between “The land is fertile,” meaning in the cultivated countryside, and “The land is not fertile,” meaning in the desert.

5. **Potentiality and actuality.** The two propositions must agree in potentiality and actuality. There is no contradiction between “Muḥammad is dead,” meaning potentially, and “Muḥammad is not dead,” meaning actually.

6. **Whole and part.** There is no contradiction between “Iraq is fertile,” meaning part of it, and “Iraq is not fertile,” meaning the whole of it.

7. **Condition.** There is no contradiction between “The student will pass at the end of the year,” meaning if he works diligently, and “The student will not pass,” meaning if he does not work diligently.

8. **Relation.** There is no contradiction between “Four is a half,” meaning in relation to eight, and “Four is not a half,” meaning in relation to ten.

Notice. These eight unities are those commonly recognized among logicians. Some add “unity of predication,” with regard to its being primary predication or common predication. This condition is necessary: contradictory propositions must agree in their predication. If predication is primary in one and common in the other, both can be true. For example, they say: “The particular is particular,” by primary predication, and “The particular is not particular,” by common predication. This is because the concept of the particular is an instance of the concept of the universal: it is predicable of many.

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1. **Source status:** Complete consecutive authorial prose in this logical division, transcribed from Rafed displayed page labels 164–165; Arabic letter-shape normalization only (Persian kaf/ya to Arabic equivalents), with source spelling anomalies retained. This displayed-page locator is verified from HTML page containers and is not silently identified with inherited printed-page metadata. English is an independent working translation, not a copied published translation.
2. **Source:** Muẓaffar, al-Manṭiq, Rafed digital presentation catalogued as Dār al-Taʿāruf 1402 AH; displayed-page text, not independently facsimile-collated · p. 164–165 https://lib.rafed.net/view.php?b_id=1604&page=164&type=c_fbook

M-C-29

READING SECTION 29

Differences in quantity, quality, and modality

الاختلاف بالكم والكيف والجهة

The differences

We said that two contradictory propositions must differ in three respects: quantity, quality, and modality.

Difference in quantity and quality. This means that if one is affirmative, the other is negative; and if one is universal, the other is particular. Accordingly:

The contradictory of the universal affirmative is the particular negative.

The contradictory of the particular affirmative is the universal negative.

If both were affirmative or both negative, they could both be true or both false. If both were universal, they could both be false, as when the subject is broader than the predicate, in the examples given above. If both were particular, they could both be true, again when the subject is broader: “Some minerals are iron” and “Some minerals are not iron.”

Difference in modality. Difference in modality is required by the nature of contradiction, just as difference in affirmation and negation is. The contradictory of anything is its removal. Thus, just as affirmation is removed by negation and negation by affirmation, a modality must be removed by a modality that contradicts it.

The modality that removes another modality may be one of the recognized modalities, in which case there is an explicit contradictory. General possibility is removed by necessity, and conversely, because possibility is the negation of necessity.

Alternatively, it may not be one of the recognized modalities for which we have a recognized name. We must then seek a recognized modality that necessarily accompanies it and apply that name to it. What we have is not an explicit contradictory, but a necessary concomitant of the contradictory.

For example, the permanent proposition is contradicted by the general actual proposition, but not by explicit contradiction: each is a necessary concomitant of the other's contradictory. If you say “The earth is always in motion,” its explicit contradictory is the negation of permanence. The negation of permanence, however, is not one of the recognized modalities. We therefore seek a modality necessarily accompanying it and say: the absence of permanence requires that motion be denied of the earth at some time; that is, “The earth is actually not in motion.” This is a general actual proposition, necessarily accompanying the contradictory of the permanent proposition.

If you say “Every human is actually writing,” its explicit contradictory is that writing is not established of a human in that manner, that is, actually. This requires permanence of negation: “Some humans are always not writing.” This is a permanent proposition, necessarily accompanying the contradictory of the general actual proposition.

There is no need to detail the contradictories of modal propositions. A student who wants them may consult the more extensive works. He has no need of them, however, and we advise him not to trouble himself with acquiring them, because they are of little benefit.

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1. **Source status:** Complete consecutive authorial prose in this logical division, transcribed from Rafed displayed page labels 165–167; Arabic letter-shape normalization only (Persian kaf/ya to Arabic equivalents), with source spelling anomalies retained. This displayed-page locator is verified from HTML page containers and is not silently identified with inherited printed-page metadata. English is an independent working translation, not a copied published translation.
2. **Digital-source anomaly:** Displayed p166 reads نقيضها السريع and لاكتابة, contextually rendered as its explicit contradictory and writing. Arabic preserves these readings; English retains the logical distinction between the explicit contradictory and its necessarily accompanying recognized modality.
3. **Logical framework:** The author's modality of actuality is technical and should not be reduced to "happening at this instant." In his example, the negation concerns actual occurrence under the stated scope and is represented by a permanent negative.
4. **Source:** Muẓaffar, al-Manṭiq, Rafed digital presentation catalogued as Dār al-Taʿāruf 1402 AH; displayed-page text, not independently facsimile-collated · p. 165–167 https://lib.rafed.net/view.php?b_id=1604&page=165&type=c_fbook

M-C-30

READING SECTION 30

Subalternation, contrariety, and subcontrariety

التداخل والتضاد والدخول تحت التضاد

Supplementary to contradiction: subalternation, contrariety, and subcontrariety

We have seen that, among the four quantified propositions, contradiction holds between the universal affirmative and the particular negative, and between the particular affirmative and the universal negative: that is, between propositions differing in both quantity and quality. It remains to consider the relations between the other pairs, which differ only in quantity or only in quality. Knowing these relations is also useful for inferring a proposition from knowledge of another proposition related to it, as will become clear.

Thus, if two quantified propositions differ in both quantity and quality, they are contradictories, already discussed. If they differ in only one of these respects, there are three cases.

1. **Subalterns** differ in quantity but not in quality: both are affirmative or both negative. They are called subalterns because one is included in the other: the particular is included in the universal.

This means that if the universal proposition is true, the particular proposition agreeing with it in quality is true; the converse does not hold. It follows that if the particular proposition is false, the universal proposition agreeing with it in quality is false; the converse does not hold.

For example, "Every piece of gold is a mineral" is true, and "Some gold is a mineral" must certainly be true with it. Again, "Some gold is black" is false, and "Every piece of gold is black" must be false with it.

2. **Contraries** differ in quality but not in quantity and are both universal. They are called contraries because, like contrary things, they cannot both be true but can both be false.

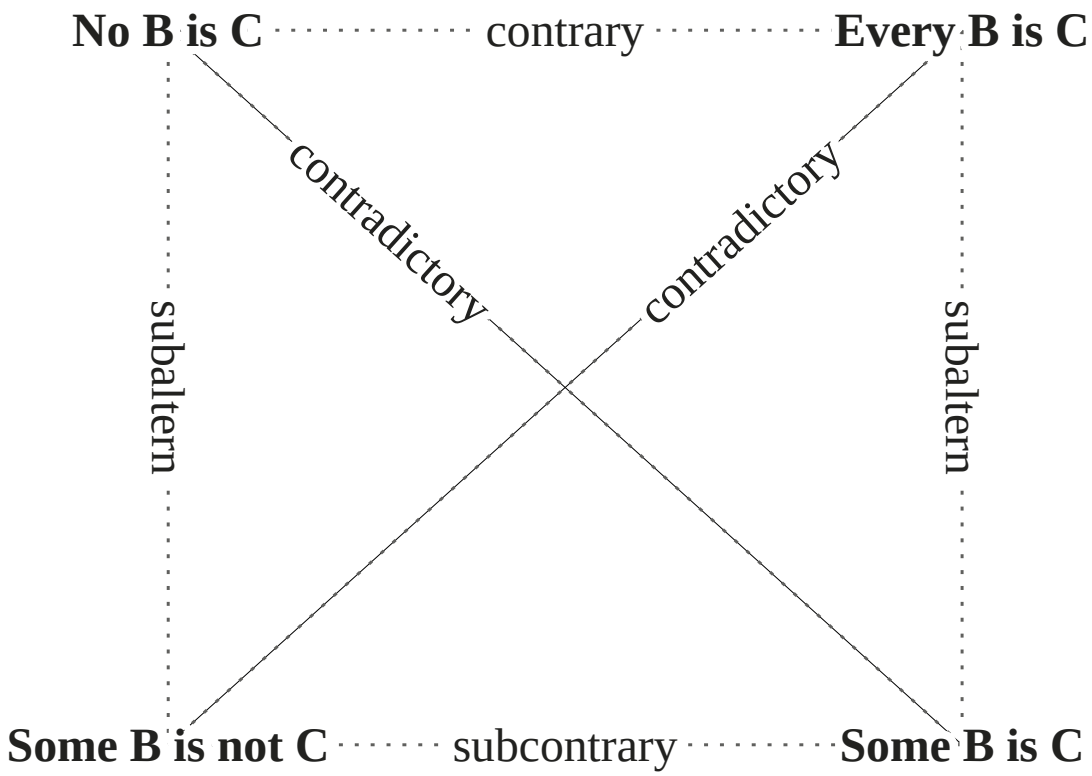
This means that if one is true, the other must be false; the converse does not hold. If one is false, the other need not be true. For example, if “Every piece of gold is a mineral” is true, “No gold is a mineral” must be false. If “Every mineral is gold” is false, however, “No mineral is gold” need not be true: in this example it is false.

3. **Subcontraries** differ in quality but not in quantity and are both particular. They are called subcontraries, first, because they fall under the two universal propositions, each under the universal agreeing with it in quality; and second, because their relation in truth and falsity is the reverse of that of contraries: they cannot both be false but can both be true.

This means that if one is false, the other must be true; the converse does not hold. If one is true, the other need not be false. For example, if “Some gold is black” is false, “Some gold is not black” must be true. If “Some minerals are gold” is true, however, “Some minerals are not gold” need not be false: in this example it is true.

Since ancient times, logicians have customarily illustrated all the relations between the quantified propositions in the following diagram.

Transcription of the source diagram. Upper right: “Every B is C”; upper left: “No B is C.” These are contraries. Lower right: “Some B is C”; lower left: “Some B is not C.” These are subcontraries. Each vertical pair is a pair of subalterns. Each diagonal pair is a pair of contradictories.



Square of opposition — redrawn from the source diagram (displayed p. 169).

1. **Source status:** Complete consecutive authorial prose in this logical division, transcribed from Rafed displayed page labels 167–169; Arabic letter-shape normalization only (Persian kaf/ya to Arabic equivalents), with source spelling anomalies retained. This displayed-page locator is verified from HTML page containers and is not silently identified with inherited printed-page metadata. English is an independent working translation, not a copied published translation.
2. **Source figure:** The square on Rafed displayed p169 (image014.gif) was visually inspected and its four formulas and six relations fully transcribed here. This is a transcription of the digital image, not independent collation of the 1968 facsimile.
3. **Logical framework:** The traditional square as presented assumes the subject domains required by its existential commitments. The text's subalternation and subcontrariety should not be reinterpreted as unrestricted laws of modern empty-domain class logic. Authorial examples about gold are retained as examples, not independently verified scientific claims.
4. **Source:** Muẓaffar, al-Manṭiq, Rafed digital presentation catalogued as Dār al-Taʿāruf 1402 AH; displayed-page text, not independently facsimile-collated · p. 167–169 https://lib.rafed.net/view.php?b_id=1604&page=167&type=c_fbook

M-C-31

READING SECTION 31

Simple conversion and its conditions

العكس المستوي وشروطه

Conversions

At the beginning of this chapter, we said that an investigator may need to establish a proposition related to the one sought, so as to infer the latter's truth from the former's truth, because they are connected by an entailment in truth. Such entailment holds between a proposition and its simple

converse, and between it and its contrapositive. We shall now discuss both kinds.

Simple conversion. Simple conversion is “the interchange of a proposition's two terms while preserving quality and truth.” That is, a proposition judged true is transformed into another proposition following the first in truth and in affirmation or negation, by interchanging its two terms: the first proposition's subject becomes the second's predicate, and its predicate the second's subject; or its antecedent becomes the consequent, and its consequent the antecedent.

The first is called the original proposition, and the second its simple converse. The term “conversion” thus has two technical uses here: the interchange itself, and the proposition in which the interchange has occurred.

To say that the converse follows the original in truth means that if the original is true, the converse must be true. It need not follow it in falsity: the original may be false while its converse is true. It follows that the original does not follow its converse in truth, but does follow it in falsity. If the converse is false, the original is false, because if the original were true, its converse would have to be true, whereas the converse is assumed false.

There are thus two rules useful in inference:

1. If the original proposition is true, its converse is true.

2. If the converse is false, its original proposition is false.

The second rule follows from the first, as you now know.

Conditions of conversion. Conversion takes place under three conditions: interchange of the terms, preservation of quality, and preservation of truth. Preservation of quantity is not required. What is required is preservation of truth, which may require quantity to be preserved in some propositions and changed in others.

What matters in the following discussion is to identify the propositions for which preserving truth in conversion requires preservation of quantity, and those for which it does not.

If the terms are interchanged and quality is preserved, but truth is not preserved, this is not called conversion; it is called a mere reversal.

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1. **Source status:** Complete consecutive authorial prose in this logical division, transcribed from Rafed displayed page labels 170–171; Arabic letter-shape normalization only (Persian kaf/ya to Arabic equivalents), with source spelling anomalies retained. This displayed-page locator is verified from HTML page containers and is not silently identified with inherited printed-page metadata. English is an independent working translation, not a copied published translation.
2. **Source:** Muẓaffar, al-Manṭiq, Rafed digital presentation catalogued as Dār al-Taʿāruf 1402 AH; displayed-page text, not independently facsimile-collated · p. 170–171 https://lib.rafed.net/view.php?b_id=1604&page=170&type=c_fbook

Conversion of universal and particular affirmatives

الموجبتان تنعكسان موجبة جزئية

Both affirmatives convert to a particular affirmative

The universal affirmative converts to a particular affirmative, and the particular affirmative converts to one of its own kind. Thus:

“Every C is B” converts to “Some B is C.”

“Some C is B” converts to “Some B is C.”

Neither converts to “Every B is C.”

Proof. (1) In the universal proposition, the predicate is either broader than the subject or coextensive with it. In either case the particular converse is certainly true, because in either case the subject is predicable of some instances of the predicate. Thus:

“Every water is liquid” gives the true converse “Some liquid is water.”

“Every human is rational” gives the true converse “Some rational being is human.”

The universal converse, however, is not true in every case. In the first case, the subject is not predicable of all instances of the predicate, since it is narrower than the predicate. Accordingly, “Every liquid is water” is false, which establishes the point sought.

(2) In the particular proposition, the predicate may be absolutely broader than the subject, absolutely narrower, overlapping it, or coextensive with it. In the first and third of these cases, its universal affirmative converse is not true. When the predicate is absolutely broader than the subject or overlaps it, the subject does not apply to all instances of the predicate. Such a universal converse is true only when the predicate is narrower or coextensive. Its particular affirmative converse, however, is true in every case:

“Some liquid is water” gives “Some water is liquid.”

“Some water is liquid” gives “Some liquid is water.”

“Some birds are white” gives “Some white things are birds.”

“Some humans are rational” gives “Some rational beings are human.”

1. **Source status:** Complete consecutive authorial prose in this logical division, transcribed from Rafed displayed page labels 171–172; Arabic letter-shape normalization only (Persian kaf/ya to Arabic equivalents), with source spelling anomalies retained. This displayed-page locator is verified from HTML page containers and is not silently identified with inherited printed-page metadata. English is an independent working translation, not a copied published translation.
2. **Digital-source anomaly:** Displayed p172 prints كل ماء مسائل. The English reads “Every water is liquid,” explicitly taking مسائل as a digital corruption of سائل on the evidence of the immediately following example كل سائل ماء. Arabic preserves مسائل; no second-witness attestation is asserted.
3. **Source:** Muẓaffar, al-Manṭiq, Rafed digital presentation catalogued as Dār al-Taʿāruf 1402 AH; displayed-page text, not independently facsimile-collated · p. 171–172 https://lib.rafed.net/view.php?b_id=1604&page=171&type=c_fbook

M-C-33

READING SECTION 33

Conversion of the universal negative and its proof

السالبة الكلية تنعكس سالبة كلية

The universal negative converts to a universal negative

Both quantity and quality are preserved. If “No animal is a tree” is true, “No tree is an animal” is true.

The proof is clear. A universal negative is true only when its subject and predicate are wholly disjoint. Wholly disjoint terms never coincide, so each can be denied of every instance of the other, whichever is made the subject.

For practice in constructing proofs by contradiction and conversion, we shall demonstrate this in the following form.

Assumption: “No B is C” is true.

Claim: “No C is B” is also true.

Proof. If “No C is B” were not true, its contradictory, “Some C is B,” would be true. Then “Some B is C,” the simple converse of that contradictory, would be true. Comparing this simple converse with the original proposition, “No B is C,” we find that it is its contradictory. If “Some B is C” were true, “No B is C” would therefore have to be false, although it is assumed true. Hence “No C is B” must be true, which was to be proved.

Further explanation. This proof shows the usefulness of contradiction and simple conversion in inference. We must retrace the proof and say:

“No B is C” is assumed true.

Its contradictory, “Some B is C,” is therefore false.

This contradictory is the converse of “Some C is B,” so the latter is also false: if the converse is false, the original is false, by the second rule.

When this original proposition, “Some C is B,” is false, its contradictory, “No C is B,” is true, which is the conclusion sought.

Thus you have used, first, the truth of an original proposition to establish the falsity of its contradictory; second, the falsity of a converse to establish the falsity of its original; and third, the falsity of an original to establish the truth of its contradictory.

You will encounter this form of inference frequently. Examine it carefully and master it.

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1. **Source status:** Complete consecutive authorial prose in this logical division, transcribed from Rafed displayed page labels 172–173; Arabic letter-shape normalization only (Persian kaf/ya to Arabic equivalents), with source spelling anomalies retained. This displayed-page locator is verified from HTML page containers and is not silently identified with inherited printed-page metadata. English is an independent working translation, not a copied published translation.
2. **Source:** Muẓaffar, al-Manṭiq, Rafed digital presentation catalogued as Dār al-Taʿāruf 1402 AH; displayed-page text, not independently facsimile-collated · p. 172–173 https://lib.rafed.net/view.php?b_id=1604&page=172&type=c_fbook

M-C-34

READING SECTION 34

Why particular negatives and disjunctives have no simple converse

السالبة الجزئية والمنفصلة لا عكس لهما

The particular negative has no converse. It never converts to either a universal or a particular proposition. Its subject may be broader than its predicate, as in “Some animals are not human.” The broader term cannot be denied of the narrower under any circumstances, either universally or particularly. Whenever the narrower applies, the broader applies with it. How, then, could the broader be denied of it? Neither “No human is an animal” nor “Some humans are not animals” is true.

The disjunctive proposition has no converse. At the beginning of this investigation we indicated that simple conversion concerns categorical and conditional propositions alike. On reflection, however, we find that conversion of a disjunctive proposition has no benefit. At most, it signifies incompatibility between antecedent and consequent. There is no natural order between them: either can be made the antecedent and the other the consequent, without any difference arising. Whether you say “Number is either even or odd” or “Number is either odd or even,” the meaning is the same.

This is why they say that a disjunctive proposition has no converse: there is no benefit in converting it.

If you transform it into a categorical proposition, however, the rules for categorical propositions apply. In this example you might say “Number is divided into even and odd.” This converts to “What is divided into even and odd is number.”

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1. **Source status:** Complete consecutive authorial prose in this logical division, transcribed from Rafed displayed page labels 174–174; Arabic letter-shape normalization only (Persian kaf/ya to Arabic equivalents), with source spelling anomalies retained. This displayed-page locator is verified from HTML page containers and is not silently identified with inherited printed-page metadata. English is an independent working translation, not a copied published translation.
2. **Digital-source anomaly:** Displayed p174 ends with وفرود, contextually rendered as “and odd.” The Arabic retains this digital reading. The author is denying a distinct inferential gain from reversing disjuncts, rather than denying commutativity of disjunction.
3. **Source:** Muẓaffar, al-Manṭiq, Rafed digital presentation catalogued as Dār al-Taʿāruf 1402 AH; displayed-page text, not independently facsimile-collated · p. 174 https://lib.rafed.net/view.php?b_id=1604&page=174&type=c_fbook

M-C-35

READING SECTION 35

Two methods of contraposition and their quantity rules

عكس النقيض وقاعدته من جهة الكم

Contraposition

This is the second transformation of a proposition whose truth is inferred from the truth of the original. It has two methods.

1. **The method of the earlier logicians**, called contraposition agreeing in quality, because it agrees with its original in quality: “Transforming a proposition into another whose subject is the contradictory of the original predicate and whose predicate is the contradictory of the original subject, while preserving truth and quality.”

Briefly, it is “interchanging the contradictories of the two terms while preserving truth and quality.” Thus “Every writer is human” becomes, by contraposition agreeing in quality, “Every nonhuman is a nonwriter.”

2. **The method of the later logicians**, called contraposition differing in quality, because it differs from its original in quality: “Transforming a proposition into another whose subject is the contradictory of the original predicate and whose predicate is the original subject itself, while preserving truth but changing quality.”

Thus “Every writer is human” becomes, by contraposition differing in quality, “No nonhuman is a writer.”

The rule of contraposition with respect to quantity. The negatives here are governed by the same rule as the affirmatives in simple conversion, and the affirmatives here by the same rule as the negatives there. That is:

1. The universal negative becomes a particular: negative in contraposition agreeing in quality, affirmative in contraposition differing in quality.

2. The particular negative likewise becomes a particular: negative in the agreeing form, affirmative in the differing form.

3. The universal affirmative becomes a universal: affirmative in the agreeing form, negative in the differing form.

4. The particular affirmative does not contrapose at all.

Proof. Each of these rules must be demonstrated. These proofs train the student to use contradiction and conversion in inference. We have adopted the style of theoretical geometry in presenting the demonstrations, so they will be easy for anyone familiar with the method of geometrical textbooks. An example has already been explained in the proof of the simple conversion of the universal negative.(1)

For brevity and clarity, in everything that follows we symbolize the contradictory of a term by putting a fatḥa above its letter:

B′, written بَ in the Arabic notation: the contradictory of the subject.

C′, written حَ in the Arabic notation: the contradictory of the predicate.

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1. **Source status:** Complete consecutive authorial prose in this logical division, transcribed from Rafed displayed page labels 175–176; Arabic letter-shape normalization only (Persian kaf/ya to Arabic equivalents), with source spelling anomalies retained. This displayed-page locator is verified from HTML page containers and is not silently identified with inherited printed-page metadata. English is an independent working translation, not a copied published translation.
2. **Notation:** English prime marks and the verbal form “non-B/non-C” represent the author’s fatḥa above the Arabic term-letter. Arabic ع denotes a particular affirmative and س a particular negative; لا denotes a universal negative. The isolated reference marker (1) on displayed p176 has no corresponding footnote text in this captured page; no note has been invented.
3. **Source:** Muẓaffar, al-Manṭiq, Rafed digital presentation catalogued as Dār al-Taʿāruf 1402 AH; displayed-page text, not independently facsimile-collated · p. 175–176 https://lib.rafed.net/view.php?b_id=1604&page=175&type=c_fbook

M-C-36

READING SECTION 36

Two proofs for the contraposition of the universal negative

برهان عكس السالبة الكلية

Proof of the contraposition of the universal negative

To establish the contraposition of the universal negative we give two proofs: one for contraposition agreeing in quality and one for contraposition differing in quality.

First. The claim is that it becomes a particular negative by contraposition agreeing in quality, and does not become a universal negative. There are therefore two conclusions sought. If “No B is C” is true, then “Some non-C is not non-B” is true, while “No non-C is non-B” does not follow.

Proof. The following are known:

1. A universal negative is true only when its two terms are wholly disjoint. This is self-evident.

2. The relation between the contradictories of two wholly disjoint terms is partial disjointness. This was demonstrated in the investigation of relations in Part One.

3. Partial disjointness amounts to two particular negatives, just as total disjointness amounts to two universal negatives. This too is self-evident.

From these three premises it follows that, if “No B is C” is true, so that the terms are wholly disjoint, both particular negatives between their contradictories are true: “Some non-B is not non-C,” and “Some non-C is not non-B.” This establishes the first conclusion sought.

The second premise also shows that the contradictories of two wholly disjoint terms are not always wholly disjoint: they may overlap. A universal negative between the contradictories of wholly disjoint terms is therefore not always true. In other words, “No non-C is non-B” is not always true. This establishes the second conclusion sought.

Second. The claim is that the universal negative becomes a particular affirmative by contraposition differing in quality, and does not become a universal affirmative. Again there are two conclusions sought. If “No B is C” is true, “Some non-C is B” is true, while “Every non-C is B” does not follow.

Proof. Since B and C are wholly disjoint, as above, one applies together with the contradictory of the other. That is, B applies together with non-C. If B and non-C coincide, at least “Some non-C is B” is true: the first conclusion sought. [In the digital formula at this point the mark over C is missing; this explicit reading restores the complement required by the preceding sentence.]

We have also seen that the contradictories of wholly disjoint terms can overlap. On that supposition, non-C occurs together with [non-B]. For those instances, non-C does not occur together with B; otherwise the contradictories B and non-B would coincide. Therefore “Every non-

C is B” is not true: the second conclusion sought. [The digital text writes B in both of the opposed lines; the first “non-B” here is an explicit conjectural reading, not an independently witnessed correction.]

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1. **Source status:** Complete consecutive authorial prose in this logical division, transcribed from Rafed displayed page labels 177–178; Arabic letter-shape normalization only (Persian kaf/ya to Arabic equivalents), with source spelling anomalies retained. This displayed-page locator is verified from HTML page containers and is not silently identified with inherited printed-page metadata. English is an independent working translation, not a copied published translation.
2. **Unresolved digital formulas:** Displayed p178 has ع حـ ب where the prose requires ع حَـ ب, and then writes ب مع حَـ in both opposed lines, where the first should read بَ مع حَـ for the stated overlap argument. Arabic preserves both digital readings. English explicitly supplies conjectural complement marks and does not present them as verified print variants. Independent early-print collation is still required.
3. **Logical framework:** The universal-negative-to-particular-affirmative contrapositive uses the traditional existential assumptions of this treatment. “Does not convert to a universal” means that the universal result is not generally entailed, not that it is false in every individual case.
4. **Source:** Muẓaffar, al-Manṭiq, Rafed digital presentation catalogued as Dār al-Taʿāruf 1402 AH; displayed-page text, not independently facsimile-collated · p. 177–178 https://lib.rafed.net/view.php?b_id=1604&page=177&type=c_fbook

M-C-37

READING SECTION 37

Proofs of the two contrapositives of the particular negative

برهان عكس السالبة الجزئية

To establish contraposition of the particular negative, we again give two proofs: one for contraposition retaining quality and one for contraposition changing quality.

First. We claim that the particular negative gives a particular negative by contraposition retaining quality, but does not give a universal. Thus two things are to be established. If “Some B are not C” is true, “Some non-C are not non-B” is true—the first conclusion—but “No non-C is non-B” need not be true—the second conclusion.

Proof. A particular negative can be true in three situations:

1. Its terms overlap without either including the other. Their complements then stand in partial exclusion, as explained in the discussion of relations.
2. Its terms are wholly disjoint. Their complements likewise stand in partial exclusion, as previously explained.
3. Its subject is strictly more extensive than its predicate. The complement of the predicate is then strictly more extensive than the complement of the subject.

In all three situations, “Some non-C are not non-B” is true. This establishes the first conclusion, whether because the complements partially exclude one another or because non-C is more extensive than non-B. In some of these situations, however, the complementary terms overlap or

one is strictly more extensive than the other. Therefore the universal negative “No non-C is non-B” is not invariably true. This establishes the second conclusion.

Second. We claim that the particular negative gives a particular affirmative by contraposition changing quality, but does not give a universal. Thus, if “Some B are not C” is true, “Some non-C are B” is true—the first conclusion—but “All non-C are B” need not be true—the second conclusion.

Proof. We have already explained that, in every possible arrangement of the subject and predicate of the particular negative, either their complements partially exclude one another or the complement of the predicate is strictly more extensive. In either case, some non-C occurs without non-B. Therefore some non-C occurs with B, since the two contradictories, B and non-B, cannot both be absent. Hence “Some non-C are B” is true, which is the first conclusion.

The complements of the subject and predicate may also overlap. “Some non-C are non-B” may therefore be true. This can be transformed into the true statement “Some non-C are not B.” The former is an affirmative with a complemented predicate; it can be made a negative with an uncomplemented predicate. With quantity unchanged, the negative with an uncomplemented predicate is more extensive than the affirmative with a complemented predicate; when the more restricted statement is true, the more extensive is certainly true. If “Some non-C are not B” is true, its contradictory, “All non-C are B,” is false. This establishes the second conclusion.

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1. **Source status:** Complete continuous authorial passage from Rafed’s digitally presented Dār al-Taʿāruf 1402 AH text, displayed page(s) 178–180; Arabic kaf/ya and paragraph/table whitespace normalized. Displayed pagination is recorded, not silently equated with a scanned print. Original symbolic readings remain in Arabic even where damaged; each conjectural English reconstruction is identified. The 1968 institutional scan has not been collated for this unit.
2. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination, not independently image-collated · p. 178–180 https://lib.rafed.net/view.php?b_id=1604&page=178&type=c_fbook

M-C-38

READING SECTION 38

Proofs of the two contrapositives of the universal affirmative

برهان عكس الموجبة الكلية

To establish contraposition of the universal affirmative, we likewise give two proofs, for the forms retaining and changing quality.

First. The universal affirmative gives a universal affirmative by contraposition retaining quality. If “All B are C” is true, as assumed, “All non-C are non-B” is true, as required.

Proof. If “All non-C are non-B” were not true, its contradictory, “Some non-C are not non-B,” would be true. Its contrapositive retaining quality, “Some B are not C,” would then be true. The contradictory of this last statement, “All B are C,” would therefore be false. This contradicts the assumption: the contradictory of that contrapositive is precisely the original statement assumed true. Consequently “All non-C are non-B” must be true, which is the required conclusion.

Second. The universal affirmative gives a universal negative by contraposition changing quality. If “All B are C” is true, as assumed, “No non-C is B” is true, as required.

Proof. If “No non-C is B” were not true, its contradictory, “Some non-C are B,” would be true. By simple conversion, “Some B are non-C” would then be true. This is a particular affirmative with a complemented predicate. As explained earlier, it can be transformed into a particular negative with an uncomplemented predicate, yielding “Some B are not C.” Its contradictory would therefore be false. **The digital formula at this point reads “All non-B are non-C”; the conclusion required by the preceding line and by the author's following words is “All B are C.”** The author continues: “This is a contradiction, since it is the original statement assumed true.” Therefore “No non-C is B” must be true, which is the required conclusion.

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1. **Source status:** Complete continuous authorial passage from Rafed’s digitally presented Dār al-Taʿāruf 1402 AH text, displayed page(s) 180–181; Arabic kaf/ya and paragraph/table whitespace normalized. Displayed pagination is recorded, not silently equated with a scanned print. Original symbolic readings remain in Arabic even where damaged; each conjectural English reconstruction is identified. The 1968 institutional scan has not been collated for this unit.
2. **Unresolved symbolic reading:** At displayed p181 the second proof prints «كل بَ حَـ» as the contradictory of «س ب حـ» and immediately identifies it with the original «كل ب حـ». The English identifies the required reading explicitly, but the Arabic is not emended. Rafed’s separate b1728 digital presentation p214 repeats the same damaged formula; this is not independent facsimile attestation.
3. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination, not independently image-collated · p. 180–181 https://lib.rafed.net/view.php?b_id=1604&page=180&type=c_fbook

M-C-39

READING SECTION 39

Why the particular affirmative has neither form of contraposition

الموجبة الجزئية لا تنعكس بعكس النقيض

To prove that the particular affirmative admits neither contraposition retaining quality nor contraposition changing quality, it is enough to show that it does not yield a particular proposition. Its failure to yield a universal follows all the more readily. We have explained that the particular is included under the universal: if the particular is false, the universal is false. We therefore say:

First. The particular affirmative does not yield a particular affirmative by contraposition retaining quality. If “Some B are C” is true, “Some non-C are non-B” need not be true.

Proof. One situation in which a particular affirmative is true is that its terms overlap without either including the other. Their complements then stand in partial exclusion, which is more general than complete disjointness or overlap. If the complements are completely disjoint, “No non-C is non-B” is true. Its contradictory, “Some non-C are non-B,” is therefore false. This is the required conclusion.

Second. The particular affirmative does not yield a particular negative by contraposition changing quality. If “Some B are C” is true, “Some non-C are not B” need not be true.

Proof. As previously explained, the complements of the terms of a particular affirmative may be completely disjoint. The resulting universal negative is then intended to be “No non-C is non-B.” **The digital formula instead omits the complement mark on B and reads “No non-C is B.”** The author continues that “All non-C are B” is true, because negation of negation is affirmation. Its contradictory, “Some non-C are not B,” is therefore false, which establishes the conclusion. The reconstruction “No non-C is non-B” is required by that stated reason; it is not an independently verified variant.

To understand why the particular affirmative does not admit contraposition, consider “Some non-humans are animals.” By contraposition retaining quality this does not yield either “Some non-animals are humans” or “All non-animals are humans”: both are false, since no non-animal is a human. By contraposition changing quality it does not yield either “Not every non-animal is a non-human” or “No non-animal is a non-human.” These too are false, since every non-animal is a non-human.

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1. **Source status:** Complete continuous authorial passage from Rafed’s digitally presented Dār al-Taʿāruf 1402 AH text, displayed page(s) 181–183; Arabic kaf/ya and paragraph/table whitespace normalized. Displayed pagination is recorded, not silently equated with a scanned print. Original symbolic readings remain in Arabic even where damaged; each conjectural English reconstruction is identified. The 1968 institutional scan has not been collated for this unit.
2. **Unresolved symbolic reading:** Displayed p182 has «لا حَـ ب» where the surrounding claim that the complements are disjoint and the subsequent double-negation inference require «لا حَـ بَ». Arabic retained; English reconstruction labelled. The b1728 digital text p215 shares the omission and supplies no independent print attestation.
3. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination, not independently image-collated · p. 181–183 https://lib.rafed.net/view.php?b_id=1604&page=181&type=c_fbook

M-C-40

READING SECTION 40

Exercises in consequence, conversion and juristic contraposition

تمرينات على العكوس

1. Suppose the proposition “Every rational person is one whom prosperity does not make arrogant” is true. Determine the truth or falsity of each of the following propositions, giving the reason:

a. Some rational people are not made arrogant by prosperity.

b. Some rational people are not among those who are not made arrogant by prosperity.

c. All those who are not made arrogant by prosperity are rational.

d. No rational person is among those who are not made arrogant by prosperity.

e. Everyone who is made arrogant by prosperity is non-rational.

f. No one who is made arrogant by prosperity is rational.

g. Some of those who are not made arrogant by prosperity are rational.

2. Suppose “Some minerals do not melt under heat” is false. Extract the true and false propositions that follow from the falsity of this proposition.

3. In his commentary *al-Īḍāḥ*, Fakhr al-Muḥaqqiqīn argued that water becomes ritually impure through an alteration by impurity assessed hypothetically. He said: “Water is overcome by impurity when the alteration is hypothetically assessed, because whenever water is not overcome, it would not be altered by it on the supposition of a difference. By contraposition this becomes: whenever water would be altered by impurity on the supposition of a difference, it is overcome.”

Explain which form of contraposition this is and how it is obtained. Notice that the proposition employed here is a connected conditional.

Author's footnote. The author of *al-Madārik* quoted this argument in his discussion of water and then raised an objection against it. Consult that discussion if you wish.

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1. **Source status:** Complete continuous authorial passage from Rafed’s digitally presented Dār al-Taʿāruf 1402 AH text, displayed page(s) 184–184; Arabic kaf/ya and paragraph/table whitespace normalized. Displayed pagination is recorded, not silently equated with a scanned print. Original symbolic readings remain in Arabic even where damaged; each conjectural English reconstruction is identified. The 1968 institutional scan has not been collated for this unit.
2. **Translation and exercise scope:** The opening Arabic «كل عاقل لا تبطره النعمة» is formally affirmative predication of the negative expression “not made arrogant by prosperity.” The English retains that structure. Item b renders the traditional particular-negative use of «ليس بعض», not a wide-scope denial of the existential quantifier. The juristic quotation and attribution have not been independently collated with al-Īḍāḥ or al-Madārik.
3. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination, not independently image-collated · p. 184
https://lib.rafed.net/view.php?b_id=1604&page=184&type=c_fbook

READING SECTION 41

Complementation and the four proofs of predicate obversion

النقض وقاعدة نقض المحمول

Among the discussions no less important than conversion for inferring the truth of a proposition from the truth of its original is **complementation** (*naqḍ*). It is therefore appropriate to append it to the discussion of conversion.

Complementation transforms a proposition into another that follows from it in truth, while leaving its terms in their original positions. It has three forms:

1. Make the complement of the first proposition's subject the subject of the second, while keeping its predicate as predicate. This transformation is called **subject-complementation**, and the transformed proposition is called **subject-complemented**.
2. Keep the first proposition's subject as subject and make the complement of its predicate the predicate. This transformation is called **predicate-complementation**, and the transformed proposition is called **predicate-complemented**.
3. Make the complement of the subject the subject and the complement of the predicate the predicate. This is called **complete complementation**, and the transformed proposition is called **complemented in both terms**.

We shall examine the rule for each form, beginning with predicate-complementation, since it provides the way to the others, as will become clear.

The rule for predicate-complementation. To obtain a true predicate-complemented proposition when its original is true, change the quality of the proposition and replace its predicate by its complement, leaving the subject and quantity unchanged. We must prove this for each of the four quantified forms.

1. Universal affirmative. Its predicate-complemented form is a universal negative. Thus “Every human is an animal” becomes “No human is a non-animal.” Symbolically, if “All B are C” is true —the assumption—“No B is non-C” is true—the required conclusion.

Proof: from “All B are C,” contraposition changing quality gives “No non-C is B.” Simple conversion gives “No B is non-C,” which is the required conclusion.

2. Particular affirmative. Its predicate-complemented form is a particular negative. Thus “Some animals are human” becomes “Not every animal is non-human.” If “Some B are C” is true—the assumption—“Some B are not non-C” is true—the required conclusion.

Proof: if “Some B are not non-C” were not true, its contradictory, “All B are non-C,” would be true. Predicate-complementation would give “No B is C.” Its contradictory, “Some B are C,” would therefore be false. But this is the original statement, contrary to the assumption. Thus “Some B are not non-C” must be true.

3. Universal negative. Its predicate-complemented form is a universal affirmative. Thus “No water is solid” becomes “All water is non-solid.” If “No B is C” is true—the assumption—“All B are non-C” is true—the required conclusion.

Proof: if “All B are non-C” were not true, its contradictory, “Some B are not non-C,” would be true. Then “Some B are C” would be true, because negation of negation is affirmation. Its contradictory, “No B is C,” would be false. But that is the original statement, contrary to the assumption. Therefore “All B are non-C” must be true.

4. Particular negative. Its predicate-complemented form is a particular affirmative. Thus “Not every mineral is gold” becomes “Some minerals are non-gold.” If “Some B are not C” is true—the assumption—“Some B are non-C” is true—the required conclusion.

Proof: from the original, “Some B are not C,” contraposition changing quality gives “Some non-C are B.” Simple conversion then gives “Some B are non-C,” which is the required conclusion.

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1. **Source status:** Complete continuous authorial passage from Rafed’s digitally presented Dār al-Taʿāruf 1402 AH text, displayed page(s) 185–187; Arabic kaf/ya and paragraph/table whitespace normalized. Displayed pagination is recorded, not silently equated with a scanned print. Original symbolic readings remain in Arabic even where damaged; each conjectural English reconstruction is identified. The 1968 institutional scan has not been collated for this unit.
2. **Terminology:** Predicate-complementation with change of quality is standard English “obversion.” Subject-complementation and complementation of both terms are kept distinct. The Arabic digital list calls the second operation «نقيض المحمول» at one point; the surrounding section consistently calls it «نقض المحمول». No substantive Arabic emendation is adopted.
3. **Logical scope:** These are the author’s traditional rules, transmitted with his examples. The water example is a stipulated logical illustration, not a general scientific claim about water in every state. The author’s treatment of negative propositions and empty subjects requires care; these translations do not independently certify all steps for modern empty-term semantics.
4. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination, not independently image-collated · p. 185–187 https://lib.rafed.net/view.php?b_id=1604&page=185&type=c_fbook

M-C-42

READING SECTION 42

Transforming the original and handling complemented predicates

تنبيهان: طريقة تحويل الأصل وتحويل معدولة المحمول

First observation: the method of transforming the original. The procedure followed in proving predicate-complementation for the universal affirmative and the particular negative is a new method of proof. We should now call it “the method of transforming the original,” before the

discussion of syllogism arrives and assigns it to one of its kinds. It stands alongside the earlier method we called “proof by establishing the falsity of the contradictory.”

You have seen that in this method we successively applied transformations already known to us: first to the original, then to its transformed result, until we reached the required conclusion. In the universal affirmative, we transformed the original into its contrapositive changing quality, which is true if the original is true. We then subjected that contrapositive to simple conversion and obtained precisely the required predicate-complemented proposition. The second transformation is true if the contrapositive—the first transformation—is true; that first transformation is true if the original is true. The second is therefore true if the original is true. This is what we intended to prove, and we reached it by a very short route.

We shall use this easy method below for subject-complementation and complete complementation. It can also be employed in proving the contrapositives by using predicate-complementation. The student should exercise skill and attend to which transformations must be used to reach the desired conclusion.

Author's footnote. This is a syllogism of equality. The predicate-complemented proposition follows from the original's contrapositive, since it is the simple converse of that contrapositive; the contrapositive follows from the original; and what follows from a consequence follows from its ground.

Second observation: transforming a complemented predicate. In contraposition and predicate-complementation we have used two truth-preserving transformations. In fact, both belong to predicate-complementation. Because they are self-evident, however, we used them as grounds of proof before proving the rule for predicate-complementation, and consequently did not then give them that name. They are:

A. Transforming an affirmative with a complemented predicate into a negative with an uncomplemented predicate, keeping the same quantity. Their import is the same; the difference is that negation is predicated in the affirmative, whereas predication is negated in the negative.

B. Transforming a negative with a complemented predicate into an affirmative with an uncomplemented predicate, keeping the same quantity. Negation of negation is affirmation. This is clear and self-evident.

1. **Source status:** Complete continuous authorial passage from Rafed’s digitally presented Dār al-Taʿāruf 1402 AH text, displayed page(s) 187–188; Arabic kaf/ya and paragraph/table whitespace normalized. Displayed pagination is recorded, not silently equated with a scanned print. Original symbolic readings remain in Arabic even where damaged; each conjectural English reconstruction is identified. The 1968 institutional scan has not been collated for this unit.
2. **Authorial note retained:** The footnote explaining the syllogism of equality on displayed p187 has been retained and translated. It belongs to the original source presentation; the later b1728 edition adds further numbered notes that are not imported into this authorial text.

3. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination, not independently image-collated · p. 187–188
https://lib.rafed.net/view.php?b_id=1604&page=187&type=c_fbook

M-C-43

READING SECTION 43

Seven exercises on obversion and methods of proof

تمرينات على نقض المحمول

1. Prove predicate-complementation of the universal affirmative by the method of establishing the falsity of its contradictory.

2. Prove predicate-complementation of the particular negative by the method of establishing the falsity of its contradictory.

3. Prove predicate-complementation of the particular negative by transforming the original: first take its contrapositive retaining quality, then continue until you obtain its predicate-complemented form.

4. Try whether predicate-complementation of the particular affirmative can be proved by transforming the original.

5. Prove predicate-complementation of the universal negative by transforming the original. Observe what the result will be, and explain what you find.

6. Prove contraposition changing quality and retaining quality for each quantified form except the particular affirmative, using the method of transforming the original. For this purpose use only the rules of predicate-complementation and simple conversion.

7. Try to prove contraposition changing quality and retaining quality for the particular affirmative by this method. Notice that you will come to a halt and be unable to reach the result; explain the reasons for that halt.

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1. **Source status:** Complete continuous authorial passage from Rafed’s digitally presented Dār al-Taʿāruf 1402 AH text, displayed page(s) 189–189; Arabic kaf/ya and paragraph/table whitespace normalized. Displayed pagination is recorded, not silently equated with a scanned print. Original symbolic readings remain in Arabic even where damaged; each conjectural English reconstruction is identified. The 1968 institutional scan has not been collated for this unit.
2. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination, not independently image-collated · p. 189
https://lib.rafed.net/view.php?b_id=1604&page=189&type=c_fbook

READING SECTION 44

Complete complementation and subject-complementation of the universals

قاعدة النقض التام ونقض الموضوع في الكليتين

To obtain a true proposition **complemented in both terms**, replace the original subject by its complement and make it the subject, and replace the original predicate by its complement and make it the predicate, changing quantity but not quality.

To obtain a true **subject-complemented** proposition, replace the original subject by its complement and make it the subject, while leaving the predicate unchanged, changing both quantity and quality.

Only the two universal propositions admit these two forms of complementation. We must give the proof for each of the quantified forms.

1. Universal affirmative. Complete complementation gives a particular affirmative; subject-complementation gives a particular negative. Thus, from “All silver is mineral,” complete complementation gives “Some non-silver is non-mineral,” and subject-complementation gives “Some non-silver is not mineral.”

For the proof, assume that “All B are C” is true. The first claim is that “Some non-B are non-C” is true; the second, that “Some non-B are not C” is true.

Proof: “All B are C” gives “All non-C are non-B” by contraposition retaining quality. Simple conversion gives “Some non-B are non-C,” which is the first conclusion. Predicate-complementation of this last proposition gives “Some non-B are not C,” which is the second conclusion.

2. Universal negative. Complete complementation gives a particular negative; subject-complementation gives a particular affirmative. Thus, from “No iron is gold,” complete complementation gives “Some non-iron is not non-gold,” and subject-complementation gives “Some non-iron is gold.”

For the proof, assume that “No B is C” is true. The first claim is that “Some non-B are not non-C” is true. The second is intended to be “Some non-B are C.” **The digital statement of this second claim reads “Some non-B are non-C”; the wording of the example and the final line of the proof instead give the uncomplemented C.**

Proof: “No B is C” gives “No C is B” by simple conversion. Its contrapositive retaining quality is “Some non-B are not non-C,” which is the first conclusion. Predicate-complementation of this last proposition gives “Some non-B are C,” which is the second conclusion.

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1. **Source status:** Complete continuous authorial passage from Rafed’s digitally presented Dār al-Taʿāruf 1402 AH text, displayed page(s) 190–191; Arabic kaf/ya and paragraph/table whitespace normalized. Displayed pagination is recorded, not silently equated with a scanned print. Original symbolic readings remain in Arabic even where damaged; each conjectural English reconstruction is identified. The 1968 institutional scan has not been collated for this unit.
2. **Unresolved symbolic reading:** Displayed p191 states the second goal of the universal-negative case as «ع بَ حَـ», while its own concluding line reads «ع بَ حـ», matching the gold example. Arabic unchanged; English marks the mismatch. The b1728 digital text p224 repeats it.
3. **Existential conditions:** The universal-to-particular steps are the author’s traditional inferences. In modern semantics allowing empty terms, the universal-affirmative inversion requires a nonempty non-C class, and the universal-negative inversion requires a nonempty C class. These requirements should not be confused with a merely mechanical change of quantifier.
4. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination, not independently image-collated · p. 190–191 https://lib.rafed.net/view.php?b_id=1604&page=190&type=c_fbook

M-C-45

READING SECTION 45

Why neither particular admits complete or subject complementation

الجزئيتان لا نقض تام لهما ولا نقض موضوع

3 and 4. The two particulars. Neither has complete complementation or subject-complementation. To prove this, it is enough to show that neither operation yields a particular proposition; failure to yield a universal then follows all the more readily, as explained when showing that the particular affirmative has no contrapositive.

The particular affirmative. Assume “Some B are C” is true. The first claim is that “Some non-B are non-C” is not invariably true. The second is intended to be that “Some non-B are not C” is not invariably true. **In stating this second goal the digital text instead prints “Some non-B are not non-C.”**

Proof: in discussing contraposition of the particular affirmative, we explained that its complementary terms can, in some cases, be completely disjoint. “No non-B is non-C” is then true. Its contradictory, “Some non-B are non-C,” is false, which establishes the first conclusion.

The predicate-complemented form of that universal negative is also true: “All non-B are C.” Its contradictory is therefore false. **Here again the digital text prints “Some non-B are not non-C”; the contradictory of the immediately preceding universal is “Some non-B are not C.”** This establishes the intended second conclusion.

The particular negative. Assume “Some B are not C” is true. The first claim is that “Some non-B are not non-C” is not invariably true. The second is that “Some non-B are C” is not invariably true.

Proof: the subject of a particular negative may be strictly more extensive than its predicate, as in “Some animals are not human.”

First, the complement of the more extensive term is strictly less extensive than the complement of the less extensive term. Therefore “All non-B are non-C” is true. Its contradictory, “Some non-B are not non-C,” is false, which establishes the first conclusion.

Second, the complement of the more extensive term is completely disjoint from the less extensive term itself. Hence “No non-B is C” is true. Its contradictory, “Some non-B are C,” is false, which establishes the second conclusion.

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1. **Source status:** Complete continuous authorial passage from Rafed’s digitally presented Dār al-Taʿāruf 1402 AH text, displayed page(s) 191–192; Arabic kaf/ya and paragraph/table whitespace normalized. Displayed pagination is recorded, not silently equated with a scanned print. Original symbolic readings remain in Arabic even where damaged; each conjectural English reconstruction is identified. The 1968 institutional scan has not been collated for this unit.
2. **Unresolved symbolic reading:** Displayed pp191–192 twice gives «س بَ حَـ» for the second goal in the particular-affirmative case, whereas subject-complementation leaves C unchanged and the preceding «كل بَ حـ» has contradictory «س بَ حـ». Both displayed occurrences remain in Arabic. The b1728 pp224–225 digital presentation repeats the same problem.
3. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination, not independently image-collated · p. 191–192 https://lib.rafed.net/view.php?b_id=1604&page=191&type=c_fbook

M-C-46

READING SECTION 46

The author’s table of relations among quantified propositions

لوح نسب المحصورات

Table of relations among the quantified propositions. The source supplies eight headings: original proposition; contradictory; simple converse; contrapositive retaining quality; contrapositive changing quality; predicate-complemented form; form complemented in both terms; and subject-complemented form.

For compact display, A(B,C) means “All B are C”; I(B,C), “Some B are C”; E(B,C), “No B is C”; and O(B,C), “Some B are not C.” Here ¬B and ¬C name the complementary terms, not negations of whole propositions.

The following table **reconstructs the intended alignment from the preceding authorial proofs**. Two complement marks missing in the digital table's universal-affirmative column are restored only in this English reconstruction; the Arabic retains the digital readings. A dash records the absence of a generally valid transform in the author's account. Several universal-to-particular entries require distinct nonempty-class premises, specified in the accompanying note; they are not valid under unrestricted empty-term semantics.

Table: The author’s table of relations among quantified propositions

Relation	Universal affirmative A(B,C)	Particular affirmative I(B,C)	Universal negative E(B,C)	Particular negative O(B,C)
Contradictory	O(B,C)	E(B,C)	I(B,C)	A(B,C)
Simple converse	I(C,B)	I(C,B)	E(C,B)	—
Contrapositive retaining quality	A(¬C,¬B)	—	O(¬C,¬B)	O(¬C,¬B)
Contrapositive changing quality	E(¬C,B)	—	I(¬C,B)	I(¬C,B)
Predicate-complemented form	E(B,¬C)	O(B,¬C)	A(B,¬C)	I(B,¬C)
Both terms complemented	I(¬B,¬C)	—	O(¬B,¬C)	—
Subject-complemented form	O(¬B,C)	—	I(¬B,C)	—

Digital-source record. In the universal-affirmative column, the predicate-complemented entry is actually printed as «لا ب حـ»—E(B,C)—and the entry for both terms complemented as «ع بَ حـ»—I(¬B,C). Both conflict with the corresponding preceding prose proofs. The particular columns also omit blank placeholders in the HTML, so their surviving entries cannot simply be aligned by row position. This remains an editorial reconstruction. A targeted comparison with the independent 1968 Najaf edition (printed p.221, NYU canvas225) confirms the row alignment and the C-complement mark in the predicate-complemented entry, but its entry for both terms complemented also lacks the C-complement mark. That second restoration remains conjectural; the 1402 printed edition has not been image-collated.

1. **Source status:** Complete continuous authorial passage from Rafed’s digitally presented Dār al-Taʿāruf 1402 AH text, displayed page(s) 193; Arabic kaf/ya and paragraph/table whitespace normalized. Displayed pagination is recorded, not silently equated with a scanned print. Original symbolic readings remain in Arabic even where damaged; each conjectural English reconstruction is identified. A targeted comparison with the independent 1968 scan is recorded below; this is not a complete collation.
2. **Table transcription and reconstruction:** The complete displayed p.193 digital table is preserved in Arabic as its original ordered columns. English supplies an explicitly editorial aligned table derived from the preceding proofs. The independent 1968 print p.221 confirms the row alignment and the C-complement mark in the universal affirmative’s predicate-obversion cell. Its both-terms-complemented cell also omits the C-complement mark; that restoration remains conjectural. Rafed b1728 p.226 repeats the defective digital table. The 1402 printed edition itself has not been image-collated.
3. **Existential conditions:** The contradictory row gives the denial of the original; the other populated cells are presented as consequences of the original. In modern set semantics allowing empty terms, the simple converse of A(B,C) and both contrapositives of E(B,C) require B to be nonempty. The forms complementing both terms or the subject alone require different existence premises: from A(B,C), non-C must be nonempty; from E(B,C), C must be nonempty. These last premises do not follow merely from B being nonempty. The other populated consequences need no additional existence premise. The proofs on displayed pp190–191 give no separate statement that the required C or non-C class exists.
4. **Targeted independent print comparison:** Visual comparison with the independent Najaf: Maṭbaʿat al-Nuʿmān, 1968 edition, printed p.221 (Princeton/NYU canvas225), confirms the table alignment and the C-complement mark in the universal-affirmative predicate-obversion cell. Its both-terms-complemented cell still reads «ع بَ حـ», without the C-complement mark, so that English restoration remains conjectural. The 1402 printed edition itself has not been image-collated. https://sites.dlib.nyu.edu/viewer/api/image/books/princeton_aco000780/225/full/!1600,2200/0/default.jpg
5. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination, not independently image-collated · p. 193 https://lib.rafed.net/view.php?b_id=1604&page=193&type=c_fbook

READING SECTION 47

The logical axiom of immediate transformation

البديهة المنطقية أو الاستدلال المباشر البديهي

All the preceding laws of propositions—contradiction, conversion and complementation—are forms of immediate inference with respect to the proposition transformed from the original: its contradictory, converse or complemented form. In the first case, one infers the falsity of one proposition from the truth of the other, and conversely. In the remaining cases, one infers the truth of the converted or complemented proposition from the truth of the original, or the falsity of the original from the falsity of its converse or complemented form. **At the first case the digital Arabic says “in complementation”; the context, the truth-to-falsity description and the contrast with the remaining cases indicate contradiction.**

We called it immediate because the mind reaches the required truth or falsity of the proposition from just one known proposition, without an intervening proposition. The proof for every type of immediate inference has already been given. One further type remains; it is self-evident and needs no more than an explanation. It may be called “the logical axiom.”

It is self-evident in mathematics that adding the same thing to two equal things does not change their relation of equality. If B = C, and a definite number, such as 4, is added to each, then B + 4 = C + 4. Likewise, subtracting a definite number from each, multiplying them by it, or dividing them by it—as with 4—does not alter equality. Thus B − 4 = C − 4, B × 4 = C × 4, and B ÷ 4 = C ÷ 4. **For the division line, the Rafed digital source displays only “B = C”; the full division equation given here is visible in the independent 1968 Najaf edition, printed p.223.**

Likewise, if B is greater or less than C, the relation remains unchanged: B + 4 is greater or less than C + 4; B − 4 is greater or less than C − 4; and similarly for the other operations.

We say something analogous about a proposition. **If it is correct to add** a word to its subject and the same word to its predicate, the relation expressed by the proposition does not change: its quantity, quality and truth are preserved. Suppose “Every human is an animal” is true. Adding “head” to both terms gives the true statement “Every human's head is an animal's head.” Or, adding the verb “loves,” the intended example is “Everyone who loves a human loves an animal.” **The digital text instead has «يجب» in its introduction and subject phrase, but «يحب» in the predicate; “loves” in the first two places is supported by the independent 1968 Najaf edition, printed p.223, which reads «يحب» in all three positions.**

If “No animal is a stone” is true, then “No animal lying down is a stone lying down” is true. If “Some mineral is not gold” is true, then “Some piece of mineral is not a piece of gold” is true.

In this way, a true proposition can be transformed into another true proposition by adding a word that can correctly be added to both subject and predicate, without changing the proposition's quantity or quality. That word may form a possessive construction, express a circumstance, be an adjective, be a verb, or be another expression of this kind.

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1. **Source status:** Complete continuous authorial passage from Rafed’s digitally presented Dār al-Taʿāruf 1402 AH text, displayed page(s) 194–195; Arabic kaf/ya and paragraph/table whitespace normalized. Displayed pagination is recorded, not silently equated with a scanned print. Original symbolic readings remain in Arabic even where damaged; each conjectural English reconstruction is identified. A targeted comparison with the independent 1968 scan is recorded below; this is not a complete collation.
2. **Digital readings and targeted print evidence:** Displayed p.194 has «في النقض» where the author describes the truth/falsity reversal of contradiction; the 1968 print also retains this wording, so its contextual interpretation remains marked. Displayed p.195 omits both divisors in the division example and has «يجب / من يجب» against «يحب» in the predicate. The independent 1968 print p.223 supplies division by 4 on both sides and «يحب» in all three positions. The English adopts these independently supported readings while the Arabic preserves the digital-source record. Rafed b1728 repeats the digital problems and is not the source of the corrected readings.
3. **Logical scope of the final rule:** The conditional «لو صح» and concluding «كلمة تصح زيادتها» restrict permissible additions. The paragraph is not translated as permission to apply every verb, intensional context or quantifier-taking operator indiscriminately to both terms. The arithmetic example uses positive 4, so it does not address inequality reversal under multiplication by a negative number.
4. **Targeted independent print comparison:** The independent Najaf: Maṭbaʿat al-Nuʿmān, 1968 edition, printed p.223 (Princeton/NYU canvas227), visibly prints «٤ ÷ حـ = ٤ ÷ ب» and «يحب / من يحب / يحب». These support the full division equation and the matching loves example in English. The preceding heading page (canvas226, between printed221 and223) retains «في النقض», so its contextual contradiction reading is still qualified. This is targeted earlier-edition evidence, not a full collation or verification of the 1402 print.
   https://sites.dlib.nyu.edu/viewer/api/image/books/princeton_aco000780/227/full/!1600,2200/0/default.jpg
5. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination, not independently image-collated · p. 194–195
   https://lib.rafed.net/view.php?b_id=1604&page=194&type=c_fbook

BOOK 5

Book V · Argument and Inference

الباب الخامس — مباحث الاستدلال

M-V-01

READING SECTION 1

Book V: argument, its aim, and the three methods of inference

الباب الخامس: الحجة وهيئة تأليفها — تصدير وطرق الاستدلال

Book V: argument and the form of its composition, or the study of inference.

Preface. The logician's highest aim and ultimate purpose is the study of argument: the known assent that is used to reach knowledge of an unknown assent. All the preceding books are, in reality, preliminaries to this purpose, including the discussion of definition, since a defining account is examined in order to help us understand the individual terms of a proposition, its subject and predicate.

An **argument** (*ḥujja*) is, in their terminology, something composed of propositions directed toward a conclusion that is obtained through them. It is called a *ḥujja* because it is advanced against an opponent to establish the conclusion. It is also called an **indication** or **proof** (*dalīl*) because it points to the conclusion. Preparing and combining it for that purpose is called **inference** (*istidlāl*).

Before anything else, one point must be noted. Not every proposition must be sought by means of an argument. Otherwise we would never arrive at knowledge of any proposition. We must eventually reach self-evident propositions, whose nature is not to be sought in that way. They are the principles of what is sought, and the capital of the enterprise of knowledge.

The methods of inference, or divisions of argument. Who among us has not learned that there is fire upon seeing smoke? Who does not expect the sound of thunder on seeing lightning in the clouds? Who does not infer that sleep restores strength, that a stone becomes wet when placed in water, and that a knife cuts soft bodies? We may judge someone to be generous because that person resembles someone we know to be generous in some characteristics, or judge a pen to be good because it resembles a pen we have tried. Thousands of such inferences pass before us every day.

In fact, these obvious inferences, from which no sentient person is entirely without a share, all return to the recognized kinds of argument we are about to explain. Usually, however, the person making the inference does not notice which of those kinds has been employed, even if that person is a scholar of logic. You might be surprised to be told that ninety-nine per cent of people are natural logicians without knowing it.

Yet human beings make many mistakes in their judgments or are unable to obtain the conclusions they seek. They therefore cannot dispense with studying systematic methods of correct thinking and productive inference.

Apart from immediate inference, which has already been discussed, the systematic methods of inference fall into three principal kinds:

1. **Syllogism** (*qiyās*): the mind uses general rules accepted as correct to move to the conclusion sought. This is the principal method.
2. **Analogy** (*tamthīl*): the mind passes from a judgment about one of two things to a judgment about the other because of a feature shared between them.
3. **Induction** (*istiqrāʾ*): the mind examines a number of particulars and derives a general judgment from them.

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1. **Source status:** Complete continuous authorial passage from Rafed's digitally presented Dār al-Taʿāruf 1402 AH text, displayed page(s) 198–199; Arabic kaf/ya and paragraph/table whitespace normalized. Displayed pagination is recorded, not silently equated with a scanned print. Original symbolic readings remain in Arabic even where damaged; each conjectural English reconstruction is identified. The 1968 institutional scan has not been collated for this unit.
2. **Authorial heading and pagination:** The Book V title is displayed separately at Rafed p196, followed by blank p197. Its preface and three methods are pp198–199; the definition of syllogism begins p200. The English heading translates the p196 title, which is also prefixed to this unit's Arabic. Do not call p198 the end of Book IV.
3. **Digital spelling and rhetorical claim:** Displayed p198 has «السحاب» corrupted to «الحساب» in the lightning example and «يستحدم» for «يستخدم»; the English renders the evident contextual sense and this note records the digital anomalies. The "ninety-nine per cent" assertion is the author's rhetoric, not a verified statistic.
4. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination, not independently image-collated · p. 198–199 https://lib.rafed.net/view.php?b_id=1604&page=198&type=c_fbook

M-V-02

READING SECTION 2

Syllogism: definition and the five qualifications

القياس: تعريفه وشرح قيوده

1. Syllogism: its definition. They define syllogism as "a discourse composed of propositions which, whenever conceded, entails in itself another discourse."

Explanation.

1. **“Discourse”** is the genus. It means a complete declarative expression, and therefore includes both a single proposition and more than one.

2. **“Composed of propositions …,” and the remainder of the definition,** is the differentia. “Propositions” is a logical plural: it includes two. This qualification excludes immediate inference, since, as already explained, that consists of a single proposition which, on the assumption that it is conceded, entails another proposition.

3. **“Whenever conceded”** comes from concession. It indicates that a syllogism need not have premises that are actually conceded. What makes it a syllogism is that, on the assumption that its propositions are conceded, another discourse must follow. The relation is like that between a proposition and its converse or its obverse: on the assumption that the proposition is true, its corresponding conversions and obversions are true. What is entailed follows that which entails it in truth alone, not in falsity, as explained under simple conversion, because the entailed proposition may be more general. From this we know that the falsity of the constituent propositions does not require the falsity of the discourse entailed by them. The falsity of that discourse does, however, require that the premises cannot all be true.

4. **“Entails”** excludes induction and analogy. Although they are composed of propositions, the further discourse does not follow them necessarily: it can fail to follow. For the most part they yield only probability, apart from certain cases of induction that will be discussed later.

5. **“In itself”** excludes the syllogism of equality, which will be discussed in its proper place. In that syllogism, another discourse follows only through an external premise, and not from the syllogism itself. For example:

B equals C; C equals D; therefore B equals D.

This follows, however, because an external premise is true: what is equal to something equal to a thing is itself equal to that thing. Thus the corresponding conclusion does not follow from “B is half of C; C is half of D.” Half of a half is not a half, but a quarter.

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1. **Source status:** Full authorial passage from Rafed’s digital presentation of the Dār al-Taʿāruf 1402 AH text, displayed page(s) 200–201. Arabic kaf/ya and paragraph/table whitespace are normalized; substantive digital readings are retained. Page numbers refer to Rafed’s displayed pg containers, not an independently collated image of that edition. Any targeted comparison with the 1968 Najaf scan is identified separately.
2. **Logical force:** The statement about a false conclusion is rendered as failure of the premises jointly: at least one premise must be false. It does not mean that every premise is false. This explicates the author’s plural «كذبها» without introducing that invalid stronger claim.
3. **Digital readings:** The Arabic retains «بقشاياه», «كون لازما», and the change from حـ to جـ in the half-of example. The English renders the evident grammatical senses and uses B, C, D consistently for the schematic terms; no independent image check of this passage is claimed.
4. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination · p. 200–201 https://lib.rafed.net/view.php?b_id=1604&page=200&type=c_fbook

General terminology of syllogism. We must first explain the general terms, leaving the terms peculiar to each kind to the appropriate place. They are as follows.

1. **The form of a syllogism:** the arrangement in which its propositions are combined.

2. **A premise:** each proposition that enters into the form of the syllogism. The premises are also called the **matter of the syllogism**.

3. **The proposition sought:** the discourse entailed by the syllogism. It is called “sought” while the mind is beginning to put the premises together.

4. **The conclusion:** the very same proposition, so called after it has been obtained through the syllogism.

5. **The terms:** the intrinsic constituents of a premise. By intrinsic constituents we mean the constituents that remain after a proposition has been analyzed. If, for example, we break a categorical proposition down into its constituents, only the subject and predicate remain, without the relation. The relation subsists in the two extremes in order to connect them; separating each from the other therefore means that the relation between them disappears. The quantifier and the modality concern the relation, so they do not remain after it has disappeared. Likewise, when a conditional proposition is analyzed into its constituents, only the antecedent and consequent remain.

The subject and predicate, or the antecedent and consequent, are thus the intrinsic constituents of premises: their **terms**.

Let us clarify these expressions by an example:

(1) A wine-drinker is morally corrupt.

(2) Every morally corrupt person has their testimony rejected.

(3) Therefore a wine-drinker has their testimony rejected.

By relating “morally corrupt” to the wine-drinker in proposition (1), and relating rejection of testimony to every morally corrupt person in proposition (2), we infer the relation between rejection of testimony and the wine-drinker in proposition (3).

Table: General terminology of syllogism

Item in the example	Logical name
Each of propositions (1) and (2)	A premise

Item in the example	Logical name
Wine-drinker, morally corrupt, and having one’s testimony rejected	Terms
Proposition (3)	The proposition sought and the conclusion
The combination of the two premises	The form of the syllogism

Notice that we have used the three-dot sign before the conclusion. This is a geometrical sign indicating the passage to the proposition sought, and it is read “therefore.” We shall use it with letters below, for brevity and clarity.

1. **Source status:** Full authorial passage from Rafed’s digital presentation of the Dār al-Taʿāruf 1402 AH text, displayed page(s) 201–202. Arabic kaf/ya and paragraph/table whitespace are normalized; substantive digital readings are retained. Page numbers refer to Rafed’s displayed pg containers, not an independently collated image of that edition. Any targeted comparison with the 1968 Najaf scan is identified separately.
2. **Terminology and notation:** “Proposition sought” translates مطلوب and “conclusion” نتيجة: these are two stages of the same proposition’s role, not two different propositions. The author’s three-dot inference sign is represented by “therefore” or ∴ in the English. The legal example is retained as his illustrative premise set.
3. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination · p. 201–202 https://lib.rafed.net/view.php?b_id=1604&page=201&type=c_fbook

M-V-04

READING SECTION 3

Syllogisms divided by matter and form

أقسام القياس بحسب مادته وهيئته

The divisions of syllogism according to its matter and arrangement. We said that the premises are called the matter of a syllogism, and their arrangement its form. Syllogism is therefore investigated in two respects.

First, in respect of its matter: the differences between its premises are considered apart from their form. The premises may be certain, probable, conceded, commonly accepted, estimative, imaginative, or of other kinds that will be explained in their place. This investigation is called **the five arts**, to which the forthcoming Book VI is devoted. Considered in this respect, syllogism is divided into demonstration, dialectic, rhetoric, poetry, and sophistry.

Second, in respect of its form: differences of arrangement are considered apart from the character of the matter. The present book investigates syllogism in this respect. It divides into two kinds, **conjunctive** and **exceptional**, according to whether the conclusion or its contradictory is explicitly stated in the premises.

The first case—where the conclusion or its contradictory is explicitly stated in the premises—is called **exceptional**, because it contains an exceptive expression. For example:

(1) If Muḥammad is a scholar, he must be respected.

(2) But he is a scholar.

(3) Therefore Muḥammad must be respected.

Conclusion (3) is itself stated in premise (1).

(1) If so-and-so were just, he would not disobey God.

(2) But he has disobeyed God.

(3) Therefore so-and-so was not just.

The contradictory of conclusion (3) is explicitly stated in premise (1).

The second case—where neither the conclusion nor its contradictory is explicitly stated in the premises—is called **conjunctive**. An example is the one given at the beginning of the discussion. Its conclusion, “A wine-drinker has their testimony rejected,” is not explicitly stated, in that form, in either premise; nor is its contradictory. It is present only potentially, inasmuch as its intrinsic constituents occur in the premises: the two terms “wine-drinker” and “having one’s testimony rejected.” Each occurs in a separate premise.

A conjunctive syllogism may consist only of categorical propositions, in which case it is called **categorical**. It may consist only of conditional propositions, or of a conditional and a categorical proposition, in which case it is called **conditional**. For example:

(1) Whenever water is flowing, it is protected from ritual defilement.

(2) Whenever it is protected in that way, it does not become ritually impure through contact with an impurity.

(3) Therefore, whenever water is flowing, it does not become ritually impure through contact with an impurity.

Both premises are connected conditional propositions.

A second example:

(1) A noun is a word.

(2) A word is either indeclinable or inflected.

(3) Therefore a noun is either indeclinable or inflected.

Premise (1) is categorical, while premise (2) is a disjunctive conditional.

We shall investigate categorical conjunctive syllogisms first, then conditional conjunctive syllogisms, and then exceptional syllogisms.

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1. **Source status:** Full authorial passage from Rafed's digital presentation of the Dār al-Taʿāruf 1402 AH text, displayed page(s) 203–204. Arabic kaf/ya and paragraph/table whitespace are normalized; substantive digital readings are retained. Page numbers refer to Rafed's displayed pg containers, not an independently collated image of that edition. Any targeted comparison with the 1968 Najaf scan is identified separately.
2. **Targeted 1968 comparison: premise number:** Rafed displayed p204 says that the contradictory conclusion is explicit in premise (٢). In the Najaf 1968 scan, canvas237, visible printed p233, the corresponding line reads (١). The Arabic base retains (٢); the English follows the directly inspected earlier print's (1). This is a targeted comparison, not full collation of either edition.
https://sites.dlib.nyu.edu/viewer/api/image/books/princeton_aco000780/237/full/!1600,2200/0/default.jpg
3. **Targeted 1968 comparison: inflected noun:** Rafed's conclusion has «معرف» where its own premise has «معربة». The 1968 scan, canvas237/printed p233, reads «معرب». The Arabic base is retained; "inflected" in the English follows the inspected print.
https://sites.dlib.nyu.edu/viewer/api/image/books/princeton_aco000780/237/full/!1600,2200/0/default.jpg
4. **Conditional terminology:** "Exceptional" translates استثنائي, referring to the explicit exceptive premise, not to an exception to logical validity. "Conjunctive" translates اقتراني, the joining of premises; it is not restricted to modern propositional conjunction.
5. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination · p. 203–204 https://lib.rafed.net/view.php?b_id=1604&page=203&type=c_fbook

M-V-05

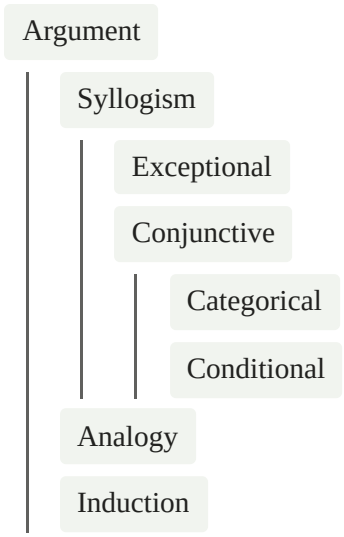
Summary of the division.

The diagram's labels and branches, transcribed from the inspected 1968 print, are:

Table: Summary diagram: argument and the kinds of syllogism

Parent	Its branches
Argument	Syllogism; analogy; induction
Syllogism	Exceptional; conjunctive
Conjunctive	Categorical; conditional

The order within each row follows the Arabic diagram from right to left.



Summary of the division — redrawn from the Najaf 1968 edition, p. 234; an explicit supplement to the missing digital figure.

1. **Source status:** Full authorial passage from Rafed's digital presentation of the Dār al-Taʿāruf 1402 AH text, displayed page(s) 205. Arabic kaf/ya and paragraph/table whitespace are normalized; substantive digital readings are retained. Page numbers refer to Rafed's displayed pg containers, not an independently collated image of that edition. Any targeted comparison with the 1968 Najaf scan is identified separately.
2. **Missing digital figure recovered from an earlier print:** Rafed displayed p205 preserves only «خلاصه التقسيم :» and no figure or inline image. The full classification tree is visible in the 1968 Najaf scan, canvas238, printed p234. The bracketed Arabic transcription and the English table give every label and branch in that inspected tree. They are an explicit supplement from this earlier witness, not a claim to have inspected the missing 1402 AH figure. https://sites.dlib.nyu.edu/viewer/api/image/books/princeton_aco000780/238/full/!1600,2200/0/default.jpg
3. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination · p. 205 https://lib.rafed.net/view.php?b_id=1604&page=205&type=c_fbook

M-V-06

READING SECTION 4

Categorical conjunctive syllogism: its three terms

الاقتراني الحملي: حدوده

Categorical conjunctive syllogism: its terms. A conjunctive syllogism must contain two premises so that they yield the proposition sought. Those premises must also contain three terms: a repeated term common to both, a term peculiar to the first, and a term peculiar to the second. The repeated common term connects the other two. It is omitted from the conclusion, which consists of those two terms, one as subject and the other as predicate. It is like a candle that consumes itself to give light to others.

Let us return to the example under general terminology, to identify its terms:

(a) “Morally corrupt” is the repeated common term that supplies the connection between

(b) “wine-drinker,” the term peculiar to the first premise, and

(c) “having one’s testimony rejected,” the term peculiar to the second premise.

The two premises yield “A wine-drinker has their testimony rejected,” with the common term omitted. Each of the three terms has been given a particular name.

(a) The middle term, or middle: the common term, so called because it mediates between its two companions in relating one to the other. It is also called the **argument**, because it is used to establish the relation between the two terms, and the **mediator in proof**, because it mediates in establishing the judgment between them. We symbolize it by M.

(b) The minor term: the term that is the subject of the conclusion. The premise containing it is called the **minor premise**, whether the term is its subject or its predicate. We symbolize it by B.

(c) The major term: the term that is the predicate of the conclusion. The premise containing it is called the **major premise**, whether the term is its predicate or its subject. We symbolize it by C. The minor and major terms together are called the **extremes**.

Thus, if we say:

All B are M;

all M are C;

we obtain: **therefore all B are C**, by omitting the repeated term M.

Author’s note. These terms apply to both kinds of conjunctive syllogism, categorical and conditional. The same applies to the general rules that follow.

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1. **Source status:** Full authorial passage from Rafed’s digital presentation of the Dār al-Taʿāruf 1402 AH text, displayed page(s) 206–207. Arabic kaf/ya and paragraph/table whitespace are normalized; substantive digital readings are retained. Page numbers refer to Rafed’s displayed pg containers, not an independently collated image of that edition. Any targeted comparison with the 1968 Najaf scan is identified separately.
2. **Notation and footnote placement:** M, B, C correspond respectively to the source’s م، ب، حـ. The authorial footnote on displayed p206 is translated at the end so that it does not interrupt the definition of the minor premise. In the Arabic transcription it remains in its source position. The malformed opening «لمقدمة» on p207 and the surplus «هنا» in its formula line are retained in Arabic.
3. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination · p. 206–207 https://lib.rafed.net/view.php?b_id=1604&page=206&type=c_fbook

M-V-07

The general rules of conjunctive syllogism. A conjunctive syllogism, whether categorical or conditional, must satisfy certain fundamental general rules if it is to yield a conclusion.

1. The middle term must recur. The very same term must be stated in the minor and major premises, without a difference. Otherwise it would not be a repeated middle term, and no connection between the extremes would be established. This is self-evident.

For example, suppose one says, “The wall has a mouse in it; every mouse has two ears.” This does not yield “The wall has two ears.” The term imagined to be the middle has not recurred: the predicate of the minor premise is “has a mouse in it,” whereas the subject of the major is only “mouse.” To make the inference productive, we could say in the major premise, “Everything that has a mouse in it has two ears,” but that premise would be false. Alternatively, we could treat only the word “mouse” as the repeated term, in which case the conclusion would be “The wall has in it something with two ears,” which is true.

A second example is “Gold is an ʿayn; every ʿayn sheds tears.” This does not yield “Gold sheds tears.” The word ʿayn is equivocal: its meaning in the minor premise differs from its meaning in the major. The middle term has therefore not recurred; only the word has recurred.

2. One premise must be affirmative. Two negative premises yield no conclusion. Their middle term does not enable us to establish a connection between the minor and major terms. A single thing can be disjoint from two things which are not disjoint from one another: horse, for example, is disjoint from human and rational. It can also be disjoint from two things which are themselves disjoint: horse is disjoint from human and bird, and human and bird are themselves disjoint.

Merely knowing that both terms are disjoint from the repeated term thus does not tell us whether they overlap outside the middle or are disjoint. Neither an affirmation nor a negation follows. If we say:

No human is a horse; no horse is rational;

we do not obtain the negative conclusion “No human is rational,” for the two extremes overlap. If we replace the second premise with “No horse is a bird,” we do not obtain the affirmative conclusion “Every human is a bird,” for the extremes are disjoint. This reasoning applies to every pair of negative premises.

3. One premise must be universal. Two particular premises yield no conclusion. Here too the middle does not enable us to establish a connection between the minor and major terms. A particular proposition indicates no more than that its two terms meet in some respect. With two particulars, we do not know whether the part of the middle that meets the minor is the very part that meets the major, or another part; either is possible. We therefore do not know whether the extremes overlap or are disjoint, and neither affirmation nor negation follows. For example:

First: “Some human is an animal; some animal is a horse.” This does not yield the affirmation “Some human is a horse.” If the second premise is replaced by “Some animal is rational,” it does not yield the negation “Some human is not rational.”

Second: “Some human is an animal; some animal is not rational.” This does not yield the negation “Some human is not rational.” If the second premise is replaced by “Some animal is not a horse,” it does not yield the affirmation “Some human is a horse.”

The same reasoning applies to every pair of particular premises, whatever the middle’s position: subject in both, predicate in both, or different in each.

4. The conclusion follows the weaker premise. If one premise is negative, the conclusion is negative, because negation is weaker than affirmation. If one is particular, the conclusion is particular, because a particular is weaker than a universal. This condition is clear: the conclusion derives from both premises together, and cannot exceed them so as to become stronger than they are.

5. A negative minor and a particular major yield no conclusion. We must suppose the minor to be universal, since otherwise the third condition would fail, and the major to be affirmative, since otherwise the second would fail.

When a syllogism is formed from a universal negative minor and a particular affirmative major, we do not know whether the minor and major terms overlap or are disjoint outside the middle. The universal negative indicates the disjointness of its terms—here, the minor and middle. The particular affirmative indicates that its terms—here, the middle and major—meet in some respect. Outside the middle, the major may therefore be disjoint from the minor, as the middle is, or it may overlap it. For example:

No crow is a human; some human is black.

This does not yield the negation “Some crow is not black.” If the second premise is replaced by “Some human is white,” it does not yield the affirmation “Some crow is white.”

In this example you may place the middle as subject in both premises, as predicate in both, or differently in each. The position makes no difference: you will find the same failure to yield a conclusion in every case.

1. **Source status:** Full authorial passage from Rafed’s digital presentation of the Dār al-Taʿāruf 1402 AH text, displayed page(s) 207–210. Arabic kaf/ya and paragraph/table whitespace are normalized; substantive digital readings are retained. Page numbers refer to Rafed’s displayed pg containers, not an independently collated image of that edition. Any targeted comparison with the 1968 Najaf scan is identified separately.
2. **Equivocal word:** In the gold example, عين can denote gold/cash and can also mean an eye. Retaining ʿayn makes the equivocation visible: “Gold is an eye” would suppress the word’s first sense and misrepresent the example.
3. **Targeted 1968 comparison: omitted conditional clause:** Rafed displayed p209 reads «واذا كانت النتيجة جزئية لان الجزئية أخس من الكلية», omitting the antecedent needed for rule4. The inspected 1968 scan, canvas242, visible printed p238, reads «واذا كانت جزئية كانت النتيجة جزئية». The Arabic base retains the omission; the English “If one is particular, the conclusion is particular” follows the complete earlier-print reading.
https://sites.dlib.nyu.edu/viewer/api/image/books/princeton_aco000780/242/full/!1600,2200/0/default.jpg

4. **Scope of the rules:** The author is stating restrictions on the standard categorical conclusion connecting the two extremes in his traditional syllogistic. His later moods that infer particular conclusions from universal premises retain traditional existential commitments; these are disclosed where the moods occur, not silently removed from his system.
5. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination · p. 207–210 https://lib.rafed.net/view.php?b_id=1604&page=207&type=c_fbook

M-V-08

READING SECTION 5

The four figures and the conditions of the first

الأشكال الأربعة والشكل الأول وشروطه

The four figures. We said that a conjunctive syllogism must have three terms: middle, minor, and major. We now add that the middle's position relative to the two extremes differs between the premises. In a categorical syllogism it may be subject in both, predicate in both, subject in the minor and predicate in the major, or the reverse. These are four arrangements, each called a **figure**. In conditional syllogisms, similarly, it occupies the positions of consequent and antecedent.

A figure, in their terminology, is therefore "a conjunctive syllogism considered in respect of how the middle is positioned relative to the two extremes." We shall discuss each of the four categorical figures, and then conditional conjunctive syllogism.

The first figure. Here the middle is predicate in the minor premise and subject in the major. The two extremes' positions relative to the middle in the premises are the same as their positions relative to one another in the conclusion. Just as the minor is subject in the conclusion, it is subject in the minor premise; just as the major is predicate in the conclusion, it is predicate in the major premise.

This explanation has a purpose. Because the minor occupies the same position in the conclusion and the minor premise, and the major the same position in the conclusion and the major premise, this figure accords with the mind's natural course. Its production of a conclusion is evident in itself and needs no further proof, unlike the other figures. They accordingly placed it first, and use it to prove the others.

Its conditions. This figure has two conditions.

1. The minor premise must be affirmative. If it were negative, we would not know whether the judgment made of the middle in the major also meets the minor outside the middle. Either is possible, so neither affirmation nor negation follows. For example:

No stone is a plant; every plant grows.

This does not yield the affirmation “Every stone grows.” If the minor premise is replaced by “No human is a plant,” it does not yield the negation “No human grows.” When the minor is affirmative, however, whatever applies to the middle in the major must apply to that to which the middle applies in the minor.

2. The major premise must be universal. If it were particular, the part of the middle to which the major is attributed might differ from the part attributed to the minor. The judgment would then not pass from the major to the minor through the middle. In fact, this condition returns to the first general rule: on that supposition the middle has not really recurred. For example:

All water is liquid; some liquid catches fire.

This does not yield “Some water catches fire.” The liquid predicated of water is specifically the portion that coincides with water; it differs from the portion that catches fire, such as oil. The middle has not recurred in meaning, although the word has recurred.

These are the conditions concerning quantity and quality. As for modality, it has been said that the minor must be an actual proposition. We have, however, undertaken not to investigate modal propositions. Their lengthy investigations would cause us to miss much that we ought to learn, while offering us little benefit.

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1. **Source status:** Full authorial passage from Rafed’s digital presentation of the Dār al-Taʿāruf 1402 AH text, displayed page(s) 211–212. Arabic kaf/ya and paragraph/table whitespace are normalized; substantive digital readings are retained. Page numbers refer to Rafed’s displayed pg containers, not an independently collated image of that edition. Any targeted comparison with the 1968 Najaf scan is identified separately.
2. **Unresolved printed wording and conjectural English sense:** Rafed p211 has «بين وضع» in the first-figure comparison. The inspected 1968 canvas244/printed p240 also reads «بين», as independently confirmed by a second visual reader; it does not attest «عين» here. The English “the same as” is an explicit contextual reconstruction supported by the following explanation’s two statements «عين وضعه», not a scan-backed correction. Arabic remains unchanged. The scan contains the complete opening without a missing diagram. The obvious digital spellings «لايجتاج» and «جعدوه» are also retained in Arabic. https://sites.dlib.nyu.edu/viewer/api/image/books/princeton_aco000780/244/full/!1600,2200/0/default.jpg
3. **Modal scope:** “Actual proposition” renders فعلية. The author explicitly sets aside the detailed modal syllogistic; no modal calculus has been supplied in its place.
4. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination · p. 211–212 https://lib.rafed.net/view.php?b_id=1604&page=211&type=c_fbook

M-V-09

Its moods. Each premise of a syllogism, taken by itself, may be any one of the four quantified propositions. Pairing the four possibilities for the minor with the four for the major gives sixteen combinations: four multiplied by four. This applies to all four figures.

A combination of the two premises is called a **mood**, a **conjunction**, or a **pairing**. Some of these sixteen combinations yield a conclusion and are called syllogisms; others do not and are called unproductive. Under the first figure’s two conditions of quantity and quality, only four moods are

productive. All the others are unproductive. The first condition eliminates eight: the two negative minors multiplied by the four majors. The second eliminates another four: the two particular majors multiplied by the two affirmative minors. Only four remain.

All four are evidently productive. Each yields one of the four quantified propositions, so all four kinds of quantified proposition are obtainable from this figure’s moods. For this reason the figure is called **complete** and **excellent**. Its moods are arranged in the order of their conclusions: universal affirmative first, then universal negative, particular affirmative, and particular negative.

First: two universal affirmatives yield a universal affirmative.

All B are M; all M are C; therefore all B are C.

Example: Every wine is intoxicating; every intoxicant is prohibited; therefore every wine is prohibited.

Second: a universal affirmative and a universal negative yield a universal negative.

All B are M; no M is C; therefore no B is C.

Example: Every wine is intoxicating; no intoxicant is beneficial; therefore no wine is beneficial.

Third: a particular affirmative and a universal affirmative yield a particular affirmative.

Some B is M; all M are C; therefore some B is C.

Example: Some people asking for alms are poor; every poor person is entitled to alms; therefore some people asking for alms are entitled to alms.

Fourth: a particular affirmative and a universal negative yield a particular negative.

Some B is M; no M is C; therefore some B is not C.

Example: Some people asking for alms are wealthy; no wealthy person is entitled to alms; therefore some people asking for alms are not entitled to alms.

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1. **Source status:** Full authorial passage from Rafed’s digital presentation of the Dār al-Taʿāruf 1402 AH text, displayed page(s) 213–214. Arabic kaf/ya and paragraph/table whitespace are normalized; substantive digital readings are retained. Page numbers refer to Rafed’s displayed pg containers, not an independently collated image of that edition. Any targeted comparison with the 1968 Najaf scan is identified separately.
2. **Order of premises:** Throughout these mood descriptions the author gives the minor premise first and the major second. The examples retain his selected substantive premises; logical productivity concerns whether the stated conclusion follows from them.
3. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination · p. 213–214 https://lib.rafed.net/view.php?b_id=1604&page=213&type=c_fbook

The second figure and its conditions

الشكل الثاني وشروطه

The second figure. The middle is predicate in both premises. The minor term is subject in both the minor premise and the conclusion, but the major changes position: it is subject in the major premise and predicate in the conclusion. For this reason this figure departs from the mind’s natural course. Its production of a conclusion is not evident, and its syllogistic force requires proof. Yet because the minor has the same position in the conclusion and the minor premise, as subject in both, this figure is closer to the natural course than the remaining figures: the subject is closer to the mind.

Its conditions. The second figure likewise has two conditions: the premises must differ in quality, and the major must be universal.

First, difference in quality. If one premise is affirmative, the other must be negative. This figure does not yield a conclusion when their qualities agree. The minor and major terms may be disjoint while sharing a predicate, or while sharing the exclusion of another predicate. They may also overlap while sharing a predicate or sharing its exclusion. Neither affirmation nor negation therefore follows.

For example, human and horse are disjoint, yet animal is predicated of both and stone is denied of both. We may say:

(a) Every human is an animal; every horse is an animal.

(b) No human is a stone; no horse is a stone.

In these examples the true relation between the extremes is negative. Human and rational likewise share the predication of animal and the denial of stone. Replace horse by rational in both examples, and the true relation between the extremes is affirmative.

When, however, the judgments in the minor and major differ in such a way that they cannot both apply to one thing, the subject of one must differ from the subject of the other. The minor and major terms are therefore disjoint, and their relation is negative. Thus the second figure’s conclusion is always negative, following the weaker premise.

Second, universality of the major. If the major were particular while the premises differed in quality, we would not know whether the minor and major terms overlap or exclude one another. A particular major and a universal minor that differ in quality indicate only the incompatibility of the minor with the part of the major mentioned in that premise. They indicate neither its incompatibility with the other, unmentioned part of the major nor its overlap with it. The cases can therefore differ.

For example:

Every ruminant has a cloven hoof; some animal does not have a cloven hoof.

This does not yield the negation “Some ruminant is not an animal.” If the major term is replaced by bird, it does not yield the affirmation “Some ruminant is a bird.”

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1. **Source status:** Full authorial passage from Rafed’s digital presentation of the Dār al-Taʿāruf 1402 AH text, displayed page(s) 214–216. Arabic kaf/ya and paragraph/table whitespace are normalized; substantive digital readings are retained. Page numbers refer to Rafed’s displayed pg containers, not an independently collated image of that edition. Any targeted comparison with the 1968 Najaf scan is identified separately.
2. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination · p. 214–216 https://lib.rafed.net/view.php?b_id=1604&page=214&type=c_fbook

M-V-11

Its moods. Under the two stated conditions, only four moods of this figure are productive. The first condition eliminates eight: pairing the two negative minors with the two negative majors eliminates four, and pairing the two affirmatives with the two affirmatives eliminates another four. The second condition eliminates four more: the two negative minors with a particular affirmative major, and the two affirmative minors with a particular negative major.

Four productive moods remain. All are proved through the first figure, as follows.

First mood: a universal affirmative and a universal negative yield a universal negative.

Example: Every ruminant has a cloven hoof; no bird has a cloven hoof; therefore no ruminant is a bird.

The proof simply converts the major and joins that converse to the original minor. The result is the first figure’s second mood, yielding the very conclusion sought. In symbols:

Assumed: all B are M; no C is M.

Claim: no B is C.

Proof: simply convert the major to “No M is C,” and join it to the minor: all B are M; no M is C. This is the second mood of the first figure, yielding “No B is C,” as required.

Second: a universal negative and a universal affirmative yield a universal negative.

Example: Nothing contingent is permanent; everything real is permanent; therefore nothing contingent is real.

The proof converts the minor, makes it the major of a new syllogism with the original major as minor, and then converts the conclusion.

Assumed: no B is M; all C are M.

Claim: no B is C.

Proof: if “No B is M” is true, its simple converse “No M is B” is true. Take that converse as major and the original major as minor: all C are M; no M is B. By the second mood of the first figure, no C is B. This simply converts to “No B is C,” as required.

Third: a particular affirmative and a universal negative yield a particular negative.

Example: Some metal is gold; no silver is gold; therefore some metal is not silver.

It is proved by the same method as the first mood.

Assumed: some B is M; no C is M.

Claim: some B is not C.

Proof: if the major “No C is M” is true, its simple converse “No M is C” is true. Joining it to the minor gives: some B is M; no M is C. By the fourth mood of the first figure, some B is not C, as required.

Fourth: a particular negative and a universal affirmative yield a particular negative.

Example: Some body is not a metal; every gold is a metal; therefore some body is not gold.

This cannot be proved by the method of conversion used in the first three moods. The minor is a particular negative and does not convert; the major converts to a particular, which cannot form a syllogism with the minor, since two particular premises yield no syllogism. We therefore turn to another method, called **reductio**.

Assumed: some B is not M; all C are M.

Claim: some B is not C.

Proof: if the conclusion “Some B is not C” were not true, its contradictory “All B are C” would be true. Make this contradictory the minor of a syllogism whose major is the original major. The first mood of the first figure results:

All B are C; all C are M; therefore all B are M.

Its contradictory, “Some B is not M,” would consequently be false. But that is precisely the minor premise assumed true. This contradicts the assumption. Hence “Some B is not C” must be true, as required.

Author’s note. The remarks following the third figure will explain that the method of conversion is called **reduction**, because conversion reduces the syllogism to the self-evident first figure in order to yield the proposition sought.

Exercise. Prove each of the first three moods by the reductio method used to prove the fourth.

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1. **Source status:** Full authorial passage from Rafed’s digital presentation of the Dār al-Taʿāruf 1402 AH text, displayed page(s) 216–219. Arabic kaf/ya and paragraph/table whitespace are normalized; substantive digital readings are retained. Page numbers refer to Rafed’s displayed pg containers, not an independently collated image of that edition. Any targeted comparison with the 1968 Najaf scan is identified separately.
2. **Targeted 1968 comparison: conclusion formula:** At displayed p218 the isolated line labeled «المدعي» reads «س ب م» (some B is not M), repeating the minor premise. The inspected 1968 scan, canvas252, visible printed p248, reads «س ب حـ» (some B is not C), also required by the subsequent reductio. The English follows that directly attested conclusion; Arabic retains the defective digital line. https://sites.dlib.nyu.edu/viewer/api/image/books/princeton_aco000780/252/full/!1600,2200/0/default.jpg
3. **Example and proof method:** حق in the second mood’s example is rendered “real”; the example’s metaphysical premise is preserved as the author supplies it. “Reductio” translates خلف, an indirect proof from the contradictory of the desired conclusion; it is distinct from immediate inference (الاستدلال المباشر).
4. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination · p. 216–219 https://lib.rafed.net/view.php?b_id=1604&page=216&type=c_fbook

M-V-12

READING SECTION 7

The third figure and its conditions

الشكل الثالث وشروطه

The third figure. The middle is subject in both premises. The major is predicate in both the major premise and the conclusion, but the minor changes position: predicate in the minor premise, subject in the conclusion. This figure therefore departs from the mind’s natural course, and does so more than the second figure. Here the change affects the conclusion’s subject, which is closer to the mind; in the second figure it affected the predicate. Yet because the major occupies the same position in the major premise and the conclusion, as in the first figure, the third is closer to the natural course than the fourth.

Its conditions. This figure also has two conditions: the minor must be affirmative, and one premise must be universal.

As for the first, if the minor were negative, we would not know whether the major, affirmed or denied of the middle, meets the minor outside the middle or is separate from it.

If the major were affirmative, the middle would be disjoint from the minor and overlap the major. A single thing can overlap one and be disjoint from another of two things that themselves overlap, or of two things that are disjoint. Rational, for example, overlaps animal and is disjoint from horse, while animal and horse overlap. It also overlaps animal and is disjoint from tree, while animal and tree are disjoint.

If the major were also negative, the middle would be disjoint from both extremes. A single thing may be disjoint from two overlapping things, or from two disjoint things: gold is disjoint from horse and animal, which overlap, and from tree and animal, which are disjoint. Thus:

(a) Nothing rational is a horse; everything rational is an animal.

This does not yield a negative conclusion. If tree is substituted for horse, it does not yield an affirmative conclusion.

(b) No gold is a horse; no gold is an animal.

As for the second condition, that one premise must be universal, the third general rule has already established that two particular premises yield no conclusion. There is nothing here requiring either one of the premises specifically to be the universal one.

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1. **Source status:** Full authorial passage from Rafed's digital presentation of the Dār al-Taʿāruf 1402 AH text, displayed page(s) 219–220. Arabic kaf/ya and paragraph/table whitespace are normalized; substantive digital readings are retained. Page numbers refer to Rafed's displayed pg containers, not an independently collated image of that edition. Any targeted comparison with the 1968 Najaf scan is identified separately.
2. **Digital repetition:** The Arabic at displayed p219 repeats «الحيوان» in «ويلاقي الحيوان الحيوان». The English renders the single intended occurrence. The final pair (b) is retained without inventing an additional conclusion that the source does not state.
3. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination · p. 219–220 https://lib.rafed.net/view.php?b_id=1604&page=219&type=c_fbook

M-V-13

Its moods. Under these two conditions, only six moods of this figure are productive. As in the first figure, the first condition eliminates eight moods. The second eliminates two: two particular affirmatives, and a particular affirmative together with a particular negative. Six remain, each requiring proof. All their conclusions are particular.

First mood: two universal affirmatives yield a particular affirmative.

Example: Every gold is a metal; every gold is expensive; therefore some metal is expensive.

It is proved by converting the minor and joining it to the original major. The third mood of the first figure then yields the desired conclusion.

Assumed: all M are B; all M are C.

Claim: some B is C.

Proof: if “All M are B” is true, its simple converse “Some B is M” is true. Join this converse to the original major: some B is M; all M are C. By the first figure’s third mood, some B is C, as required.

The mood does not yield a universal conclusion, because B may be more general than C, even by partial overlap, as in the example.

Second: two universals, with the major negative, yield a particular negative.

Example: Every gold is a metal; no gold is silver; therefore some metal is not silver.

It is proved by converting the minor, as in the first mood.

Assumed: all M are B; no M is C.

Claim: some B is not C.

Proof: convert the minor to “Some B is M” and join it to the major: some B is M; no M is C. By the first figure’s fourth mood, some B is not C, as required.

Third: two affirmatives, with the minor particular, yield a particular affirmative.

Example: Some bird is white; every bird is an animal; therefore some white thing is an animal.

Again the proof converts the minor.

Assumed: some M is B; all M are C.

Claim: some B is C.

Proof: convert the minor to “Some B is M” and join it to the major: some B is M; all M are C. By the first figure’s third mood, some B is C, as required.

Fourth: two affirmatives, with the major particular, yield a particular affirmative.

Example: Every bird is an animal; some bird is white; therefore some animal is white.

The proof converts the major, takes it as a new minor with the original minor as major, and then converts the conclusion.

Assumed: all M are B; some M is C.

Claim: some B is C.

Proof: convert the major to “Some C is M” and make it minor to the original minor: some C is M; all M are B. The first figure’s third mood gives “Some C is B.” Its simple converse is “Some B is C,” as required.

Fifth: a universal affirmative and a particular negative yield a particular negative.

Example: Every animal is sentient; some animal is not a human; therefore some sentient thing is not a human.

Conversion cannot prove this mood. The particular negative does not convert, while the universal affirmative converts to a particular, and two particulars do not form a syllogism. It is therefore proved by reductio.

Assumed: all M are B; some M is not C.

Claim: some B is not C.

Proof: if “Some B is not C” were not true, its contradictory “All B are C” would be true. Make that the major of a syllogism with the original minor: all M are B; all B are C. By the first mood of the first figure, all M are C. Its contradictory, “Some M is not C,” would be false. But this is precisely the major premise assumed true. That is a contradiction; hence “Some B is not C” must be true, as required.

Sixth: a particular affirmative and a universal negative yield a particular negative.

Example: Some gold is a metal; no gold is iron; therefore some metal is not iron.

The proof converts the minor and joins it to the original major. The fourth mood of the first figure then yields the required conclusion.

Assumed: some M is B; no M is C.

Claim: some B is not C.

Proof: convert the minor to “Some B is M” and join it to the major: some B is M; no M is C. By the first figure’s fourth mood, some B is not C, as required.

1. **Source status:** Full authorial passage from Rafed’s digital presentation of the Dār al-Taʿāruf 1402 AH text, displayed page(s) 220–223. Arabic kaf/ya and paragraph/table whitespace are normalized; substantive digital readings are retained. Page numbers refer to Rafed’s displayed pg containers, not an independently collated image of that edition. Any targeted comparison with the 1968 Najaf scan is identified separately.
2. **Existential condition:** The first and second moods infer a particular from universal premises by converting “All M are B” to “Some B is M”. This uses the traditional existential commitment for M. With modern empty-class semantics, add that at least one M exists. The English preserves the author’s inference and examples rather than silently presenting these two moods as valid without that condition.
3. **Proof display:** The digital table on p223 separates «فيكذب نقيضها» from the line «س م حـ» that it governs. The English restores sentence order while preserving every logical step. The Arabic retains the source row order and its spelling «صمه» for the act of joining.
4. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination · p. 220–223 https://lib.rafed.net/view.php?b_id=1604&page=220&type=c_fbook

M-V-14

Remarks: the reductio method.

1. Each mood of the third figure, like each mood of the second, can be proved by reductio.

Reductio is an indirect inference that proves the falsity of the contradictory of the proposition sought, and thereby establishes the truth of that proposition. In its application to the figures, one takes the contradictory of the conclusion to be established and says: if that conclusion were not true, its contradictory would be true. Supposing the contradictory true, one joins it to one of the premises already assumed true, so that they form a mood of the first figure. They then yield a proposition contradicting the other premise, which is true by assumption. This is a contradiction, so that new conclusion must be false. Its falsity must arise from the falsity of the contradictory of the proposition sought. The truth of the proposition sought is thus established. Examples have already been given.

The student should apply this method to every mood to sharpen the mind, observing which premise of the original syllogism must be chosen so that it and the contradictory form a productive mood.

1. **Source status:** Full authorial passage from Rafed’s digital presentation of the Dār al-Taʿāruf 1402 AH text, displayed page(s) 224. Arabic kaf/ya and paragraph/table whitespace are normalized; substantive digital readings are retained. Page numbers refer to Rafed’s displayed pg containers, not an independently collated image of that edition. Any targeted comparison with the 1968 Najaf scan is identified separately.
2. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination · p. 224 https://lib.rafed.net/view.php?b_id=1604&page=224&type=c_fbook

M-V-15

2. Proof by assumption. Some moods of the third or second figure that have a particular premise can also be proved by **assumption**. There is no harm in explaining it to illuminate students’ thinking, although conversion and reductio make it dispensable. It has three stages.

First, the assumption. We suppose a special name for the part that is the subject of the judgment in the particular proposition: let it be D. For in saying, for example, “Some animal is not a human,” the “some” must refer to something determinate of which human can correctly be denied —a horse, ape, bird, or the like. We stipulate that this determinate thing is called D. In a proposition such as “Some M is B,” D is therefore another expression for the relevant “some M.”

Second, extracting two propositions that are true after the assumption. After making this assumption, we can obtain two certainly true propositions:

(1) A universal affirmative whose subject is the assumed name D and whose predicate is the subject of the particular proposition. In the example, “All D are M” is true, because D is some M by assumption, and the more general is certainly predicated of every instance of the more specific.

(2) A universal proposition, affirmative or negative according to the particular’s quality, whose subject is D and whose predicate is the particular’s predicate. In the example, “All D are B” is true, because D is the part that is entirely B. If the particular is negative, such as “Some M is not B,” then “No D is B” is true, because D is the part of which B is denied.

Third, combinations yielding the desired conclusion. Extracting these two propositions increases our stock of knowledge. We use them, together with the two original premises assumed true, to form useful combinations and derive the conclusion whose truth we wish to establish.

Now that we understand these stages, let us try the method on the fifth mood of the third figure.

Assumed: all M are B; some M is not C.

Claim: some B is not C.

Proof by assumption: in the particular negative “Some M is not C,” suppose the part of M which is not C to be D. We extract the two true propositions:

(1) All D are M.

(2) No D is C.

Take (1) as minor to the original minor: all D are M; all M are B. By the first mood of the first figure, all D are B. Make this conclusion minor to (2): all D are B; no D is C. By the second mood of the third figure, some B is not C, as required.

Let us try the method again on the fourth mood of the second figure.

Assumed: some B is not M; all C are M.

Claim: some B is not C.

Proof by assumption: in the particular negative “Some B is not M,” suppose the part of B which is not M to be D. We extract the two true propositions:

(1) All D are B.

(2) No D is M.

Take (2) as minor to the original major: no D is M; all C are M. By the second mood of the second figure, no D is C. Next convert (1) to “Some B is D,” and take this converse as minor to that last conclusion: some B is D; no D is C. By the fourth mood of the first figure, some B is not C, as required.

You see that we used the simple converse of (1) along the way. Without it, we could have formed a syllogism only in the third figure, which comes after the second. We similarly use this converse when proving the second figure’s third mood by assumption. For further practice, the student should use the method on the other moods, not treated here, that have a particular premise.

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1. **Source status:** Full authorial passage from Rafed’s digital presentation of the Dār al-Taʿāruf 1402 AH text, displayed page(s) 224–226. Arabic kaf/ya and paragraph/table whitespace are normalized; substantive digital readings are retained. Page numbers refer to Rafed’s displayed pg containers, not an independently collated image of that edition. Any targeted comparison with the 1968 Najaf scan is identified separately.
2. **Existence in the assumption method:** D is stipulated from the witness already asserted by a particular premise. Consequently the later conversion of “All D are B” to “Some B is D” has an explicit nonempty basis; the method is not introducing an arbitrary empty class.
3. **Digital duplication:** At the start of displayed p225, «فانه بعد الفرض» is duplicated. The Arabic retains the duplication; the English renders the sentence once. The spelling defects «لاستخراج», «تأليبف», and «الصادقيتن» are also retained in Arabic and rendered by their evident senses.
4. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination · p. 224–226 https://lib.rafed.net/view.php?b_id=1604&page=224&type=c_fbook

M-V-16

3. Reduction. Another method for proving the productivity of the three figures other than the first is **reduction**: transforming the figure into the first figure. This may be done by interchanging the two premises in the fourth figure, or by replacing one premise with its simple converse. In the second figure one converts the major in some moods where it is convertible; in the third one converts the minor in some such moods, as explained above.

In some moods we may need to use predicate-obversion or contraposition when simple conversion is unavailable, in order to reach a first-figure syllogism yielding the very conclusion sought.

For practice, the student should carefully apply this procedure to all the moods of the two figures.

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1. **Source status:** Full authorial passage from Rafed’s digital presentation of the Dār al-Taʿāruf 1402 AH text, displayed page(s) 227. Arabic kaf/ya and paragraph/table whitespace are normalized; substantive digital readings are retained. Page numbers refer to Rafed’s displayed pg containers, not an independently collated image of that edition. Any targeted comparison with the 1968 Najaf scan is identified separately.

2. **Terminology:** رد is “reduction”; نقض المحمول is predicate-obversion; عكس النقيض is contraposition. These names follow the distinctions established in Book IV and are not interchangeable.

3. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination · p. 227 https://lib.rafed.net/view.php?b_id=1604&page=227&type=c_fbook

M-V-17

READING SECTION 8

The fourth figure: conditions, moods, and proofs

الشكل الرابع: شروطه وضروبه

The fourth figure. The middle is subject in the minor and predicate in the major, the reverse of the first figure. Both the minor and major terms therefore occupy positions in the conclusion different from their positions in the premises. This figure departs furthest from the mind’s natural course, and its productivity is obscure to the mind. A number of logicians accordingly omit it from their works and content themselves with the first three figures.

Its conditions. This figure must satisfy the three general conditions previously stated for every figure: it must not consist of two negatives, of two particulars, or of a negative minor and particular major. It also has two special conditions:

1. Neither premise may be a particular negative.

2. When both premises are affirmative, the minor must be universal. Thus, if the minor is a particular affirmative, the major cannot be affirmative; it must be a universal negative.

Its moods. Under these five conditions, only five moods are productive. The first eliminates four: the two negative minors paired with the two negative majors. The second eliminates three pairs of particulars, whether both affirmative or different in quality. The third eliminates one: a universal negative minor with a particular affirmative major. The fourth eliminates two: a particular negative, as minor or major, paired with a universal affirmative. The fifth eliminates one: a particular affirmative minor with a universal affirmative major.

Five moods remain, which we now prove.

First mood: two universal affirmatives yield a particular affirmative.

Example: Every human is an animal; everything rational is a human; therefore some animal is rational.

It is proved by reduction: interchange the premises to obtain the first figure, then convert its conclusion to obtain the proposition sought.

Assumed: all M are B; all C are M.

Claim: some B is C.

Proof: interchanging the premises gives all C are M; all M are B. By the first mood of the first figure, all C are B. Its converse is “Some B is C,” as required.

This mood does not yield a universal conclusion, because the minor term may be more general than the major, as in the example.

Second: a universal affirmative and a particular affirmative yield a particular affirmative.

Example: Every human is an animal; some bearer of live young is a human; therefore some animal is a bearer of live young.

It too is proved by reduction, interchanging the premises and then converting the conclusion. It does not yield a universal conclusion, because the minor term may be more general.

Third: a universal negative and a universal affirmative yield a universal negative.

Example: Nothing contingent is permanent; every substrate of originated events is contingent; therefore nothing permanent is a substrate of originated events.

This too is proved by reduction, interchanging the premises and then converting the conclusion.

Fourth: a universal affirmative and a universal negative yield a particular negative.

Example: Every liquid evaporates; no iron is a liquid; therefore some evaporating thing is not iron.

This cannot be proved by reduction through interchanging the premises, because the first figure does not yield a conclusion from a negative minor. It can, however, be proved by converting both premises and combining the converses into a first-figure syllogism.

Assumed: all M are B; no C is M.

Claim: some B is not C.

Proof: convert the premises to “Some B is M” and “No M is C.” By the fourth mood of the first figure, some B is not C, as required.

Fifth: a particular affirmative and a universal negative yield a particular negative.

Example: Some liquid evaporates; no iron is a liquid; therefore some evaporating thing is not iron.

For the same reason, this cannot be proved by interchanging the premises. It can be proved by converting both premises, exactly as in the preceding mood.

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1. **Source status:** Full authorial passage from Rafed's digital presentation of the Dār al-Taʿāruf 1402 AH text, displayed page(s) 227–230. Arabic kaf/ya and paragraph/table whitespace are normalized; substantive digital readings are retained. Page numbers refer to Rafed's displayed pg containers, not an independently collated image of that edition. Any targeted comparison with the 1968 Najaf scan is identified separately.
2. **Existential conditions:** The first mood requires an existing C when "All C are B" is converted to "Some B is C". The fourth requires an existing M when "All M are B" is converted. These are the author's traditional existential commitments; with modern empty-class semantics they must be stated as additional conditions. The other particular conclusions here have a particular premise supplying a witness.
3. **Targeted 1968 comparison: missing proof premise:** Rafed displayed p228's proof table retains «كل م ب» but drops the other premise needed after interchange. The inspected 1968 scan, canvas264, visible printed p260, displays both «كل حـ م» and «كل م ب». The English follows that complete first-figure pair, "All C are M; all M are B"; Arabic retains the incomplete digital row.
   https://sites.dlib.nyu.edu/viewer/api/image/books/princeton_aco000780/264/full/!1600,2200/0/default.jpg
4. **Example terminology:** محل للحوادث is "substrate of originated events", retaining the classical metaphysical example. ولود is rendered "bearer of live young" in its biological context.
5. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination · p. 227–230 https://lib.rafed.net/view.php?b_id=1604&page=227&type=c_fbook

M-V-18

Exercises.

1. Prove the second and then the fifth mood of the fourth figure by the method of assumption.

2. Prove the third and then the fourth mood of the fourth figure by reductio.

3. Prove the fourth mood of the second figure by reduction: take the predicate-obverse of each premise, then the simple converse of the obverted major, to obtain the proposition sought.

4. Prove the fifth mood of the third figure by reduction: take the predicate-obverse of each premise, then the simple converse of the obverted major, to form a first-figure syllogism. Then take the same-quality contrapositive of its conclusion to obtain the proposition sought.

5. Prove the first and then the second mood of the second figure by reduction, taking the predicate-obverse of each premise. Complete the rest of the proof yourself. In both moods you will need simple conversion to reach the desired conclusion; success depends on your attention and skill in choosing where to use it.

6. Try to prove the third mood of the second figure by reduction, taking the predicate-obverse of each premise. If you cannot reach the conclusion, explain why.

7. Prove two moods of the third figure by reductio, choosing whichever you wish.

In all these exercises, the student would do well to keep concrete examples of the moods being proved at hand. Examples will make the matter clearer.

1. **Source status:** Full authorial passage from Rafed's digital presentation of the Dār al-Taʿāruf 1402 AH text, displayed page(s) 231. Arabic kaf/ya and paragraph/table whitespace are normalized; substantive digital readings are retained. Page numbers refer to Rafed's displayed pg containers, not an independently collated image of that edition. Any targeted comparison with the 1968 Najaf scan is identified separately.
2. **Exercises retained:** All seven authorial exercises and the closing study instruction are translated. No solutions have been substituted for the exercises. "Same-quality contrapositive" corresponds to عكس النقيض الموافق as distinguished in Book IV.
3. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination · p. 231 https://lib.rafed.net/view.php?b_id=1604&page=231&type=c_fbook

M-V-19

READING SECTION 9

Conditional conjunctive syllogism: terms, divisions, and connected premises

الاقتراني الشرطي: تعريفه وحدوده وأقسامه والمؤلف من المتصلات

Conditional conjunctive syllogism: its definition and terms. The meaning and terms of categorical conjunctive syllogism have already been explained. Conditional conjunctive syllogism differs only in containing a conditional proposition, either as both premises or as one of them. Its terms are therefore the same as those of categorical syllogism: middle, minor, and major. A term may, however, be the antecedent or consequent of a conditional; the middle in particular may also be part of an antecedent or consequent, as will be explained.

We may therefore define it as **a conjunctive syllogism in which some or all of the premises are conditional propositions**.

Its divisions. Conditional conjunctive syllogism has two divisions.

1. By its premises. It may consist of two connected conditionals, two disjunctive conditionals, one connected and one disjunctive conditional, a categorical and a connected conditional, or a categorical and a disjunctive conditional. These are five kinds.

2. By whether the middle is a complete or an incomplete component. A conditional consists of two sides. Two propositions may share a complete component—the whole antecedent or consequent of each—or an incomplete component—part of the antecedent or consequent of each. They may also share a component complete in one premise and incomplete in the other. These give three kinds.

First: the premises share a complete component of each. For example:

Whenever a person is rational, that person is content with what suffices;

whenever that person is content with what suffices, that person is free of need;

therefore, whenever a person is rational, that person is free of need.

Second: the premises share an incomplete component of each. For example:

If the Qurʾān is a miracle, the Qurʾān endures;

if endurance means persistence, what endures does not change;

therefore, if the Qurʾān is a miracle, then, if endurance means persistence, the Qurʾān does not change.

Observe carefully that the minor's consequent, "The Qurʾān endures," and the major's consequent, "What endures does not change," form a categorical conjunctive syllogism in the first figure, yielding "The Qurʾān does not change." We make that conclusion the consequent of a conditional whose antecedent is the major's antecedent. We then make this whole conditional the consequent of another conditional whose antecedent is the minor's antecedent. This last conditional is the required conclusion.

That is how a conclusion is obtained from this kind when its premises are connected conditionals. We shall be content with this explanation, without enumerating its divisions or conditions: their discussion would be lengthy, and the form departs from the mind's customary course.

Third: the premises share a component complete in one and incomplete in the other. This kind is conceivable in a syllogism composed of a categorical and a conditional, whose explanation and conditions will follow. With exclusively conditional premises, however, one must suppose one conditional to be simple and the other originally composed of a categorical and a conditional, so that the shared term is a complete component of the first and an incomplete component of the second. For example:

If prophethood is from God, then, if Muḥammad is a prophet, he does not leave his community without direction;

if he does not leave his community without direction, he must appoint a guide;

therefore, if prophethood is from God, then, if Muḥammad is a prophet, he must appoint a guide.

Observe that the minor's consequent and the major together form a conditional syllogism of the first kind, in which the premises share a complete component. In the manner of the first figure, they yield "If Muḥammad is a prophet, he must appoint a guide." We then make that conclusion the consequent of a conditional whose antecedent is the original minor's antecedent. This new conditional is the required conclusion.

That is how a conclusion is obtained from this third kind when it consists of two connected conditionals. We shall be content with this explanation for exclusively conditional premises, for the reason already given concerning the second kind.

What remains is the first kind, where the shared component is complete in both premises, and the third kind when composed of a categorical and a conditional. Since these accord with the mind’s customary course, we shall discuss them at somewhat greater length. As already stated, there are five kinds according to whether the premises are connected, disjunctive, or mixed. We shall proceed in that order.

1. A syllogism composed of connected conditionals. When its premises share a complete component, this kind corresponds exactly to categorical conjunctive syllogism: in forming the four figures, in their conditions of quantity and quality, in their conclusions, and in proving them by conversion, reductio, and assumption. There is no need to repeat the details. The student need only replace the categorical propositions with connected conditionals.

There is, however, a condition: both must be entailment conditionals. This is a general condition for every combination involving connected conditional premises. Coincidental conditionals have no force in producing a conclusion, because the relation between their terms is not intrinsic. They form only the appearance of a syllogism, not a genuine one.

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1. **Source status:** Full authorial passage from Rafed’s digital presentation of the Dār al-Taʿāruf 1402 AH text, displayed page(s) 232–234. Arabic kaf/ya and paragraph/table whitespace are normalized; substantive digital readings are retained. Page numbers refer to Rafed’s displayed pg containers, not an independently collated image of that edition. Any targeted comparison with the 1968 Najaf scan is identified separately.
2. **Targeted 1968 comparison: omitted negation:** Rafed displayed p233 reads «واذا يترك أمته سدي» in the second premise, omitting لم and breaking the shared middle. The inspected 1968 scan, canvas269/visible printed p265, reads «واذا لم يترك أمته سدى». The Arabic base retains the defect; the English “if he does not leave” follows the directly inspected print. https://sites.dlib.nyu.edu/viewer/api/image/books/princeton_aco000780/269/full/!1600,2200/0/default.jpg
3. **Conditional and theological scope:** The religious examples retain the author’s stated premises and conditional nesting. They illustrate formal composition, rather than independently establish the Qurʾān’s permanence or the appointment of a guide. لزومية is rendered “entailment conditional” and اتفاقية “coincidental conditional”; treating both as bare modern material implication would erase a distinction on which the author expressly relies.
4. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination · p. 232–234 https://lib.rafed.net/view.php?b_id=1604&page=232&type=c_fbook

M-V-20

READING SECTION 10

Disjunctive premises: converting affirmative disjunctions to connected conditionals

المؤلف من المنفصلات: تمهيد وتحويل المنفصلة الموجبة

2. A syllogism composed of disjunctive conditionals.

Preliminary discussion. A disjunctive indicates the opposition between its sides in truth and falsity. If it is joined to another disjunctive sharing a complete or incomplete component, a connection between the extremes sufficient to obtain a fixed conclusion may not be apparent. The opposition of one thing to two things entails neither opposition nor lack of opposition between those two themselves. This parallels what we said about two negatives under the second general rule: one thing's disjointness from two others entails neither their mutual disjointness nor its absence. Hence two disjunctives would yield no conclusion, and there would be no syllogism composed of them.

This is correct to a point if we insist on leaving the disjunctives just as they stand. But a disjunctive entails a connected conditional and can be transformed into it. Transforming both disjunctives produces a syllogism of two connected conditionals, yielding a connected conclusion. If we insist on a disjunctive conclusion, a connected conditional can in turn be transformed into a disjunctive entailed by it. We then obtain the disjunctive conclusion we want.

Before investigating this kind of syllogism, therefore, we must know how to transform a disjunctive into an entailed connected conditional, and conversely.

Transforming an affirmative disjunctive into a connected conditional. We explained in the appropriate place that disjunctives have three kinds.

1. Genuine disjunction. This entails four connected conditionals agreeing with it in quantity and quality, and it may be transformed into any of them. In two, the antecedent is one of the original sides and the consequent is the contradictory of the other. A genuine disjunction excludes the joint truth of its sides, so the realization of either entails the other's absence. In the other two, the antecedent is the contradictory of one side and the consequent is the other side itself. A genuine disjunction also excludes their joint absence, so the absence of either entails the realization of the other.

Thus, if "A number is either even or odd" is true as a genuine disjunction, the following four connected conditionals are true:

(1) If the number is even, it is not odd.

(2) If the number is odd, it is not even.

(3) If the number is not even, it is odd.

(4) If the number is not odd, it is even.

2. A disjunction excluding joint truth. It entails only the first two conditionals, in each of which the antecedent is one original side and the consequent the contradictory of the other. Like genuine disjunction, it excludes joint truth, but it does not exclude joint absence.

Thus, if “The thing is either a tree or a stone” is true as a disjunction excluding joint truth, these are true:

(1) If the thing is a tree, it is not a stone.

(2) If the thing is a stone, it is not a tree.

These are not true:

(3) If the thing is not a tree, it is a stone.

(4) If the thing is not a stone, it is a tree.

3. A disjunction excluding joint falsity. It entails only the last two conditionals, in each of which the antecedent is the contradictory of one side and the consequent the other side itself. Like genuine disjunction, it excludes joint absence, but it does not exclude joint truth.

Thus, if “Zayd is either in the water or does not drown” is true as a disjunction excluding joint falsity, these are true:

(3) If Zayd is not in the water, he does not drown.

(4) If Zayd drowns, he is in the water.

The following two conditionals are not true:

(1) If Zayd is in the water, he drowns.

(2) If Zayd drowns, he is not in the water.

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1. **Source status:** Full authorial passage from Rafed’s digital presentation of the Dār al-Taʿāruf 1402 AH text, displayed page(s) 235–237. Arabic kaf/ya and paragraph/table whitespace are normalized; substantive digital readings are retained. Page numbers refer to Rafed’s displayed pg containers, not an independently collated image of that edition. Any targeted comparison with the 1968 Najaf scan is identified separately.
2. **Kinds of disjunction:** حقيقية excludes both joint truth and joint falsity; مانعة الجمع excludes joint truth only; مانعة الخلو excludes joint falsity only. English “either … or” therefore always needs the stated qualification; it is not uniformly exclusive.
3. **Unresolved printed mismatch in the drowning example:** The final item (2) on Rafed p237 reads «اذا غرق زيد فهو ليس في الماء». The inspected 1968 canvas272/visible printed p268 has the same reading. It is indeed a non-entailed conditional, but it is not the second of the two templates just specified: that template would have the antecedent “Zayd does not drown”. The printed example is retained in both Arabic and English; no unmarked negation has been inserted. https://sites.dlib.nyu.edu/viewer/api/image/books/princeton_aco000780/272/full/!1600,2200/0/default.jpg
4. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination · p. 235–237 https://lib.rafed.net/view.php?b_id=1604&page=235&type=c_fbook

M-V-21

Transforming a negative disjunctive into a connected conditional. A negative disjunctive, universal or particular, is transformed into a particular negative connected conditional. A genuine disjunctive gives four, in the same pattern as the affirmative; a disjunction excluding joint truth or joint falsity gives two, likewise following the affirmative pattern.

If, in the manner of a genuine disjunction, we say:

It is never the case that a noun is either inflected or nominative;

the following four connected conditionals are true:

(1) Sometimes it is not the case that, if the noun is inflected, it is not nominative.

(2) Sometimes it is not the case that, if the noun is nominative, it is not inflected.

(3) Sometimes it is not the case that, if the noun is not inflected, it is nominative.

(4) Sometimes it is not the case that, if the noun is not nominative, it is inflected.

Some of these conditionals are not true universally in this example. If, for instance, we make (1) universal—"It is never the case that, if the noun is inflected, it is not nominative"—it is false, because its contradictory is true: "Sometimes, if the noun is inflected, it is not nominative."

Negative disjunctions excluding joint truth or joint falsity are transformed in the same way. The student should supply examples of them.

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1. **Source status:** Full authorial passage from Rafed's digital presentation of the Dār al-Taʿāruf 1402 AH text, displayed page(s) 237. Arabic kaf/ya and paragraph/table whitespace are normalized; substantive digital readings are retained. Page numbers refer to Rafed's displayed pg containers, not an independently collated image of that edition. Any targeted comparison with the 1968 Najaf scan is identified separately.
2. **Negative scope and retained authorial claim:** The negation applies to the conditional relation, while "sometimes" expresses particular quantity; it is not a mere negation of the consequent. The general claim that every negative genuine disjunctive yields all four particular negatives is preserved. It should not be identified without qualification with negation of a modern Boolean exclusive-or: that negation alone does not supply counterinstances for every displayed conditional. The range of circumstances and the author's stronger relational interpretation matter.
3. **Targeted print comparison:** The four grammatical examples and the claim about their particular quantity are present in the inspected 1968 scan, canvas273/visible printed p269. This check confirms the wording, not an independent proof of the rule under another logical semantics. https://sites.dlib.nyu.edu/viewer/api/image/books/princeton_aco000780/273/full/!1600,2200/0/default.jpg
4. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination · p. 237 https://lib.rafed.net/view.php?b_id=1604&page=237&type=c_fbook

M-V-22

Transforming a connected conditional into a disjunctive. An affirmative entailment conditional entails both a disjunction excluding joint truth and a disjunction excluding joint falsity, each agreeing with it in quantity and quality. It may therefore be transformed into either.

First, the disjunction excluding joint truth. It consists of the original antecedent and the contradictory of the consequent. Since the antecedent entails the consequent, it certainly cannot be jointly true with the consequent's contradictory; otherwise the consequent and its contradictory would both be true.

Thus, if "Whenever Zayd drowns, he is in the water" is true, then "Always: either Zayd has drowned or he is not in the water" is true as a disjunction excluding joint truth.

Second, the disjunction excluding joint falsity. Conversely, it consists of the contradictory of the antecedent and the original consequent. Since the antecedent cannot be jointly true with the contradictory of the consequent, the contradictory of the antecedent and the original consequent cannot both be absent. If both were absent, the antecedent's contradictory would be removed by the antecedent, and the consequent by its contradictory. The antecedent and the contradictory of the consequent could then be jointly true. That is a contradiction.

In the foregoing example, therefore, this must be true: "Always: either Zayd has not drowned or he is in the water," as a disjunction excluding joint falsity.

The negative is treated on the pattern of the affirmative when transformed into disjunctions excluding joint truth or joint falsity that agree with it in quantity and quality.

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1. **Source status:** Full authorial passage from Rafed's digital presentation of the Dār al-Taʿāruf 1402 AH text, displayed page(s) 238. Arabic kaf/ya and paragraph/table whitespace are normalized; substantive digital readings are retained. Page numbers refer to Rafed's displayed pg containers, not an independently collated image of that edition. Any targeted comparison with the 1968 Najaf scan is identified separately.
2. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination · p. 238 https://lib.rafed.net/view.php?b_id=1604&page=238&type=c_fbook

M-V-23

Combining disjunctives and its conditions. After these preliminaries, we turn to the subject itself. Since the antecedent and consequent of a disjunctive have no inherent distinction, the two disjunctive premises likewise have no natural priority: either may correctly be made the minor. Accordingly, this kind does not itself form the four figures.

Once the two disjunctives have been transformed into connected conditionals, however, we must attend to the form of their combination, identify its figure, and observe that figure's conditions. We may therefore be obliged to make one premise in particular the minor, so that the resulting figure satisfies its conditions.

Logicians have said much and disagreed considerably about this kind's conditions. Their disagreement apparently arises from failing to observe the need to transform each disjunctive into a connected conditional, and instead trying to take the conclusion directly from the disjunctives.

Some, or most, consequently state among its conditions that both premises must be affirmative, and that they must not both exclude joint truth or both be genuine disjunctions.

Yet if we transform the disjunctives into connected conditionals, we find that they can yield a conclusion even if one is negative, both exclude joint truth, or both are genuine disjunctions. The connected conditionals must simply form a syllogism in one of the four figures satisfying its conditions. For instance, if both premises exclude joint truth, they must be transformed into connected conditionals forming the third figure, as an example will show. In another figure they would not yield a conclusion, since its conditions would not be satisfied.

We may therefore state one general condition: **the two disjunctives must be transformable into connected conditionals forming a syllogism in one of the four figures, with that figure's conditions satisfied.** The student should strive to extract every connected conditional entailed by each premise, then compare them to find a productive syllogism yielding the desired conclusion.

How to obtain the conclusion. From the foregoing we know the required procedure. As stipulated, we are discussing only the first kind, where the premises share a complete component. Proceed as follows.

1. Transform each disjunctive into every connected conditional into which it can be transformed. A genuine disjunction gives four; each of the other two kinds gives two.

2. Compare the conditionals obtained from one premise with those obtained from the other. Select a pair with a repeated middle term and an arrangement satisfying a figure's conditions. Usually more than one pair will be productive; selecting one that yields the proposition sought is enough.

3. Obtain a connected conclusion. If desired, transform it into an entailed disjunctive excluding joint truth or joint falsity.

For example, if both premises are genuine disjunctions, transform each into four connected conditionals. Comparing the four with the four gives sixteen combinations. Examination shows that eight have no repeated middle and so form no syllogism. Of the remaining eight, some yield entailment between the original sides of the two disjunctions, and others between their contradictories, using different figures. The student should choose whichever is most closely suited to the conclusion sought.

For practice, let us test an example. Suppose a judge is presented with a person accused of murder. There is a red stain on the person's garment, which the accused claims is ink. To begin confirming or refuting that claim, the judge says:

This stain is either blood or ink—a disjunction excluding joint truth;

and it is either blood or does not wash out—a disjunction excluding joint falsity.

Transform the first into:

(1) Whenever the stain is blood, it is not ink.

(2) Whenever it is ink, it is not blood.

Transform the second into:

(3) Whenever the stain is not blood, it does not wash out.

(4) Whenever the stain washes out, it is blood.

Comparing (1) and (2) with (3) and (4) gives four combinations. Two have no repeated middle: (3) with (1), and (4) with (2).

The combination of (4) with (1), taking (4) as minor, is in the first figure. It yields:

Whenever the stain washes out, it is not ink.

This connected conclusion may be transformed into either of these disjunctives:

Either the stain washes out or it is ink—excluding joint truth;

either it does not wash out or it is not ink—excluding joint falsity.

The combination of (3) with (2) is also in the first figure and yields:

Whenever the stain is ink, it does not wash out.

It too may be transformed into two disjunctives:

Either the stain is ink or it washes out—excluding joint truth;

either it is not ink or it does not wash out—excluding joint falsity.

Notice that these two disjunctives are the very same as those obtained from the first conclusion. Only the positions of the antecedent and consequent have been interchanged. This makes no difference in a disjunctive, since its components have no natural priority, as repeatedly explained.

1. **Source status:** Full authorial passage from Rafed's digital presentation of the Dār al-Taʿāruf 1402 AH text, displayed page(s) 238–241. Arabic kaf/ya and paragraph/table whitespace are normalized; substantive digital readings are retained. Page numbers refer to Rafed's displayed pg containers, not an independently collated image of that edition. Any targeted comparison with the 1968 Najaf scan is identified separately.
2. **Example assumptions:** The stain's behavior in washing is an assumed premise of the author's illustration. The translation preserves it and the resulting inference; it is not an independently established forensic generalization.
3. **Scope of the blanket conversion procedure:** This procedure depends on the preceding conversion rules, including the author's account of negative disjunctives. The qualification noted at M-V-21 remains relevant; the translation does not recast every disjunction as the same Boolean connective.

4. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination · p. 238–241 https://lib.rafed.net/view.php?b_id=1604&page=238&type=c_fbook

M-V-24

READING SECTION 11

A connected and a disjunctive premise

المؤلف من المتصلة والمنفصلة

3. A syllogism composed of a connected and a disjunctive conditional.

Its varieties. This kind also divides into the three kinds already described. As stipulated, we consider only the first, in which the premises share a complete component.

It has four varieties. The connected conditional may be minor or major, and in either case the shared term may be its antecedent or consequent. In the disjunctive there is no difference between the shared term’s being antecedent or consequent, because its two sides have no inherent distinction.

Conditions and the method of obtaining a conclusion. A connected and a disjunctive conditional yield a conclusion only by reducing the disjunctive to a connected conditional. The syllogism then consists of two connected conditionals and returns, in its conditions and productivity, to the first kind. If this transformation permits a productive syllogism in one of the four figures satisfying its conditions, it yields a conclusion; otherwise it is unproductive.

Some stipulate that the disjunctive must not be negative. This condition is correct to a point, because a negative disjunctive transforms only into a particular negative connected conditional. A particular negative has a productive role in only two moods among all the figures: the fifth of the third figure, consisting of a universal affirmative and a particular negative, and the fourth of the second figure, consisting of a particular negative and a universal affirmative. These two moods are uncommon.

If, however, a negative disjunctive can be transformed into a particular negative connected conditional which, together with the original connected premise, forms either of those moods, the syllogism is productive. The proposed condition is therefore not universally correct.

For example, suppose we say:

It is never the case that this is either a human or a horse—a negative disjunction excluding joint falsity;

whenever this is a human, it is an animal.

These do not yield a conclusion. The disjunctive excludes joint falsity and transforms into:

(1) Sometimes it is not the case that, if this is not a human, it is a horse.

(2) Sometimes it is not the case that, if this is not a horse, it is a human.

Joining (1) to the original connected premise gives no repeated middle. Joining (2) to it gives the first or fourth figure, neither of which is productive with a particular negative.

If, however, we replace the original connected premise with “Whenever this is rational, it is a human,” it combines with (2) into the fourth mood of the second figure and yields:

Sometimes it is not the case that, if this is not a horse, it is rational.

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1. **Source status:** Full authorial passage from Rafed’s digital presentation of the Dār al-Taʿāruf 1402 AH text, displayed page(s) 241–243. Arabic kaf/ya and paragraph/table whitespace are normalized; substantive digital readings are retained. Page numbers refer to Rafed’s displayed pg containers, not an independently collated image of that edition. Any targeted comparison with the 1968 Najaf scan is identified separately.
2. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination · p. 241–243 https://lib.rafed.net/view.php?b_id=1604&page=241&type=c_fbook

M-V-25

4. A syllogism composed of a categorical and a connected conditional.

Its varieties. In this kind, as previously indicated, the shared component must be complete in the categorical and incomplete in the connected conditional. It therefore admits only this one of the three divisions. A categorical’s component is a simple term, whereas a conditional’s component was originally a proposition; the shared component cannot be complete in both or incomplete in both. This is clear.

There are four varieties: the connected conditional may be minor or major, and in either case the shared component may occur in its antecedent or consequent. The two varieties closest to the mind’s natural course are those sharing a component in the consequent, whether the conditional is minor or major.

How to obtain the conclusion. In all four varieties, proceed as follows.

1. Compare the categorical with the side of the connected conditional in which the shared component occurs. Form from them a categorical syllogism in one of the four figures, satisfying that figure’s conditions and yielding a categorical proposition.

2. Take this categorical conclusion together with the other side of the connected conditional, which contains no shared component. Form the connected conclusion from them. The unshared side keeps its original position, antecedent or consequent; the other side is the categorical conclusion

just obtained.

For example:

Whenever a metal is gold, it is rare;

every rare thing is valuable;

therefore, whenever a metal is gold, it is valuable.

We formed a categorical syllogism from the conditional's consequent and the categorical premise. In the first figure it yielded "The metal is valuable." We then made this the consequent of the connected conclusion, keeping the original antecedent, the side in which no component was shared.

A second example:

No free person is abject;

whenever the government is oppressive, everyone present in the country is abject;

therefore, whenever the government is oppressive, no free person is present in the country.

We formed a categorical syllogism from the categorical premise and the conditional's consequent. In the second figure it yielded "No free person is present in the country." We made that the consequent of a conditional whose antecedent is the original conditional's unshared side.

Conditions. Of the four varieties, we shall state only the conditions of the two nearest the mind's natural course: those sharing a component in the consequent, whether the conditional is minor or major, as in the examples.

First, the categorical premise and the consequent must form a figure satisfying the conditions stated for categorical syllogism.

Second, the connected conditional must be affirmative. If it is negative, transform it into an affirmative entailed by it through predicate-obversion. The categorical syllogism then consists of the original categorical and the contradictory of the conditional's consequent, with the conditions of its figure satisfied.

For example:

It is never the case that, if the state is oppressive, some people are free;

every happy person is free.

The universal negative conditional is transformed by predicate-obversion into the universal affirmative:

Whenever the state is oppressive, no people are free.

Joining this to the categorical yields, by the second figure and the method already described:

Whenever the state is oppressive, no people are happy.

Remark. This kind, composed of a categorical and a connected conditional, is very important in inference. In particular, reductio syllogism can be analyzed into one of its two naturally ordered varieties. Keep this in mind; we shall explain later how reductio is analyzed in that way.

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1. **Source status:** Full authorial passage from Rafed's digital presentation of the Dār al-Taʿāruf 1402 AH text, displayed page(s) 243–245. Arabic kaf/ya and paragraph/table whitespace are normalized; substantive digital readings are retained. Page numbers refer to Rafed's displayed pg containers, not an independently collated image of that edition. Any targeted comparison with the 1968 Najaf scan is identified separately.
2. **Scope of the account:** The author outlines a general procedure for four varieties, then expressly states conditions only for the two in which the shared component lies in the consequent. The translation preserves this limitation; it does not independently validate unrestricted substitution into a conditional antecedent.
3. **Digital duplication and example vocabulary:** Rafed p243 duplicates «النوع أن» in its opening. The Arabic retains it and the English gives the sentence once. ذليل is "abject", and حر "free"; the political examples remain the author's stipulated premise sets.
4. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination · p. 243–245 https://lib.rafed.net/view.php?b_id=1604&page=243&type=c_fbook

M-V-26

5. A syllogism composed of a categorical and a disjunctive conditional. As in the preceding kind, the shared component must be complete in the categorical and incomplete in the disjunctive. The reason has already been given.

Here, however, the categorical may share a component with every part of the disjunctive—the form nearest the mind's natural course—or only with some parts. In either case the categorical may be minor or major, making four varieties.

For example:

(1) Three is a number.

(2) A number is either even or odd.

(3) Therefore three is either even or odd.

This is the first variety: a categorical minor sharing a component with every part of the disjunctive. The disjunctive is understood as "Always: either the number is even or the number is odd." The word "number," shared between the premises, is thus present in both parts.

You see how the conclusion was obtained: we removed the shared term “number” and put the categorical’s remaining component in its place. The conclusion is also disjunctive. This follows the pattern of the first figure in categorical syllogism. We proceed in the same way to obtain conclusions of this kind, and shall be content with this much explanation.

Conclusion. We have discussed conditional combinations at greater length than is customary in the logic books normally taught, because of their great usefulness and the need for them. Most scientific demonstrations rest on conditional combinations. Even so, we have omitted many investigations that this short work cannot accommodate, restricting ourselves to the most important kinds, which are most readily grasped by the mind’s natural disposition.

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1. **Source status:** Full authorial passage from Rafed’s digital presentation of the Dār al-Taʿāruf 1402 AH text, displayed page(s) 245–246. Arabic kaf/ya and paragraph/table whitespace are normalized; substantive digital readings are retained. Page numbers refer to Rafed’s displayed pg containers, not an independently collated image of that edition. Any targeted comparison with the 1968 Najaf scan is identified separately.
2. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination · p. 245–246 https://lib.rafed.net/view.php?b_id=1604&page=245&type=c_fbook

M-V-27

READING SECTION 12

Exceptional syllogism: composition, division, and conditions

القياس الاستثنائي: تعريفه وتأليفه وتقسيمه وشروطه

Exceptional syllogism: its definition and composition. This syllogism and its definition were mentioned earlier. It belongs to the complete syllogisms: those whose production of a conclusion does not depend on an additional premise of the kind needed in the syllogism of equality and similar cases, to be discussed among the appendices.

We said that either the conclusion itself or its contradictory is explicitly stated in an exceptional syllogism. It cannot, however, be stated as an independent premise whose truth is conceded. That would make the inference a begging of the question. Saying that the conclusion or its contradictory is explicitly stated therefore means that it occurs as part of a premise.

Since it is itself a proposition, yet forms part of a proposition, the premise containing it must be conditional: a conditional is originally composed of two propositions. One premise of this syllogism must accordingly be conditional. The other is the **exceptive premise**, containing the exceptive expression from which the syllogism takes its name. It asserts one of the conditional’s sides or its contradictory, so as to yield the other side or its contradictory, as will be explained.

Its division. The conditional may be connected or disjunctive. Accordingly, exceptional syllogism divides into the connected and the disjunctive kinds.

Its conditions. Three things are required:

1. One premise must be universal: two particular premises yield no conclusion.

2. The conditional must not be coincidental.

3. The conditional must be affirmative. For connected conditionals in particular, this means that a negative is transformed into an affirmative entailed by it, which is put in its place.

Each of the two kinds has its own rule of productivity, which we shall now explain in detail.

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1. **Source status:** Full authorial passage from Rafed's digital presentation of the Dār al-Taʿāruf 1402 AH text, displayed page(s) 247–248. Arabic kaf/ya and paragraph/table whitespace are normalized; substantive digital readings are retained. Page numbers refer to Rafed's displayed pg containers, not an independently collated image of that edition. Any targeted comparison with the 1968 Najaf scan is identified separately.
2. **Equality comparison: retained printed wording and interpretation:** Both Rafed p247 and inspected 1968 canvas284 contain «لا يتوقف فيها على الانتاج مقدمة أخرى كقياس المساواة ونحوه». Taken as a loose comparison, «كقياس» could misleadingly make equality an example of a complete syllogism. The English construes it as referring to the additional premise needed in equality, in accord with the explicit definition on pp200–201 and the later equality analysis. The Arabic is unchanged. Canvas284 has no visible folio; printed p280 is inferred from adjacent numbered pages, not asserted as visible. https://sites.dlib.nyu.edu/viewer/api/image/books/princeton_aco000780/284/full/!1600,2200/0/default.jpg
3. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination · p. 247–248 https://lib.rafed.net/view.php?b_id=1604&page=247&type=c_fbook

M-V-28

The rule for the connected kind. There are two ways to obtain a conclusion from a connected exceptional syllogism.

1. Assert the antecedent itself, to yield the consequent itself. Whenever that which entails is realized, what it entails is certainly realized, whether the entailed is more general than or coextensive with it. Asserting the consequent, however, need not yield the antecedent: the consequent may be more general, and the realization of the more general does not entail the realization of the more specific.

For example:

Whenever water is flowing, it is protected from ritual defilement. But this water is flowing. Therefore it is protected.

If instead we said, "But it is protected," this would not yield "Therefore it is flowing," because it may be protected while stagnant, if present in a large quantity.

2. Assert the contradictory of the consequent, to yield the contradictory of the antecedent. If what is entailed is absent, that which entails it is certainly absent, even when the entailed is more general. Asserting the contradictory of the antecedent, however, does not yield the contradictory of

the consequent, because the consequent may be more general. Denial of the more specific does not entail denial of the more general: the contradictory of the more specific is more general than the contradictory of the more general.

For example:

Whenever water is flowing, it is protected from ritual defilement. But this water is not protected. Therefore it is not flowing.

If instead we said, “But it is not flowing,” this would not yield “It is not protected.” It may be non-flowing yet protected because of its large quantity.

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1. **Source status:** Full authorial passage from Rafed’s digital presentation of the Dār al-Taʿāruf 1402 AH text, displayed page(s) 248–249. Arabic kaf/ya and paragraph/table whitespace are normalized; substantive digital readings are retained. Page numbers refer to Rafed’s displayed pg containers, not an independently collated image of that edition. Any targeted comparison with the 1968 Najaf scan is identified separately.
2. **Targeted 1968 comparison: missing negation:** Rafed displayed p248 reads «وثبوت الاعم يلزم منه ثبوت الاخص», reversing the surrounding argument by omitting لا. The inspected 1968 scan, canvas285/visible printed p281, reads «وثبوت الأعم لا يلزم منه ثبوت الأخص». The English follows that directly attested negation; the Arabic retains the digital omission. https://sites.dlib.nyu.edu/viewer/api/image/books/princeton_aco000780/285/full/!1600,2200/0/default.jpg
3. **Legal example:** معتصم is water protected from becoming ritually impure merely through contact with impurity, as used in the author’s legal example. The logical point is that the consequent may have a wider extension than the antecedent.
4. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination · p. 248–249 https://lib.rafed.net/view.php?b_id=1604&page=248&type=c_fbook

M-V-29

The rule for the disjunctive kind. There are three ways to obtain a conclusion from a disjunctive exceptional syllogism.

1. If the conditional is a genuine disjunction, asserting either side yields the contradictory of the other, and asserting the contradictory of either yields the other side itself. If you say, “A number is either even or odd,” the exceptive premise can take four forms:

(a) But this number is even; therefore it is not odd.

(b) But this number is odd; therefore it is not even.

(c) But this number is not even; therefore it is odd.

(d) But this number is not odd; therefore it is even.

This is straightforward. It applies when the disjunctive has two parts. It may instead have three or more, as in “A word is either a noun, a verb, or a particle.” If you assert one part—for example, “But it is a noun”—you obtain as many categorical conclusions as there are remaining parts: “It is not a verb and it is not a particle.”

If you assert the contradictory of one part—for example, “But it is not a noun”—you obtain a disjunction of the remaining parts: “This word is either a verb or a particle.” You may then use this conclusion as a premise in another exceptional syllogism, asserting one of its parts or its contradictory, until one determinate part remains.

The same method can be used when a disjunction has more than three parts. Continue exhausting the alternatives until only one remains and the matter is confined to it. This is sometimes called the **method of circulation and alternation**, the **proof by examination and division**, or the **proof by exhaustive enumeration**. We already used it in Part One to establish the relation between contradictories in the discussion of relations. It is very useful in disputation and dialectic.

2. If the conditional excludes joint falsity, asserting the contradictory of either side yields the other side itself. Asserting one side does not yield the contradictory of the other, because the assumption permits their joint truth. One side’s truth therefore does not require the other’s falsity.

3. If the conditional excludes joint truth, asserting either side yields the contradictory of the other. Asserting the contradictory of one does not yield the other side itself, because the assumption permits both sides to be absent. The falsity of one therefore does not require the truth of the other. This and the preceding case are clear.

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1. **Source status:** Full authorial passage from Rafed’s digital presentation of the Dār al-Taʿāruf 1402 AH text, displayed page(s) 249–250. Arabic kaf/ya and paragraph/table whitespace are normalized; substantive digital readings are retained. Page numbers refer to Rafed’s displayed pg containers, not an independently collated image of that edition. Any targeted comparison with the 1968 Najaf scan is identified separately.
2. **Method names:** The three names are الدوران والترديد، السبر والتقسيم، الاستقصاء. Here they designate successive elimination within an exhaustive disjunction, not empirical induction. The disjunction must actually cover the alternatives for the final elimination to establish the remaining one.
3. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination · p. 249–250 https://lib.rafed.net/view.php?b_id=1604&page=249&type=c_fbook

M-V-30

READING SECTION 13

Implicit syllogism and enthymeme

خاتمة في لواحق القياس: القياس المضمر أو الضمير

Conclusion: appendices to syllogism.

Implicit syllogism, or enthymeme. In much of our speech and writing we use syllogisms without necessarily noticing them. Usually, however, we do not adhere to their logical form. We may omit a premise or the conclusion because it is obvious, because we rely on the listener’s intelligence, or through inattention. We may state the conclusion before the premises or depart from their natural order. It can therefore be difficult to restore our words to a complete syllogistic form.

A syllogism whose conclusion or one premise is omitted is called an **implicit syllogism**. One whose major premise alone is omitted is called an **enthymeme**. For example, “This is a human because it is rational” originally has this form:

This is rational—the minor premise;

everything rational is a human—the major premise;

therefore this is a human—the conclusion.

The major has been omitted and the conclusion placed first.

You may instead say, “This is a human because everything rational is a human,” omitting the minor while placing the conclusion first.

You may also say, “This is rational because everything rational is a human,” giving only the two premises and omitting the conclusion because it is known. Apply the same analysis to the examples you encounter.

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1. **Source status:** Full authorial passage from Rafed’s digital presentation of the Dār al-Taʿāruf 1402 AH text, displayed page(s) 251. Arabic kaf/ya and paragraph/table whitespace are normalized; substantive digital readings are retained. Page numbers refer to Rafed’s displayed pg containers, not an independently collated image of that edition. Any targeted comparison with the 1968 Najaf scan is identified separately.
2. **Unresolved printed connective:** The last example reads «هذا ناطق لأن كل ناطق انسان» in Rafed p251 and in the inspected 1968 canvas288. The author describes it as two premises with the conclusion omitted, but لأن (“because”) misleadingly makes the first proposition sound like a conclusion. Both Arabic and English retain that connective, with this explicit qualification. No silent substitution of “and” is made. The canvas has no visible folio; p284 is inferred from the adjacent sequence. https://sites.dlib.nyu.edu/viewer/api/image/books/princeton_aco000780/288/full/!1600,2200/0/default.jpg
3. **Terminological restriction:** Here “enthymeme” is the narrower ضمير with the major premise omitted; the author’s broader قياس مضمر also includes an omitted minor or conclusion.
4. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination · p. 251 https://lib.rafed.net/view.php?b_id=1604&page=251&type=c_fbook

M-V-31

Acquiring premises by analysis. You will remember that, at the beginning of the book, we described five stages through which the mind passes to reach the unknown. The last three, we said, constitute **thought**. At the end of Part One we applied these stages to acquiring a definition. Having now studied the kinds of syllogism, we can apply them to acquiring knowledge by assent. Let us recall and clarify the five stages.

1. Confronting the problem. This is indispensable for someone seeking premises in order to obtain something unknown. Unless one has noticed and confronted an unknown problem, and been perplexed by ignorance of it, one will not think about how to solve it. This stage is therefore a preliminary to thought, not thought itself.

2. Identifying the kind of problem. In respect of form, one must identify whether it is a categorical proposition, a connected or disjunctive conditional, affirmative or negative, with a negative term or with simple terms, modal or nonmodal, and so forth. In respect of matter, one must identify which sciences, branches of knowledge, rules, or theories it concerns.

This identification is required before thinking and acquiring premises. Otherwise one would remain where one was, cast into a sea of information that only increased one's ignorance. The mind would become dull and unable even to turn to its knowledge, much less organize that knowledge and solve the problem with it. This stage too is indispensable to thought but is its preliminary, not part of thought itself.

3. The mind's movement from the problem to what is known. This is the first stage and movement of thought. Once someone has confronted the problem and identified its kind, the mind turns to the way of solving it. One returns to stored knowledge, searching through it to seize what will help solve the problem. This turning and returning is the mind's movement from the unknown to the known. It begins thinking and is therefore its first stage.

4. The mind's movement among what is known. This is the second stage of thought, the most important and difficult of its movements. It distinguishes thinkers; here feet slip and the overconfident become entangled. Whoever can examine what is known well, return to self-evident truths, and find what really leads to the problem's solution has received a great share of knowledge.

There are no precisely fixed rules for examining what is known and obtaining premises that lead to the solution and uncover the unknown. We do, however, have a general method on which we can rely for acquiring premises. We call it **analysis**, and it is the reason for this section.

When we confront a problem, it must be a proposition of some kind. Suppose it is categorical and we wish to solve it through categorical conjunctive syllogism. Proceed as follows.

First, analyze the proposition sought into subject and predicate. Its subject must be the minor term of the syllogism, and its predicate the major. Set out each term separately.

Second, seek everything that can be predicated of either extreme, and everything of which either extreme can be predicated, whether genus, species, differentia, property, or common accident. Also seek everything that can be denied of either extreme, and everything of which either extreme can be denied. This gives us a collection of affirmative and negative categorical propositions.

Third, examine the information obtained. Match propositions in which the minor is subject or predicate against propositions in which the major is subject or predicate. If we can match a pair so that they form a figure satisfying its conditions, we have succeeded and can reach the proposition sought. Otherwise we must seek another route.

The very same method is followed when the proposition sought is conditional. We form our stock of knowledge from conditional propositions, unless we choose to reduce the conditional to a categorical entailed by it.

If we wish to solve the problem through exceptional syllogism, proceed as follows.

First, search for everything that entails the proposition sought and everything entailed by it; then everything that entails its contradictory and everything entailed by that contradictory.

Second, search for everything opposed to its contradictory in both truth and falsity, in truth alone, or in falsity alone.

Third, from the first search, form connected conditionals wherever suitable materials are found. Try asserting the antecedent or the contradictory of the consequent of each. Whichever is correct forms a connected exceptional syllogism through which we pass to the proposition sought.

Alternatively, from the second search form genuine disjunctions, or either of their two sister kinds, wherever suitable materials are found. Try asserting the other part, which is opposed to the proposition sought or its contradictory, and try asserting the contradictory of the other part in the propositions formed. Whichever is correct forms a disjunctive exceptional syllogism through which we pass to the proposition sought.

5. The mind’s movement from what is known to the unknown. This is thought’s final stage. Once a productive syllogism has been completely formed, the mind must pass from it to the conclusion: the proposition sought and the solution to the problem.

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1. **Source status:** Full authorial passage from Rafed’s digital presentation of the Dār al-Taʿāruf 1402 AH text, displayed page(s) 251–254. Arabic kaf/ya and paragraph/table whitespace are normalized; substantive digital readings are retained. Page numbers refer to Rafed’s displayed pg containers, not an independently collated image of that edition. Any targeted comparison with the 1968 Najaf scan is identified separately.
2. **Method and restrictions:** The search instructions propose candidate premises; “whichever is correct” retains the need for true or conceded premises and a valid syllogistic arrangement. The compressed disjunctive instruction on p254 does not cancel the distinct permissible assertions for genuine disjunction, exclusion of joint truth, and exclusion of joint falsity established immediately before it.
3. **Term formation:** معدولة and محصلة are rendered as propositions with a negative term and propositions with simple terms, preserving the distinction between negation inside a term and negative propositional quality. The source’s minor spelling defects remain in the Arabic transcription.
4. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination · p. 251–254 https://lib.rafed.net/view.php?b_id=1604&page=251&type=c_fbook

Compound syllogisms: definition and two forms

القياسات المركبة: تمهيد وتعريف وأقسام

Compound syllogisms: preliminary explanation and definition. To establish the proposition sought, analysis must eventually reach self-evident premises whose knowledge requires no acquisition or inquiry. Otherwise analysis would continue without end and the proposition sought could never be obtained.

This termination in self-evident premises occurs in two ways. Sometimes analysis immediately yields two self-evident premises, stops there, and obtains the proposition sought from them. Together they form a **simple syllogism**, because it alone obtains the conclusion. This is the supposition underlying the kinds and divisions of syllogism discussed so far.

At other times analysis initially reaches two premises of which one or both require acquisition. The process cannot then stop. A premise requiring acquisition becomes another proposition sought, for which we must again acquire premises. We form another syllogism whose conclusion is that very premise: the second syllogism's conclusion becomes a premise of the first. If both premises require acquisition, two syllogisms must be formed to obtain them.

If all the premises assembled at this second stage are self-evident, acquisition stops with them. If some or all require acquisition, further syllogisms must be formed for each of them. We continue until we arrive at self-evident premises requiring no acquisition or inquiry.

Such ordered combinations, in which the conclusion of one becomes a premise of another and all lead to one original proposition sought, are called a **compound syllogism**, because they consist of two or more syllogisms. Its definition is therefore: **that which is composed of two or more syllogisms to obtain a single proposition sought**.

We often use compound syllogisms. Consequently, a demonstration may contain many premises, more than two, all directed toward one conclusion. An inexperienced person may suppose it to be one syllogism. In fact, it resolves into several coordinated syllogisms in the manner just explained. Intermediate conclusions or some premises have merely been omitted, following the previously explained practice of implicit syllogism. Restoring the original form may require intelligence and practice.

The divisions of compound syllogism. It divides into connected and disconnected forms.

1. Connected: the intermediate conclusions are not suppressed. Each is stated once as the conclusion of one syllogism and again as the premise of another. For example:

(a) Every poet is sensitive.

(b) Every sensitive person suffers.

Therefore every poet suffers.

Take this conclusion as a premise of another syllogism to yield the original proposition for which the first syllogism was introduced. Beginning again:

(a) Every poet suffers.

(b) Everyone who suffers has strong feeling.

Therefore every poet has strong feeling.

2. Disconnected: the intermediate conclusions are detached and suppressed rather than stated. Using the same example:

(a) Every poet is sensitive.

(b) Every sensitive person suffers.

(c) Everyone who suffers has strong feeling.

Therefore every poet has strong feeling.

This is the same conclusion as in the connected form. The disconnected form is more commonly used in the sciences: the intermediate conclusions are omitted because they are clear.

Some compound syllogisms receive special names because of particular features. There is no harm in discussing some of them to illuminate the mind.

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1. **Source status:** Full authorial passage from Rafed's digital presentation of the Dār al-Taʿāruf 1402 AH text, displayed page(s) 255–257. Arabic kaf/ya and paragraph/table whitespace are normalized; substantive digital readings are retained. Page numbers refer to Rafed's displayed pg containers, not an independently collated image of that edition. Any targeted comparison with the 1968 Najaf scan is identified separately.
2. **Two senses of connectedness:** Here موصول and مفصول concern whether intermediate conclusions are explicitly repeated. They must not be confused with متصل and منفصل, the connected and disjunctive types of conditional proposition.
3. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination · p. 255–257 https://lib.rafed.net/view.php?b_id=1604&page=255&type=c_fbook

Reductio syllogism. We have mentioned reductio twice: first in the initial remarks following the third figure, where we called it the reductio method and explained it to some extent, using it to prove certain moods of the second and third figures; and again at the end of the fourth kind of conditional conjunctive syllogism, composed of a connected conditional and a categorical, where we noted that reductio resolves into such a conditional syllogism. The student would do well to return to those two discussions before entering the following details.

A researcher may be unable to establish the proposition sought directly and may devise an indirect approach. One seeks a proof invalidating its contradictory in order to establish the proposition itself, because two contradictories cannot both be false. Invalidating the contradictory to establish the proposition sought is called **reductio syllogism**. This is why, in the remarks following the third figure, we described reductio as indirect inference.

We may therefore define it as **a compound syllogism that establishes the proposition sought by invalidating its contradictory**.

It is compound because it consists of two syllogisms: a conditional conjunctive syllogism made from a connected conditional and a categorical, and an exceptional syllogism.

Its procedure. To establish a proposition by invalidating its contradictory, we use the method now to be explained. First return to our earlier uses of reductio. Take the fourth mood of the second figure as the example:

Assumed true: (1) some B is not M; (2) all C are M.

Claimed conclusion: some B is not C.

The proof in outline is this: if the proposition sought were not true, its contradictory would be true. But its contradictory is not true, because its truth would entail a contradiction. Therefore the proposition sought must be true. This is an exceptional syllogism whose major premise is established by showing the resulting contradiction. That contradiction is shown through a conditional conjunctive syllogism composed of a connected premise—whose antecedent denies the proposition sought and whose consequent asserts its contradictory—and a categorical assumed true.

The detailed proof has the following stages.

1. Take the contradictory of the proposition sought, “All B are C,” and join it to a premise assumed true, namely the major, “All C are M.” They form a first-figure syllogism:

All B are C; all C are M; therefore all B are M.

2. Compare this result with the other premise assumed true, “Some B is not M.” They are contradictories. Either “Some B is not M” is false, contrary to its assumed truth—a contradiction—or the newly obtained “All B are M” is false. The latter is therefore determined.

3. The falsity of that conclusion must arise from the falsity of one premise, since by assumption there is no defect in the syllogism’s arrangement. The premise assumed true cannot be false. The other premise, which is the contradictory of the proposition sought—“All B are C”—must therefore be false. The proposition sought, “Some B is not C,” is established.

4. Finally, set out the inference as follows.

(a) A conditional conjunctive syllogism:

(1) Minor: if “Some B is not C” is not true, then all B are C.

(2) Major, assumed true: all C are M.

By the method of obtaining a conclusion from the fourth kind of conditional syllogism, this yields:

If “Some B is not C” is not true, then all B are M.

(b) An exceptional syllogism:

(1) Minor: the preceding conditional conclusion—if “Some B is not C” is not true, then all B are M.

(2) Major: but “All B are M” is false, because its contradictory, “Some B is not M,” is true by assumption.

Therefore “Some B is not C” must be true, which is the proposition sought.

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1. **Source status:** Full authorial passage from Rafed’s digital presentation of the Dār al-Taʿāruf 1402 AH text, displayed page(s) 257–259. Arabic kaf/ya and paragraph/table whitespace are normalized; substantive digital readings are retained. Page numbers refer to Rafed’s displayed pg containers, not an independently collated image of that edition. Any targeted comparison with the 1968 Najaf scan is identified separately.
2. **Order and indirectness:** The final exceptional syllogism follows the author’s naming of its conditional as minor and its exceptive premise as major. The indirect proof uses the incompatibility and exhaustiveness of contradictories; it does not assert that a false conclusion makes both original premises false.
3. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination · p. 257–259 https://lib.rafed.net/view.php?b_id=1604&page=257&type=c_fbook

M-V-34

The syllogism of equality. Among the difficult syllogisms that can be reduced to a compound syllogism is the syllogism of equality. Its name comes from the familiar original example: “A equals B; B equals C; therefore A equals C.” It can also involve likeness, resemblance, and similar

relations. Examples include “A human comes from a sperm-drop; the sperm-drop comes from the elements; therefore a human comes from the elements,” and “Body is a constituent of animal; animal is a constituent of human; therefore body is a constituent of human.”

The correctness of the syllogism of equality depends on a true external premise that has been omitted: for example, an equal of an equal is equal; a constituent of a constituent is a constituent; a thing like what is like a thing is itself like that thing; and so forth. It does not yield the proposed conclusion when that external premise is false. Thus “Two is half of four; four is half of eight” does not yield “Two is half of eight,” because half of a half is not a half.

Analyzing this syllogism. As you see, its arrangement differs from a familiar productive syllogism: the entire middle is not shared. The second premise’s subject, B, is only part of the first premise’s predicate, “equal to B.” We must therefore analyze it and restore it to a properly arranged syllogism by adding the omitted external premise to its two premises.

At first sight, merely adding that premise does not solve the difficulty. It is not apparent how the premises form a syllogism sharing the whole middle, or what kind of syllogism it is. It has therefore been regarded as difficult to resolve into the ordered terms of a syllogism yielding this conclusion. Some count it among simple syllogisms; others among compound ones.

It is more correct to count it as compound. We say that it consists of two syllogisms.

The first syllogism:

Minor: the original first premise, “A is equal to B.”

Major: “Everything equal to B is equal to an equal of C.”

This major is true and is taken from the original second premise, “B is equal to C.” According to that premise, “an equal of C” is another expression for B. “Everything equal to B is equal to B” is a self-evidently true proposition. Substituting the expression “an equal of C” for B gives “What is equal to B is equal to an equal of C.” The first syllogism is accordingly a categorical syllogism in the first figure, whose middle is “equal to B.”

It yields: **A is equal to an equal of C.**

The second syllogism:

Minor: the preceding conclusion, “A is equal to an equal of C.”

Major: the external premise, “What is equal to an equal of C is equal to C.”

This too forms a categorical syllogism in the first figure, with “equal to an equal of C” as middle.

It yields: **A is equal to C**, which is the proposition sought.

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1. **Source status:** Full authorial passage from Rafed's digital presentation of the Dār al-Taʿāruf 1402 AH text, displayed page(s) 259–260. Arabic kaf/ya and paragraph/table whitespace are normalized; substantive digital readings are retained. Page numbers refer to Rafed's displayed pg containers, not an independently collated image of that edition. Any targeted comparison with the 1968 Najaf scan is identified separately.
2. **Targeted 1968 comparison: quantified major:** Rafed displayed p260 gives the first new major as «مساو لب مساو لمساوي ج». The inspected 1968 canvas298/visible printed p294 explicitly has «كل مساو لب مساو لمساوي ج». The English "Everything equal to B …" follows the quantified earlier print; Arabic retains the digital form. https://sites.dlib.nyu.edu/viewer/api/image/books/princeton_aco000780/298/full/!1600,2200/0/default.jpg
3. **Description and transitivity:** "An equal of C" designates B as supplied by the second premise; it is not a claim that only one thing equals C. The external premise supplies the needed transitivity of the relevant relation. Similarity or resemblance is not silently treated as transitive in every interpretation: the author expressly makes the inference depend on the appropriate external premise being true.
4. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination · p. 259–260 https://lib.rafed.net/view.php?b_id=1604&page=259&type=c_fbook

M-V-35

READING SECTION 15

Induction: definition, complete and incomplete

الاستقراء: تعريفه وأقسامه

2. Induction: its definition. We previously defined induction as **the mind's examination of a number of particulars, from which it derives a general judgment**. Suppose, for instance, that we examine several kinds of animal and find that each moves its lower jaw when chewing. We infer the general rule that every animal moves its lower jaw when chewing.

Induction is the basis of all our universal judgments and general rules. We obtain a general rule or universal judgment only after examining and surveying particulars. If we find them united in a judgment, we abstract the general rule or universal judgment from them. Induction is therefore inference from the particular to the general. Its reverse is syllogism, inference from the general to the particular: a syllogism must contain a universal premise whose general judgment is applied to the subject of the conclusion.

Its divisions. Induction is either complete or incomplete, according to whether it surveys all the particulars or only some.

The first, complete induction, yields certainty. It has been said that it returns to the syllogism by division used in demonstrations. For example: every figure is either spherical or polygonal; every spherical figure is finite; every polygonal figure is finite; therefore every figure is finite.

Author's note. Syllogism by division belongs to the kind composed of a disjunctive and categorical premises, but it has as many categoricals as the disjunctive has parts. The disjunctive is not transformed into a connected conditional; it remains as it is. The argument appears to resolve into as many categorical syllogisms as the disjunctive has parts.

The second, incomplete induction, examines only some particulars. An example is the judgment, obtained by surveying most kinds of animal, that an animal moves its lower jaw when chewing. They say it yields only probability, since one of its particulars may lack the judgment. It has, for instance, been said that the crocodile moves its upper jaw when chewing.

1. **Source status:** Full authorial passage from Rafed's digital presentation of the Dār al-Taʿāruf 1402 AH text, displayed page(s) 261–262. Arabic kaf/ya and paragraph/table whitespace are normalized; substantive digital readings are retained. Page numbers refer to Rafed's displayed pg containers, not an independently collated image of that edition. Any targeted comparison with the 1968 Najaf scan is identified separately.
2. **Examples and epistemic scope:** The zoological claims are retained as historical examples, including the author's "it has been said" qualification about the crocodile; they are not independently verified biological findings. The following objection and reply qualify the opening claim that all universal rules come from induction, explicitly excluding self-evident judgments from genuine induction.
3. **Authorial note and digital headings:** The authorial footnote on syllogism by division is translated in full beside its first occurrence. Rafed's headings/spellings «ناقض» ,«الاسقراء», and «بأساها» are retained in the Arabic; the English follows the divisions that the text itself immediately names as complete and incomplete.
4. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination · p. 261–262 https://lib.rafed.net/view.php?b_id=1604&page=261&type=c_fbook

M-V-36

A difficult objection. Syllogism is the principal proof of philosophical propositions and yields certainty. Yet it always depends on a universal premise. Its basis must therefore be induction, since we said that no universal rule is obtained except by examining its particulars.

Most general rules undoubtedly have infinitely many instances, so complete induction is impossible for them. It follows that most rules on which we rely in constructing syllogisms would be merely probable. Most of our syllogisms would then be probable, and most of our arguments in every science and art would be nondemonstrative. No one supposes this to be the case.

Can we therefore claim that incomplete induction yields certain knowledge, in opposition to all the ancient logicians? This claim may seem plausible. We are certain of general matters without having surveyed all their instances. We judge decisively that the whole is greater than the part, although it is impossible to survey every whole and every part; that two is half of four, although it is impossible to survey every two and every four; that every fire burns, and every human dies, although it is impossible to survey every instance of fire and human. There are countless such self-evident rules, quite apart from theoretical ones.

1. **Source status:** Full authorial passage from Rafed's digital presentation of the Dār al-Taʿāruf 1402 AH text, displayed page(s) 262. Arabic kaf/ya and paragraph/table whitespace are normalized; substantive digital readings are retained. Page numbers refer to Rafed's displayed pg containers, not an independently collated image of that edition. Any targeted comparison with the 1968 Najaf scan is identified separately.
2. **Argumentative role:** This unit presents the author's objection, not his final account of induction. The solution follows immediately and distinguishes observation, causal explanation, self-evidence, and complete similarity.
3. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination · p. 262 https://lib.rafed.net/view.php?b_id=1604&page=262&type=c_fbook

The solution to the objection. We answer by distinguishing several ways in which induction proceeds.

1. On observation alone. Someone observes some or most particulars sharing a feature and infers that the feature belongs to all particulars. Surveying some animals and finding that they move their lower jaw when chewing is an example. This inference is open to counterexample, so its judgment is not certain. It is this kind alone that the ancient logicians considered in their discussion.

2. On causal explanation as well. Someone who has observed some particulars investigates the cause of the feature's presence and discovers that it belongs to those observed particulars because of a cause or characteristic present in their kind. Reason has no doubt that an effect never fails to follow its cause. The observer conducting the induction then decisively judges that the feature belongs to every particular of that kind, including those not observed.

For example, a researcher observes a certain medicinal substance producing purgation, investigates the cause of this effect, and analyzes the substance into its elements. Discovering that their effect on the body under ordinary conditions is purgation, the researcher judges with certainty that this substance always produces that effect.

All scientific discoveries and many of our judgments about observed things belong to this kind. These judgments are not open to counterexample and are therefore certain. We judge that water descends from a higher place, for example, without doubt, even though we have observed only a tiny fraction of its instances. This is because we know the explanation of that descent. If, however, a researcher discovers that what was taken to be the cause was mistaken and that the feature has another cause, the judgment and knowledge must change.

3. On rational self-evidence. An example is our judgment that the whole is greater than the part. Conceiving the whole, the part, and the meaning of "greater" suffices for this judgment. This is not really induction, because it does not depend on observation. Conceiving the subject and predicate suffices even if not one instance has been observed.

4. On complete similarity among the particulars. For example, we test some instances of a kind of fruit and find them pleasant-tasting. We then judge decisively that all instances of that kind have this feature. Similarly, if we prove that the angles of a particular triangle equal two right angles, we decisively judge that every triangle is so. Examining one particular is enough. This is because the particulars are alike and similar in constitution, so a feature of one is a feature of all, without difference.

With these four kinds explained, it is clear that incomplete induction does not invariably fail to yield certainty. It does so when based on bare observation alone. The second kind, induction based on causal explanation, is called in modern logic the **method of deduction** or the **method of scientific inquiry**. It has investigations that this book cannot accommodate.

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1. **Source status:** Full authorial passage from Rafed's digital presentation of the Dār al-Taʿāruf 1402 AH text, displayed page(s) 262–264. Arabic kaf/ya and paragraph/table whitespace are normalized; substantive digital readings are retained. Page numbers refer to Rafed's displayed pg containers, not an independently collated image of that edition. Any targeted comparison with the 1968 Najaf scan is identified separately.
2. **Authorial certainty claims:** The statements that causal induction cannot be overturned and that complete similarity justifies certainty are the author's epistemological claims. He himself adds that discovering a mistaken causal attribution requires revising the judgment. The translation preserves both the claim and that qualification, rather than recasting his theory as contemporary fallibilism.
3. **Conditions behind the examples:** The medicinal example explicitly says "under ordinary conditions". The fruit example depends on the additional claim of complete similarity; tasting a sample alone does not establish that premise. The triangle example presupposes the geometry in which the cited theorem holds. These conditions are disclosed, not supplied as new words in the Arabic.
4. **Digital duplication:** Rafed p263 duplicates «وان للوصف علة». The Arabic retains the duplicated phrase; the English renders the intended statement once. "Method of deduction" translates the author's label طريق الاستنباط and does not assimilate every modern use of deduction to his causal induction.
5. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination · p. 262–264 https://lib.rafed.net/view.php?b_id=1604&page=262&type=c_fbook

M-V-38

READING SECTION 16

Analogy: definition and its four components

التمثيل: تعريفه وأركانه

3. Analogy: its definition. This is the third kind of argument, with which the discussions of Book V conclude. As previously defined, analogy is **the mind's passage from a judgment concerning one of two things to a judgment concerning the other because of a feature shared between them**. In other words, it is **establishing a judgment in one particular because it is established in another particular resembling it**.

Analogy is what jurists call **qiyās**, which Sunnis include among the proofs of legal rulings. The Imāmīs deny its probative authority and regard acting on it as effacing religion and wasting the sacred law.

For example, suppose it is established for us that nabīdh resembles wine in intoxicating its drinker, and that wine is prohibited. We may infer that nabīdh too is prohibited, or at least possibly prohibited, because they share the feature of intoxication.

Its components. Analogy has four components:

1. **The original case:** the first particular, in which the judgment is known to be established—wine in the example.

2. **The new case:** the second particular, in which we seek to establish the judgment—nabīdh in the example.

3. **The common feature:** the respect in which the original and new cases resemble one another—intoxication in the example.

4. **The judgment:** what is known to be established in the original and is to be established in the new case—prohibition in the example.

When these components are present, an analogy is formed. If the original case's judgment is unknown, or it lacks the common feature, no analogy is obtained. This is clear.

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1. **Source status:** Full authorial passage from Rafed's digital presentation of the Dār al-Taʿāruf 1402 AH text, displayed page(s) 265–266. Arabic kaf/ya and paragraph/table whitespace are normalized; substantive digital readings are retained. Page numbers refer to Rafed's displayed pg containers, not an independently collated image of that edition. Any targeted comparison with the 1968 Najaf scan is identified separately.
2. **Authorial school attribution:** The contrast between Sunni and Imāmī legal method, including the polemical characterization of analogy, is Muẓaffar's account. The English preserves the attribution and does not present the characterization as an independent judgment by the translator. His discussion of an expressly stated cause in the next unit supplies an important qualification.
3. **Legal and logical terminology:** Nabīdh is retained as the named drink in the hypothetical legal example. Juristic qiyās here means analogy between cases; it must not be confused with the syllogism discussed under the same Arabic word earlier in Book V. الأصل، الفرع، الجامع، الحكم are rendered original case, new case, common feature, and judgment.
4. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination · p. 265–266 https://lib.rafed.net/view.php?b_id=1604&page=265&type=c_fbook

M-V-39

Its epistemic value. For all its simplicity, analogy is among the arguments that yield only possibility. Two things' resemblance in one respect, or even in several, does not require resemblance in every respect. If you see someone resembling another person in height, features, or habits, and one is certainly a criminal, you may not judge the other a criminal merely because they resemble one another in some qualities or actions.

When the points of resemblance between the original and new cases become stronger and more numerous, however, the possibility becomes more likely, approaches certainty, and amounts to probability. Physiognomic inference belongs here. We may judge someone virtuous or wicked merely on seeing them because we previously knew a person who closely resembled them in features or habits and was virtuous or wicked. Yet none of this avails anything toward truth.

We may, however, know that the common feature—the respect of resemblance—is a **complete cause** of the judgment's presence in the original case. We then infer with certainty that the judgment is established in the new case, because its complete cause is present there. An effect cannot fail to follow its complete cause.

Everything turns, however, on establishing that the common feature is indeed the complete cause of the judgment. This requires investigation and examination that are not easy to accomplish even in natural matters. In this respect analogy belongs to the kind of induction based on causal explanation already mentioned; indeed, it is that very kind.

As for establishing the common feature as the complete cause of a legal ruling, we have no route to it except through the Lawgiver himself. Thus, when the cause is expressly stated by the Lawgiver, all jurists agree on using it to establish the ruling in the new case. An example is the saying, peace be upon him: “Well-water is ample; nothing spoils it … because it has a source supplying it.” From this one infers that every water with such a source—bathhouse water or water from a supply tap—is ample and not spoiled by anything.

In fact, an analogy in which the common feature is known to be a complete cause belongs to demonstrative syllogism, yielding certainty. The common feature is then the middle term, the new case the minor, and the judgment the major. In the water example we say:

(1) Bathhouse water has a source supplying it.

(2) Every water with a source supplying it is ample and not spoiled by anything, according to the cause stated in the report.

Therefore bathhouse water is ample and not spoiled by anything.

It thereby ceases to be called analogy, or qiyās in the jurists’ disputed sense.

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1. **Source status:** Full authorial passage from Rafed’s digital presentation of the Dār al-Taʿāruf 1402 AH text, displayed page(s) 266–267. Arabic kaf/ya and paragraph/table whitespace are normalized; substantive digital readings are retained. Page numbers refer to Rafed’s displayed pg containers, not an independently collated image of that edition. Any targeted comparison with the 1968 Najaf scan is identified separately.
2. **Complete cause and legal report:** The move to certainty depends on knowing a complete cause, not merely finding a shared property. The quoted report is reproduced as quoted and abbreviated by the author, including its ellipsis; this unit does not independently collate the report or adjudicate the legal scope of «لا يفسده شيء». “Source supplying it” renders مادة in the water example, not philosophical prime matter.
3. **Probability and resemblance:** احتمال is rendered “possibility” or “likelihood” according to context, and ظن “probability”. The source explicitly rejects inferring criminality or moral character from mere resemblance. Its sharper standard is causal sufficiency, which then changes the argument’s classification.
4. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination · p. 266–267 https://lib.rafed.net/view.php?b_id=1604&page=266&type=c_fbook

M-V-40

READING SECTION 17

Final exercises on syllogisms and the close of Part Two

تمرينات على الأقيسة ونهاية الجزء الثاني

Exercises on syllogisms.

1. Someone argued against mental existence as follows: if quiddities existed in the mind, the mind would be hot and cold when conceiving heat and cold, straight and round, and so on. The consequent is false, so the antecedent is likewise false. Arrange this discourse as a logical syllogism and identify its kind.

2. Someone argued that God Most High knows, because what lacks a thing cannot give it, and He, glorified be He, has created knowledge in us. Therefore He knows. Identify the kind of inference and arrange it in order.

3. It is reported that the learned are the heirs of the prophets. Since they did not inherit money or landed property from them, they inherited knowledge and moral character. Is this a logical inference? Identify its kind.

4. In support of mental existence they argued: “We undoubtedly make affirmative judgments about some impossible things, as when we judge that the conjunction of contradictories differs from the conjunction of contraries. An affirmative proposition requires the existence of its subject. Since this existence is not external, it is in the mind.” How would you arrange this proof according to the logical rules? Identify its kind and whether it is simple or compound, bearing in mind that “Since this existence is not external …” expresses an exceptional syllogism.

5. They argued that the subject of an affirmative proposition must exist because a thing’s being established for something requires the existence of that for which it is established. How would you arrange this as a logical syllogism?

6. Put the following propositions into syllogistic form and identify the kind and figure: “The holder of a demonstrative argument is not overcome,” because “that person is in the right” and “everyone in the right is not overcome.” If the first proposition is instead conditional—“If the argument is demonstrative, its holder is not overcome”—how would you compose the premises to obtain this conditional as their conclusion? What kind of syllogism would it then be?

7. Put the following propositions into syllogistic form and identify its kind: “Only the learned among His servants fear God”; “Since Khālid did not fear God, glorified be He, he is not among the learned.”

8. Which figure yields all four quantified propositions?

9. Investigate why the third figure yields only a particular conclusion.

10. In which figure may the major premise be particular while the syllogism remains productive?

11. If one premise of a syllogism is particular, why must the other be universal?

12. If a syllogism’s minor premise is negative, may its major be particular? Why?

13. How do we obtain a conclusion from these two disjunctives: “A person is either learned or ignorant,” a genuine disjunction; and “A person is either ignorant or happy,” a disjunction excluding joint falsity?

14. Can a productive syllogism be formed from these two disjunctives: “Either the student makes an effort or does not pass the examination,” excluding joint falsity; and “The student either makes an effort or is negligent,” excluding joint truth?

15. A beggar approached someone and pressed the request insistently. From that insistence, the person approached inferred that the beggar did not deserve assistance. This inference uses an exceptional syllogism. How would you extract it?

16. Recast the proofs of the rule of predicate-obversion, on source pages 186–190, as logical syllogisms according to what you have learned about simple and compound syllogism.

17. Try also to apply the rules of syllogism to the proofs of contraposition.

18. The proof of predicate-obversion for the universal affirmative, on source page 186, can be reduced to a syllogism of equality and to a conditional syllogism of two connected premises. How? The same applies to its counterparts.

End of Part Two.

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1. **Source status:** Full authorial passage from Rafed’s digital presentation of the Dār al-Taʿāruf 1402 AH text, displayed page(s) 268–270. Arabic kaf/ya and paragraph/table whitespace are normalized; substantive digital readings are retained. Page numbers refer to Rafed’s displayed pg containers, not an independently collated image of that edition. Any targeted comparison with the 1968 Najaf scan is identified separately.
2. **Exercise wording and cross-references:** All eighteen exercises are retained without supplied solutions. Page references 186–190 and 186 are those in the Rafed 1402 AH digital base, not page numbers in this translation or the 1968 scan. The closing notice is displayed separately at Rafed p270.
3. **Targeted 1968 comparison: missing imperative:** Rafed p269’s exercise16 begins abruptly «إلى قياسات … البراهين». The inspected 1968 canvas308/visible printed p304 reads «أرجع البراهين … إلى قياسات». The English “Recast the proofs” follows that directly attested imperative; the Arabic retains the digital omission. The same image includes exercises17–18 and the Part Two closing notice, but uses different internal page cross-references. https://sites.dlib.nyu.edu/viewer/api/image/books/princeton_aco000780/308/full/!1600,2200/0/default.jpg
4. **Retained exercise problems:** The questions invite logical analysis of the stated wording; the translation does not assume every proposed argument is already sound. In particular, exercise7’s “only” and the direction of its inference should be analyzed, not silently reversed. The source spellings «اللع»، «يحشي», and «الخ بالطلب» are retained in Arabic and rendered by their evident intended words in English.
5. **Source:** Muẓaffar, al-Manṭiq, Dār al-Taʿāruf 1402 AH, Rafed digital presentation; displayed pagination · p. 268–270 https://lib.rafed.net/view.php?b_id=1604&page=268&type=c_fbook

Glossary

Editorial explanations of terms in this reading edition. The online dictionary adds related discussions in the other books and keeps automatic word matches separate.

عرضي · Accidental predicate

ʿaraḍī

A predicate outside a thing's constitutive essence, whether or not it happens to be inseparable from that thing.

تصديق · Assent

taṣdīq

Cognition involving judgment that a proposition is true.

علة الوجود · Cause of being

ʿillat al-wujūd

What explains the thing's existence, distinguished from what merely explains our assent to a proposition about it.

دور · Circularity

dawr

A defect in explanation or definition in which knowing the proposed definition already requires knowing what it purports to define.

طرد وعكس · Coextensiveness

ṭard wa-ʿaks

The requirement that a defining expression apply to every instance of what is defined and to no other instance; necessary for adequacy, not sufficient for essentiality.

عرض عام · Common accident

ʿaraḍ ʿāmm

A nonessential predicate shared by more than one species; it does not by itself provide a distinctive account.

رسم تام · Complete descriptive definition

rasm tāmm

A descriptive account combining the proximate genus with a proprium; the proprium does not become part of the essence.

حد تام · Complete essential definition

ḥadd tāmm

A definition that gives both proximate genus and proximate differentia.

هل المركبة · Composite predicative question

hal murakkaba

Asks whether a predicate belongs to a subject whose existence is presupposed.

تصوّر · Conception

taṣawwur

An intelligible content considered without necessarily assenting to a judgment.

دلالة وضعية · Conventional signification

dalāla waḍʿiyya

The link between sign and signified depends upon an instituted or agreed convention. M-L-02

برهان · Demonstration

burhān

A rigorous inference yielding scientific certainty on the basis of appropriate premises.

برهان إنّي · Demonstration through an effect or sign

burhān innī

Reasoning from what is known to the fact concluded without thereby providing the relevant causal explanation of why it is so.

برهان لمّي · Demonstration through the cause

burhān limmī

Reasoning that explains the conclusion through a cause appropriate to the fact demonstrated; compare demonstration from a sign or effect.

رسم · Descriptive definition

rasm

Description using nonessential but distinctive attributes.

فصل · Differentia

faṣl

The essential constituent by which a species is distinguished within a genus; it is not merely any observable distinctive feature.

حدّ · Essential definition

ḥadd

A specification of essential constituents; not to be confused with delimitation (ḥadd) in ontology.

ذاتي · Essential predicate

dhātī

What enters the essence or its constitutive account in the relevant logical division; not simply something permanently accompanying a subject.

شرح الاسم · Explanation of the name

sharḥ al-ism

An account of what a name signifies, prior to determining whether the named thing exists; an account may later function as a real definition.

ما الشارحة والحقيقية · Explanatory and real 'what'

mā shāriḥa wa-ḥaqīqiyya

The same formal account may answer 'what?' before existence is established (explanatory) and after it is established (real); the epistemic setting differs.

ما الشارحة · Explanatory what?

mā shāriḥa

The question concerning the quiddity before the thing's existence has been established.

جنس · Genus

jins

An essential universal predicable of species differing specifically, supplying the common element in an essential account.

علة التصديق · Ground of assent

ʿillat al-taṣdīq

What justifies the intellect's assent to a claim, which need not be identical with the cause of the thing's existence.

دلالة التزامية · Implicative signification

dalāla iltizāmiyya

The verbal sign leads the mind to a concomitant meaning outside the instituted meaning itself. M-L-03

دلالة تضمنية · Inclusive signification

dalāla taḍammuniyya

The verbal sign signifies a component contained within the whole instituted meaning. M-L-03

رسم ناقص · Incomplete descriptive definition

rasm nāqiṣ

A description using a proprium without the proximate genus, or pairing it with a more remote genus.

حد ناقص · Incomplete essential definition

ḥadd nāqiṣ

An essential account giving less than both proximate genus and proximate differentia, normally by a differentia alone or with a remote genus.

دلالة طبعية · Natural signification

dalāla ṭabʿiyya

A sign indicates something through a naturally disposed or customary reaction that may vary among persons. M-L-02

خاصة · Proprium

khāṣṣa

An attribute distinctive of a species without entering its essential definition, as opposed to the constitutive differentia.

الفصل القريب · Proximate differentia

faṣl qarīb

The immediate essential distinction by which the species is specified.

الجنس القريب · Proximate genus

jins qarīb

The nearest genus included in the subject's essential account.

ماهية · Quiddity

māhiyya

What a thing is, considered as the answer to the question 'What is it?' This need not itself establish its external existence.

دلالة عقلية · Rational signification

dalāla ʿaqliyya

Cognition proceeds from sign to signified through an intrinsic causal or existential concomitance. M-L-02

ما الحقيقية · Real what?

mā ḥaqīqiyya

The same question of quiddity when existence is already established.

جنس بعيد · Remote genus

jins baʿīd

A genus above the proximate genus in a classificatory hierarchy; by itself it leaves more of the specific nature unspecified.

دال · Sign

dāll

The item whose apprehension leads a knower to something else by reason of a known connection. M-L-02

دلالة · Signification

dalāla

An expression or other sign has a relation such that awareness of it leads the mind to another intelligible object. M-L-02

مدلول · Signified

madlūl

What the mind reaches through awareness of a sign and of the relevant connection. M-L-02

هل بسيطة · Simple 'whether'

hal basīṭa

The interrogative asking whether a thing exists at all, rather than whether it possesses some further attribute.

هل البسيطة · Simple existential question

hal basīṭa

Asks whether the subject exists.

دلالة مطابقية · Total signification

dalāla muṭābaqiyya

The verbal sign signifies the entire meaning for which it was instituted. M-L-03

تعريف لفظي · Verbal definition

taʿrīf lafẓī

An explanation of a word through another expression whose customary signification is already understood; it does not yet explain the essence of the thing.

140 authorial reading passages; 140 with English. AI-assisted reading translation; consult the edition note for its coverage and review status.